David L. Nelson and Michael M. Cox

LEHNINGER PRINCIPLES OF BIOCHEMISTRY

Fifth Edition

CHAPTER 21

Lipid Biosynthesis

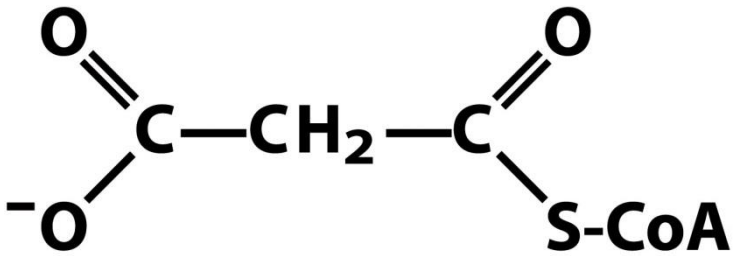
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Biosynthesis of fat



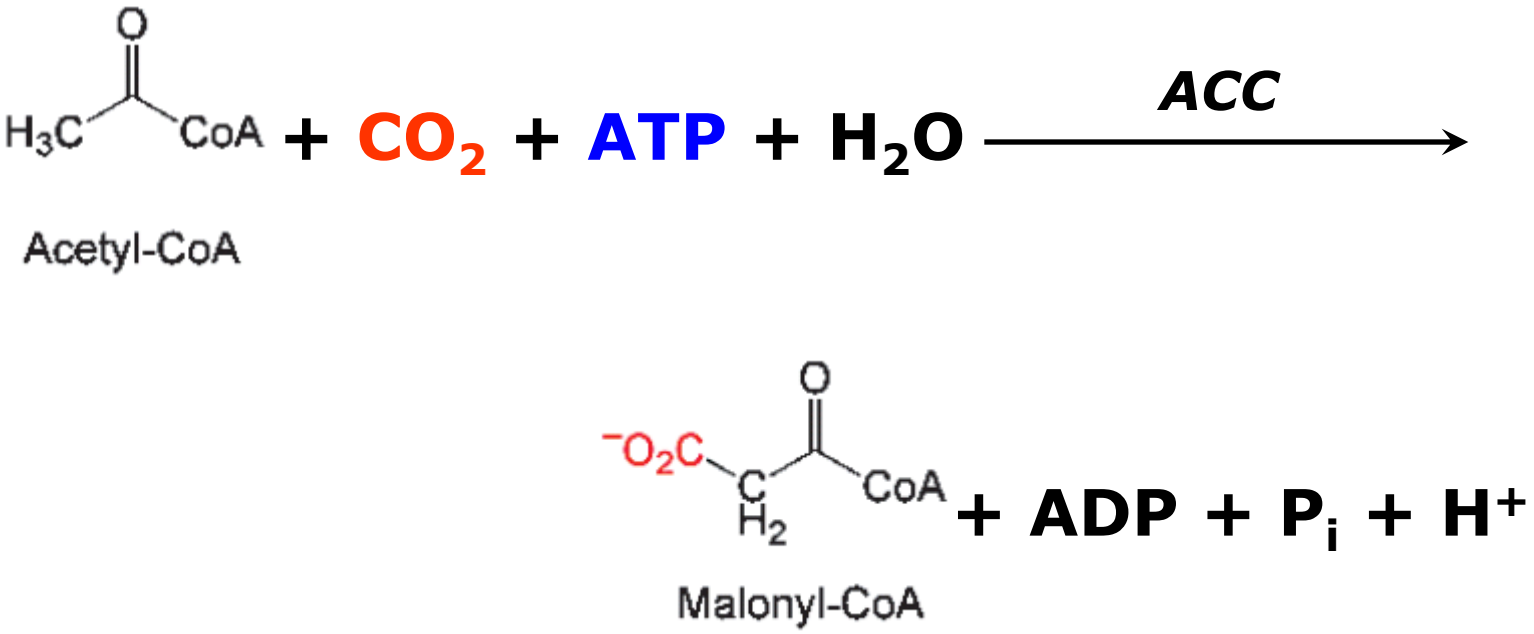
Biosynthesis and breakdown of fatty acids occur by different pathways

1. Where ?
   (cytosol / mitochondria)
2. Enzymes ?
   (fatty acid synthase / β-oxidation enzymes)
3. Intermediate ?

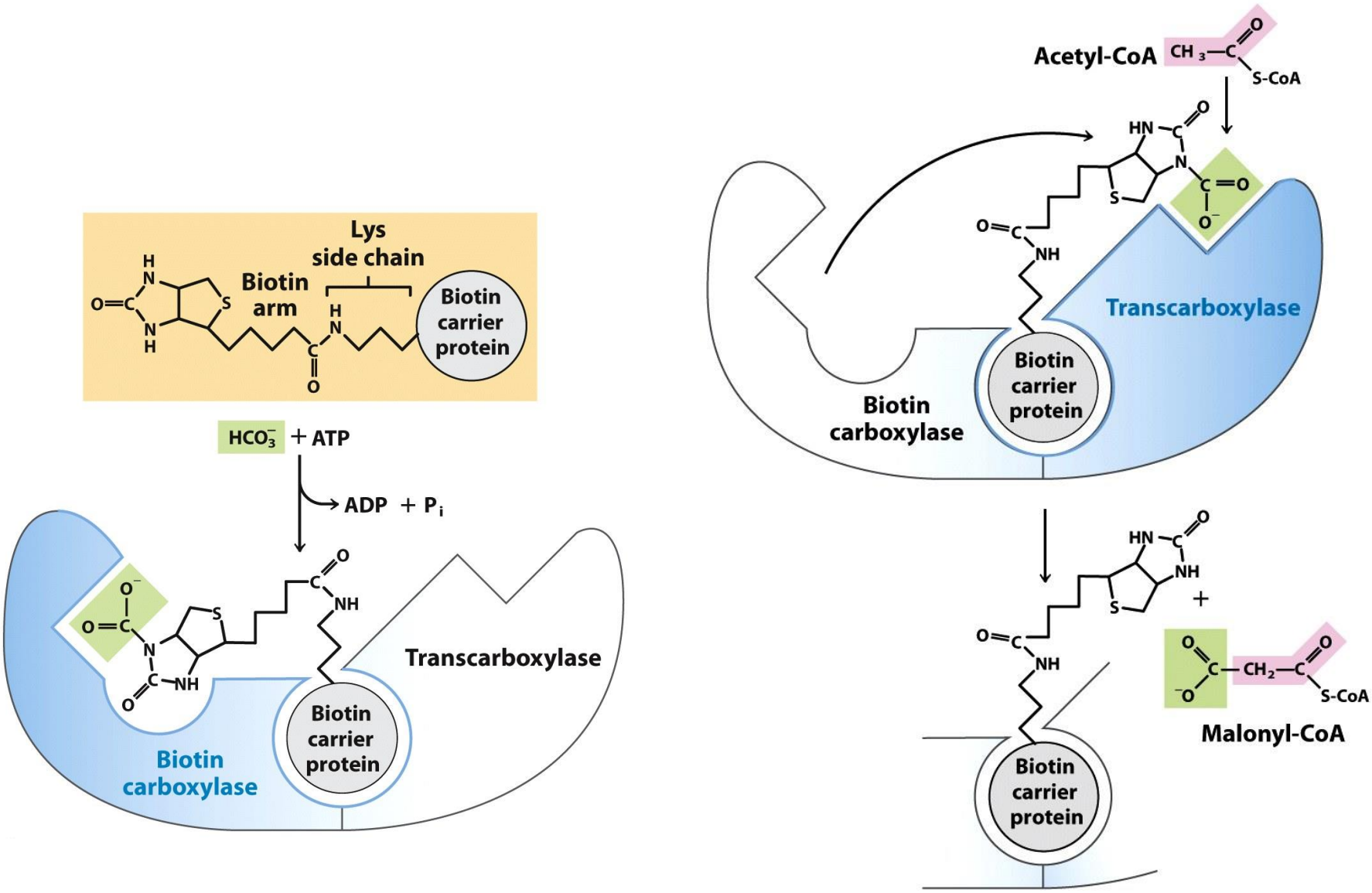


Malonyl-CoA

ACC (acetyl-CoA carboxylase)



The acetyl-CoA carboxylase reaction



fatty acid synthase

FAS I

vertebrate &
fungi(growth)

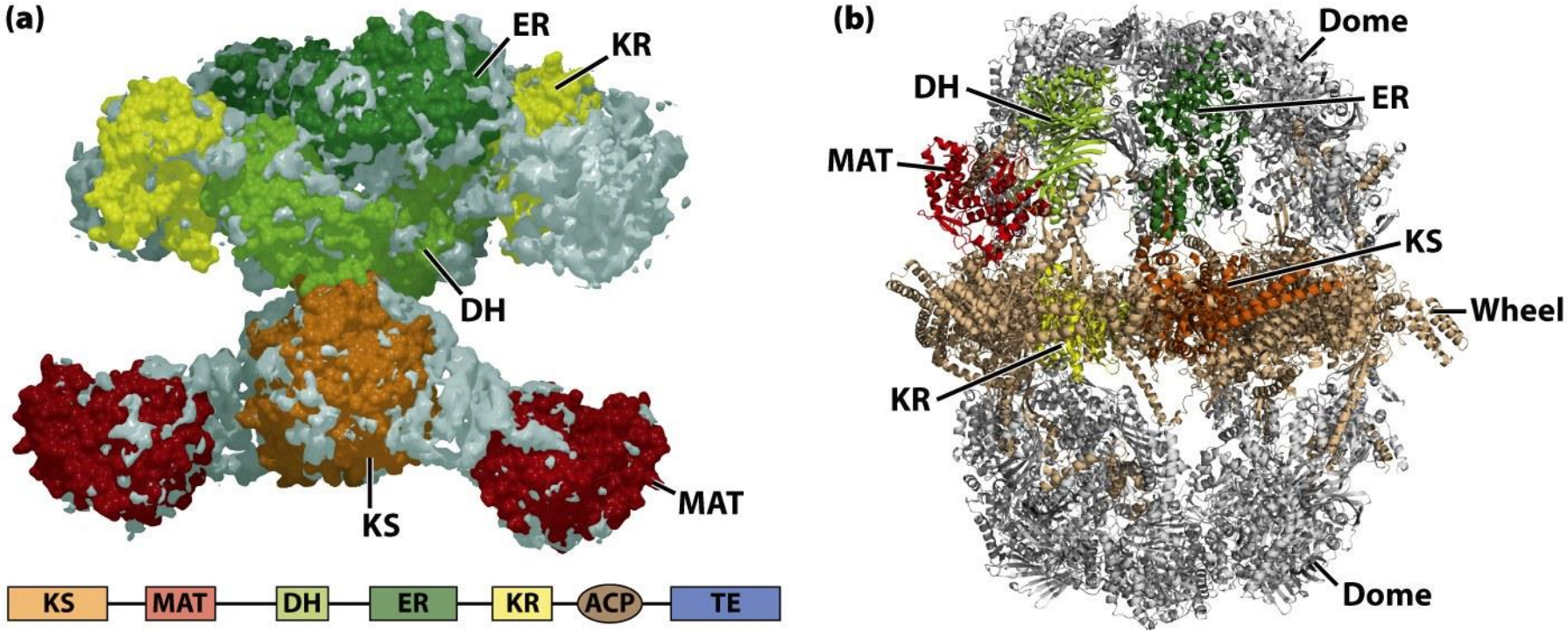
single polypeptide
homodimer or
heterododecamer

FAS II

plants &
bacteria

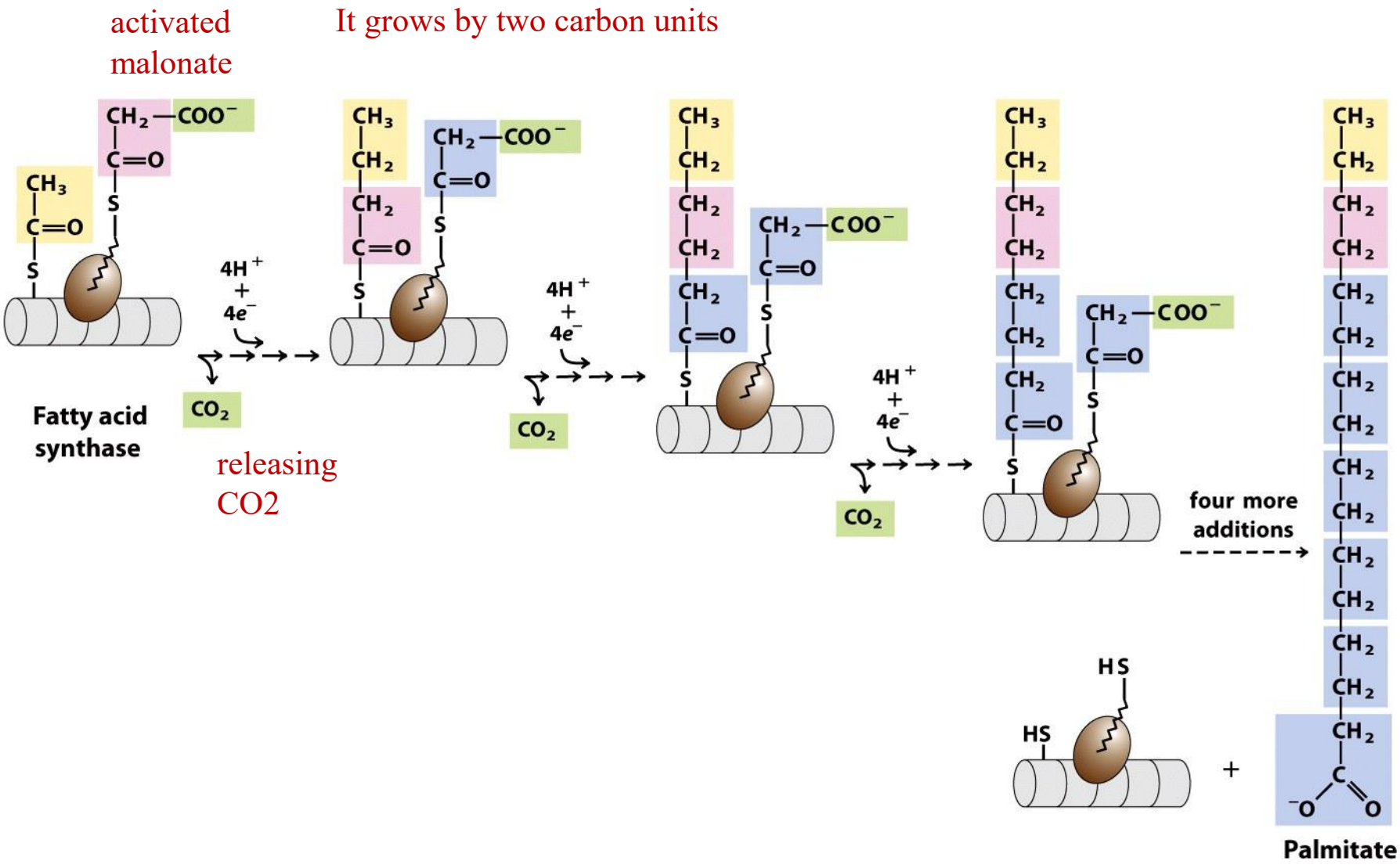
separate enzymes
for individual steps

The structure of fatty acid synthase type I systems

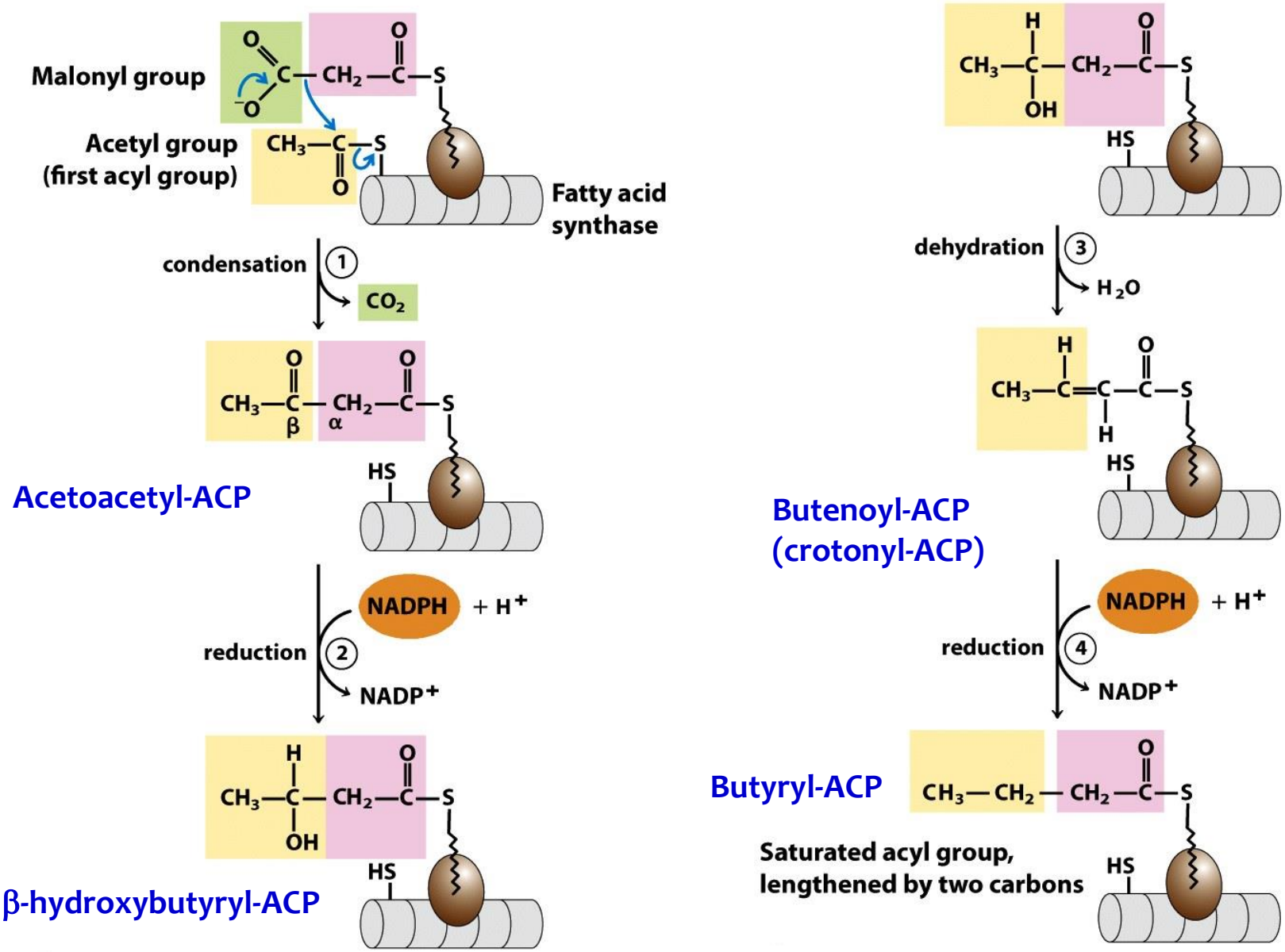


β-ketoacyl-ACP synthase (KS), malonyl/acetyl-CoA—ACP transferase (MAT), β-hydroxyacyl-ACP dehydratase (DH), enoyl-ACP reductase (ER), and β-ketoacyl-ACP reductase (KR). ACP is the acyl carrier protein. The seventh domain (TE) is a thioesterase that releases the palmitate product from ACP when the synthesis is completed.

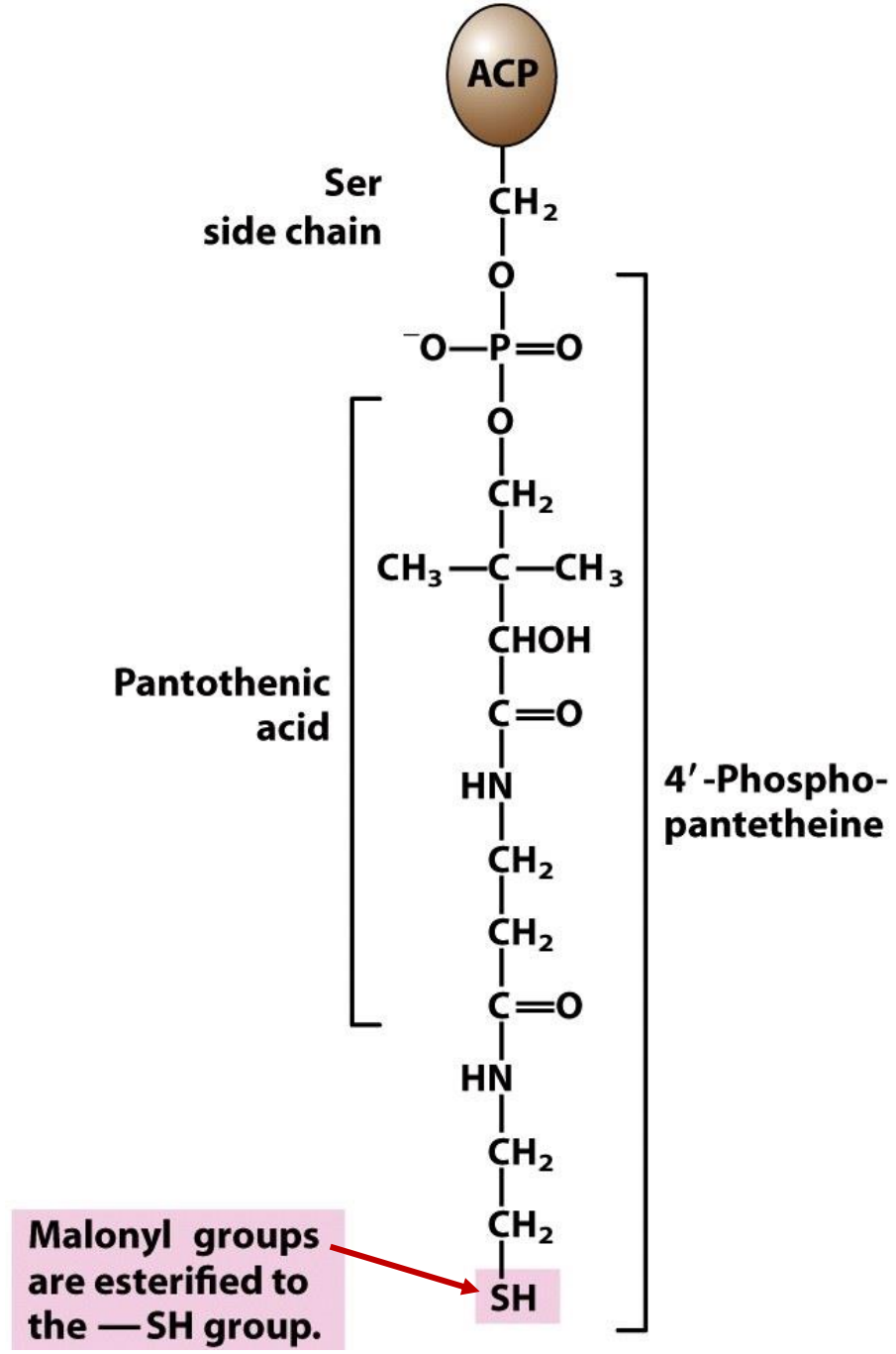
The overall process of palmitate synthesis



Addition of two carbons to a growing fatty acyl chain: a four-step sequence



Acyl carrier protein (ACP). The prosthetic group is 4′-phosphopantetheine, which is covalently attached to the hydroxyl group of a Ser residue in ACP.



ACP is flexible, then attached to KS
Acetyl group binds to KS

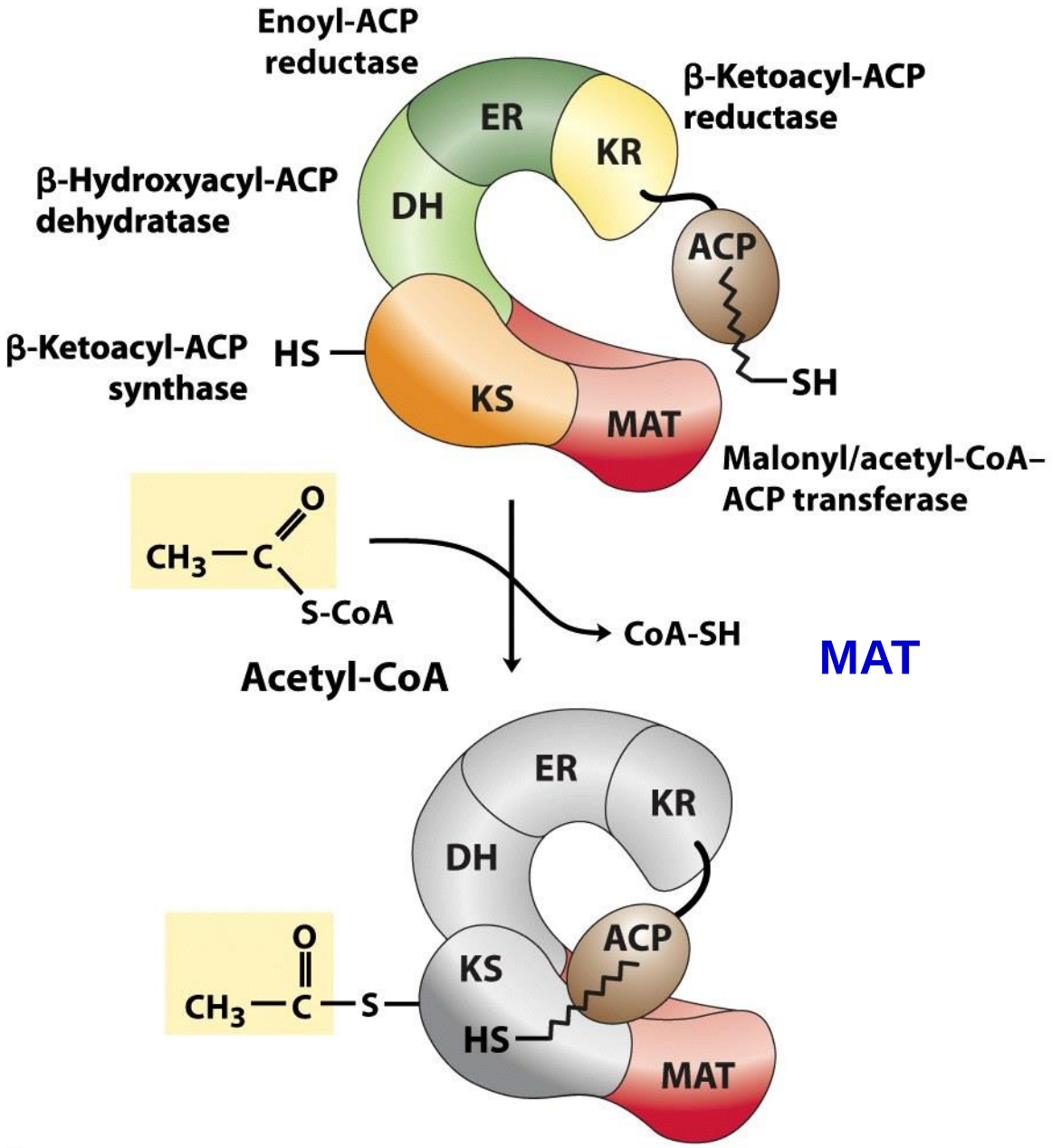
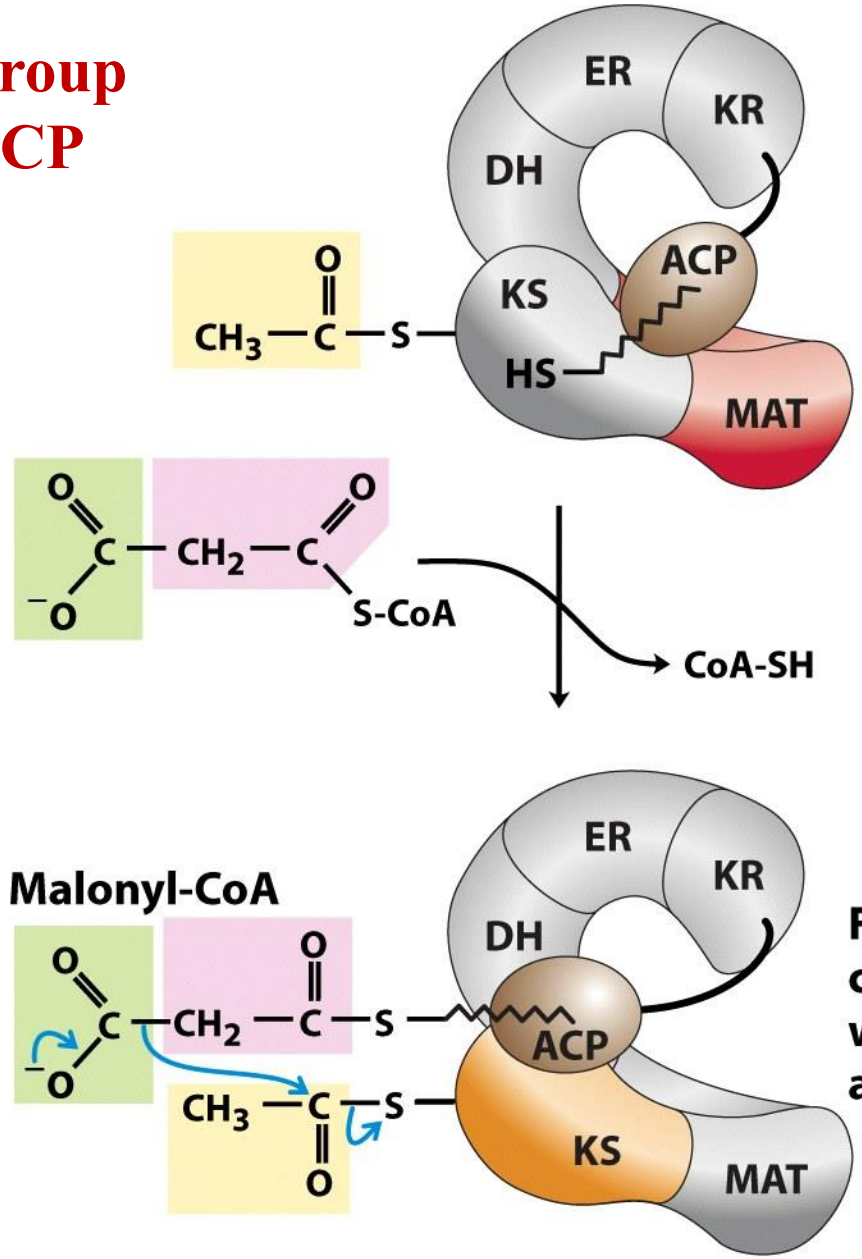


Figure 21-6 part 1
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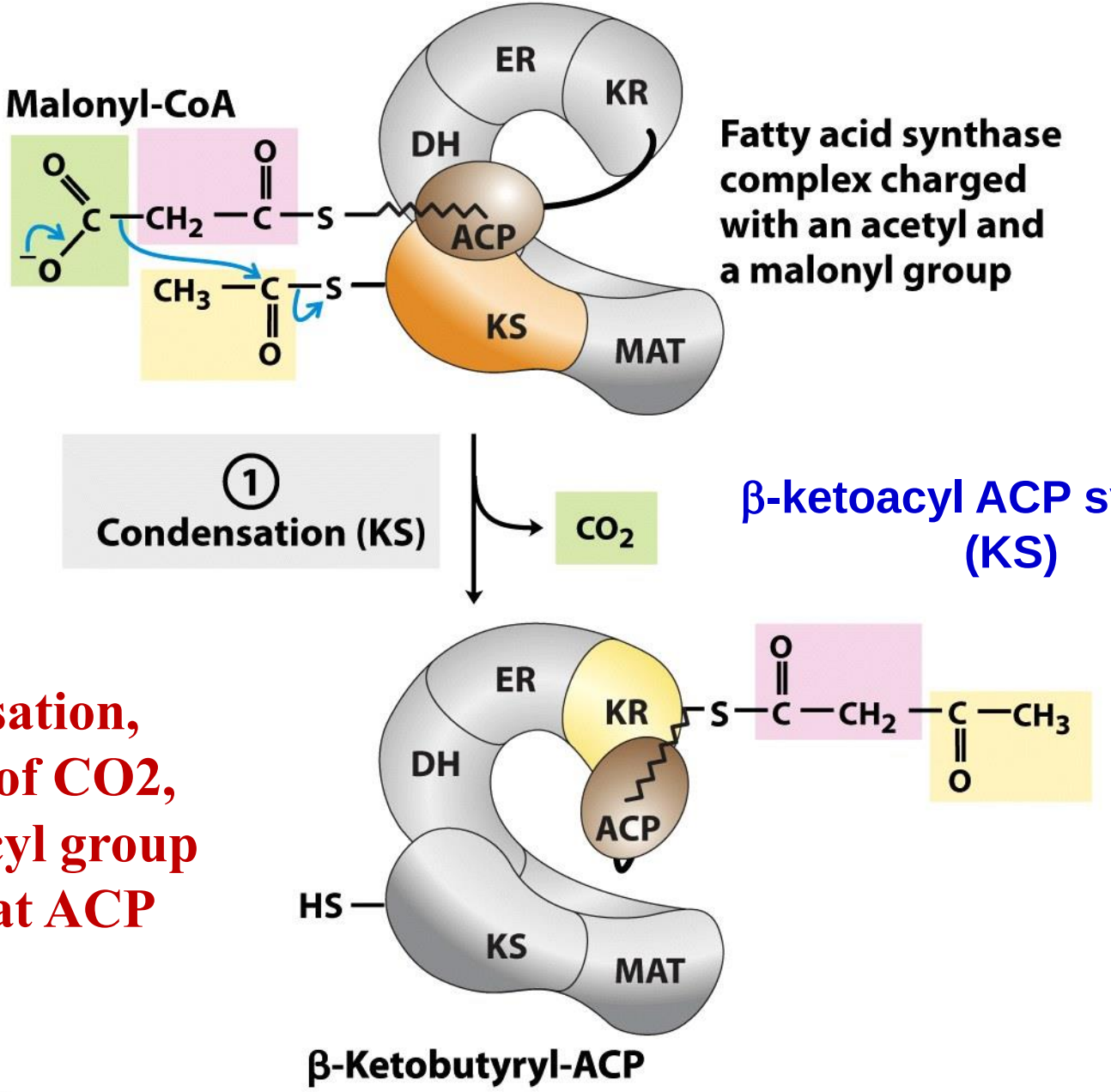
Malonyl group binds to ACP



Malonyl-acetyl-CoA ACP transferase (MAT)

Fatty acid synthase complex charged with an acetyl and a malonyl group

Figure 21-6 part 2
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Condensation, Release of CO2, β-ketoacyl group formed at ACP

Figure 21-6 part 3
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Reduction of βKB to βHB

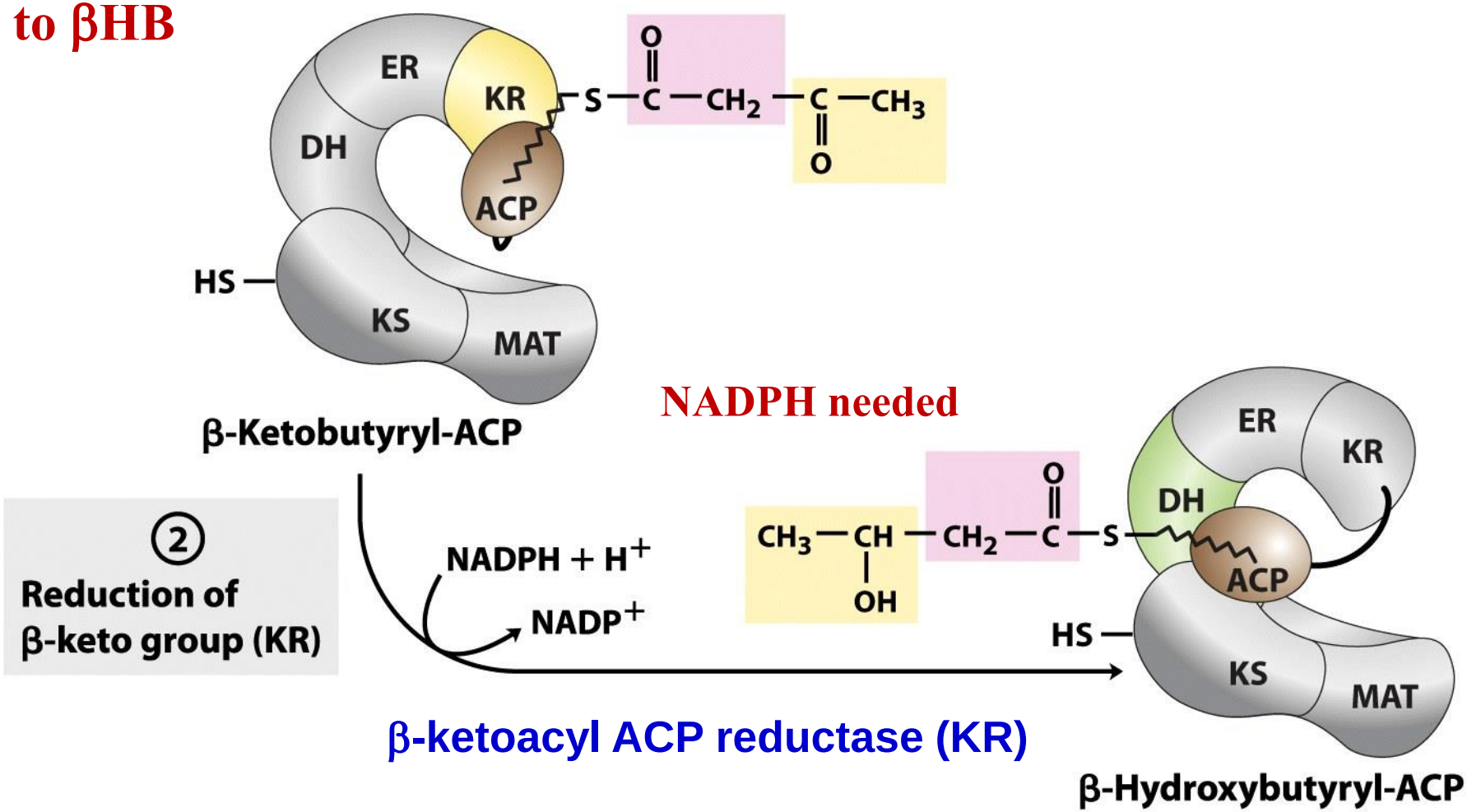


Figure 21-6 part 4
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Dehydration of βHB to tButenoyl

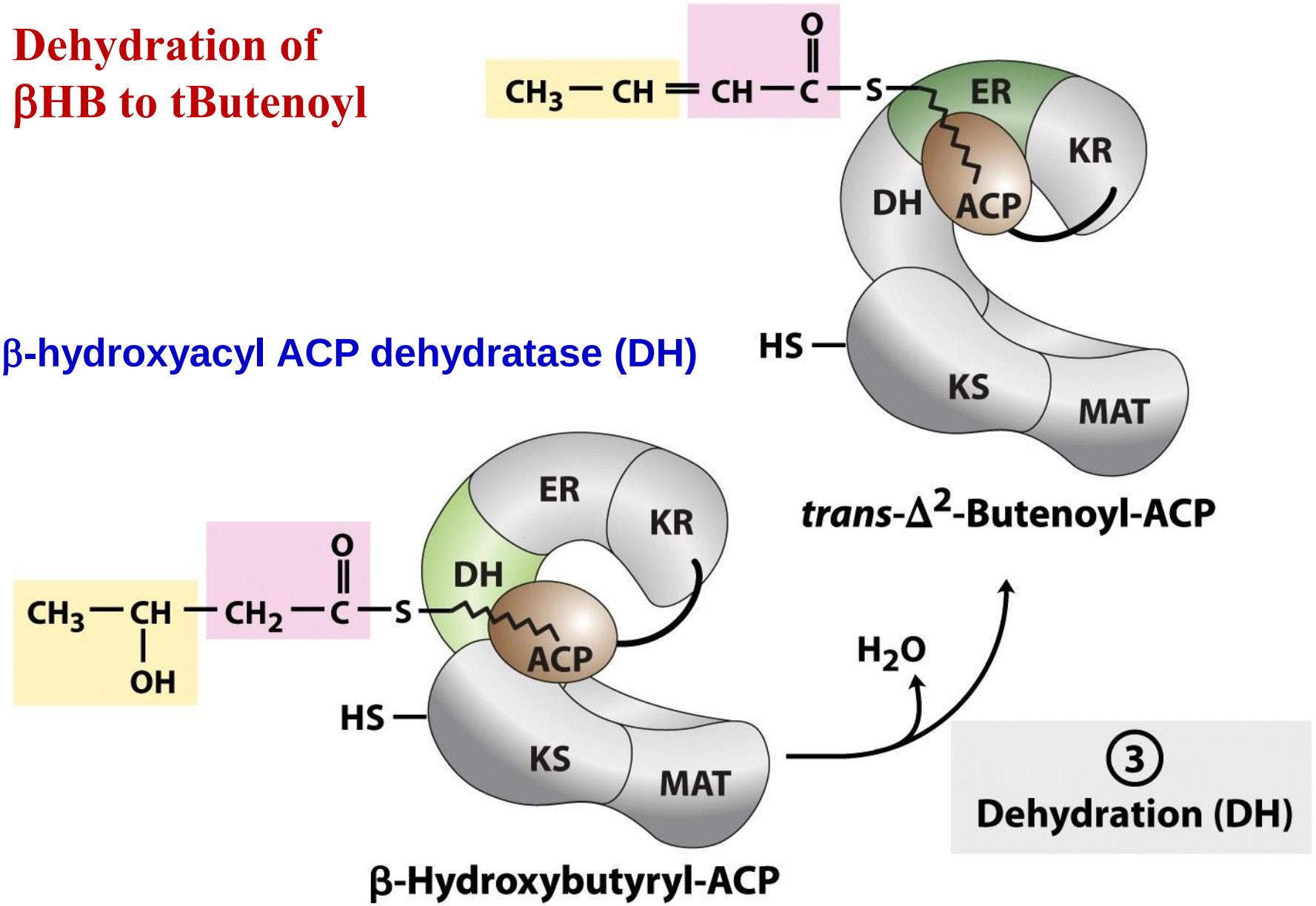


Figure 21-6 part 5
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Reduction of tButenoyl to Butyryl

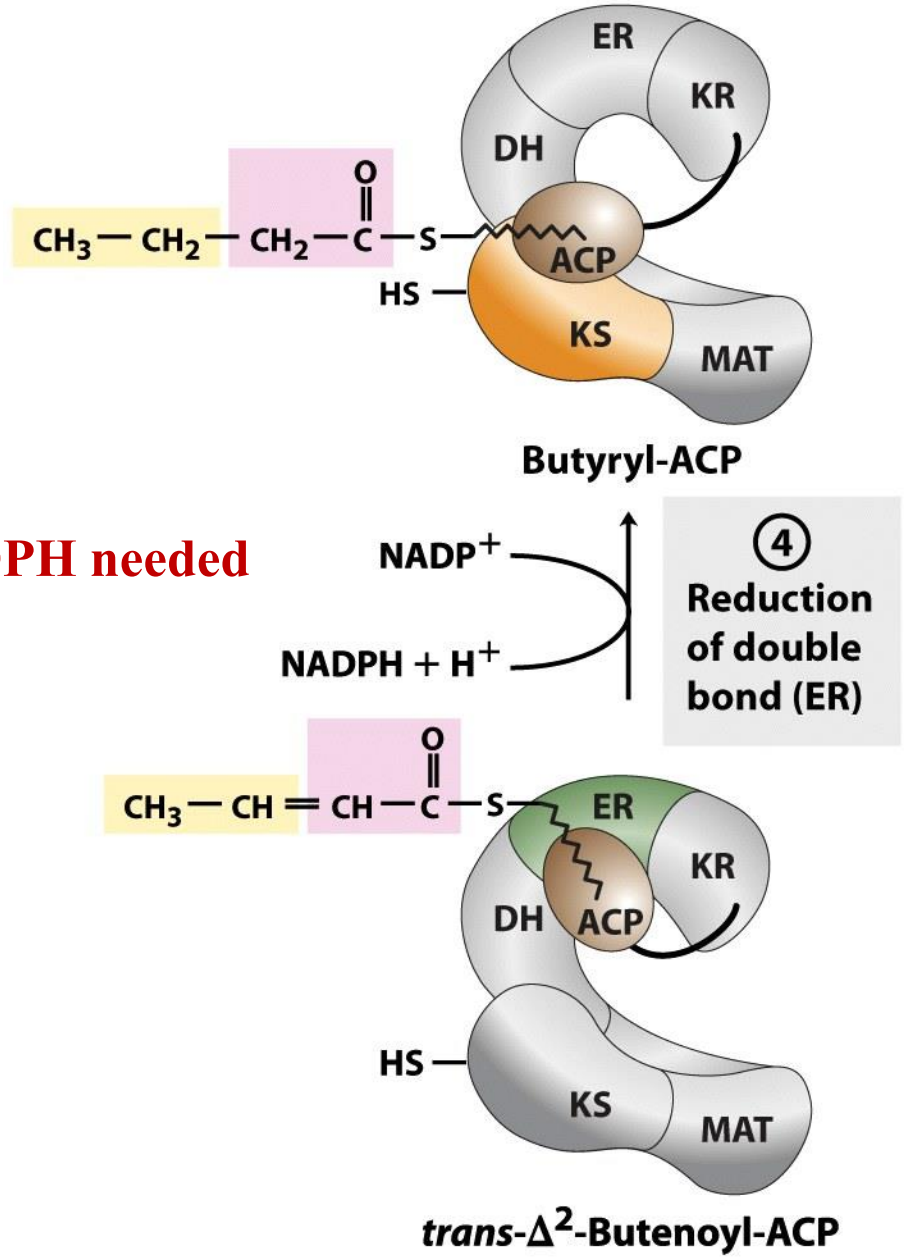


Figure 21-6 part 6
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Translocation of Butyryl to KS

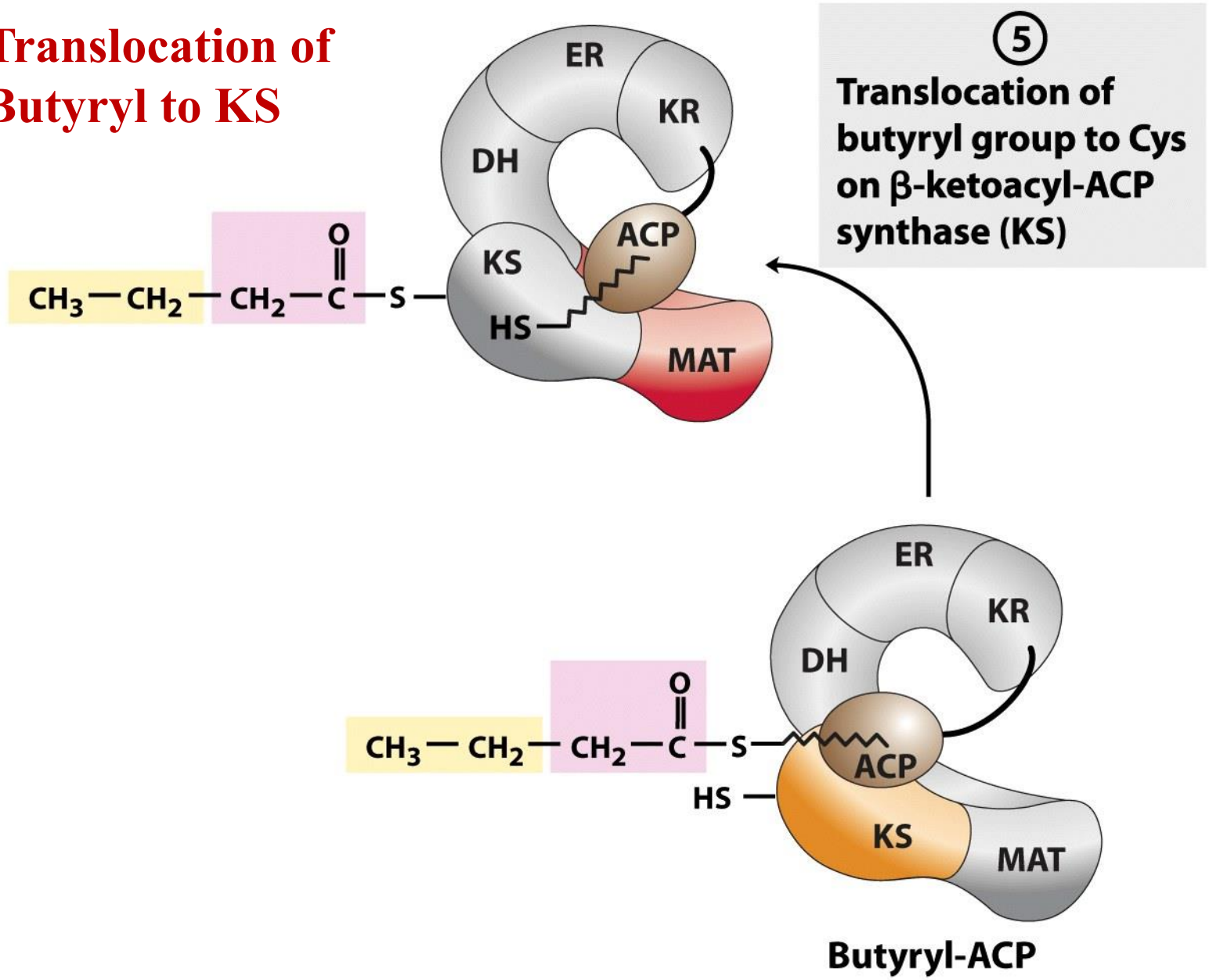


Figure 21-6 part 7
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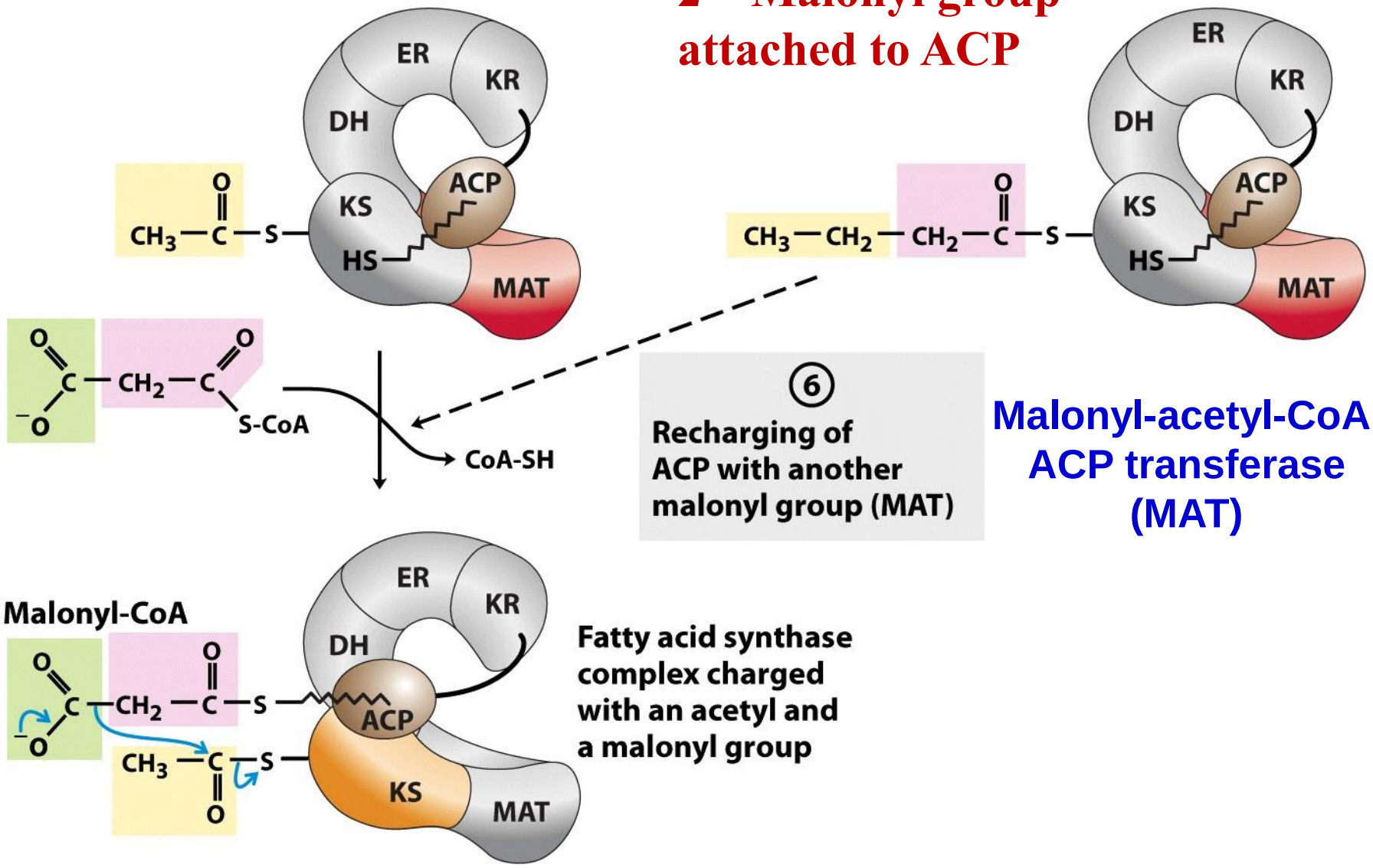
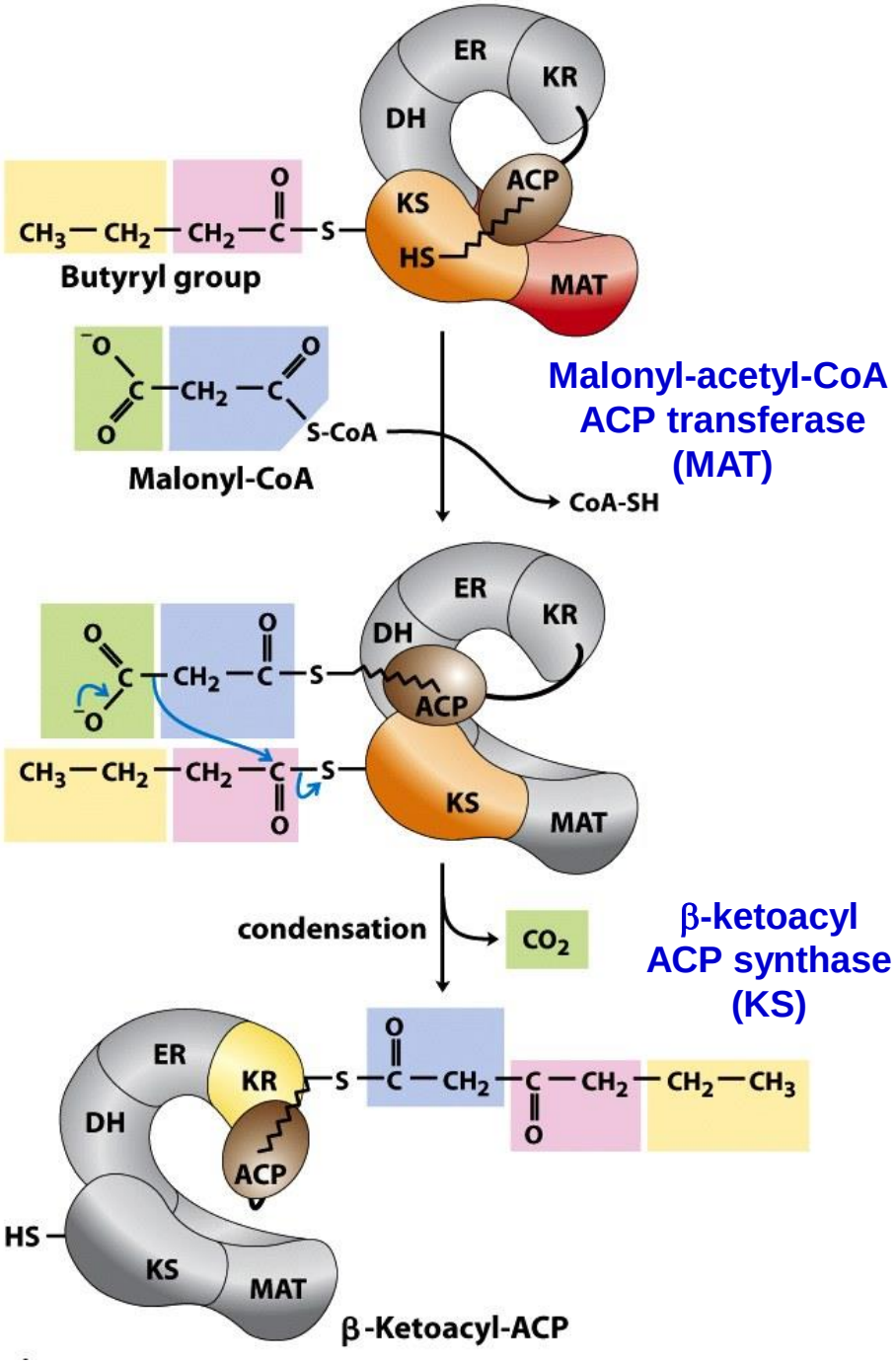


Figure 21-6 part 8

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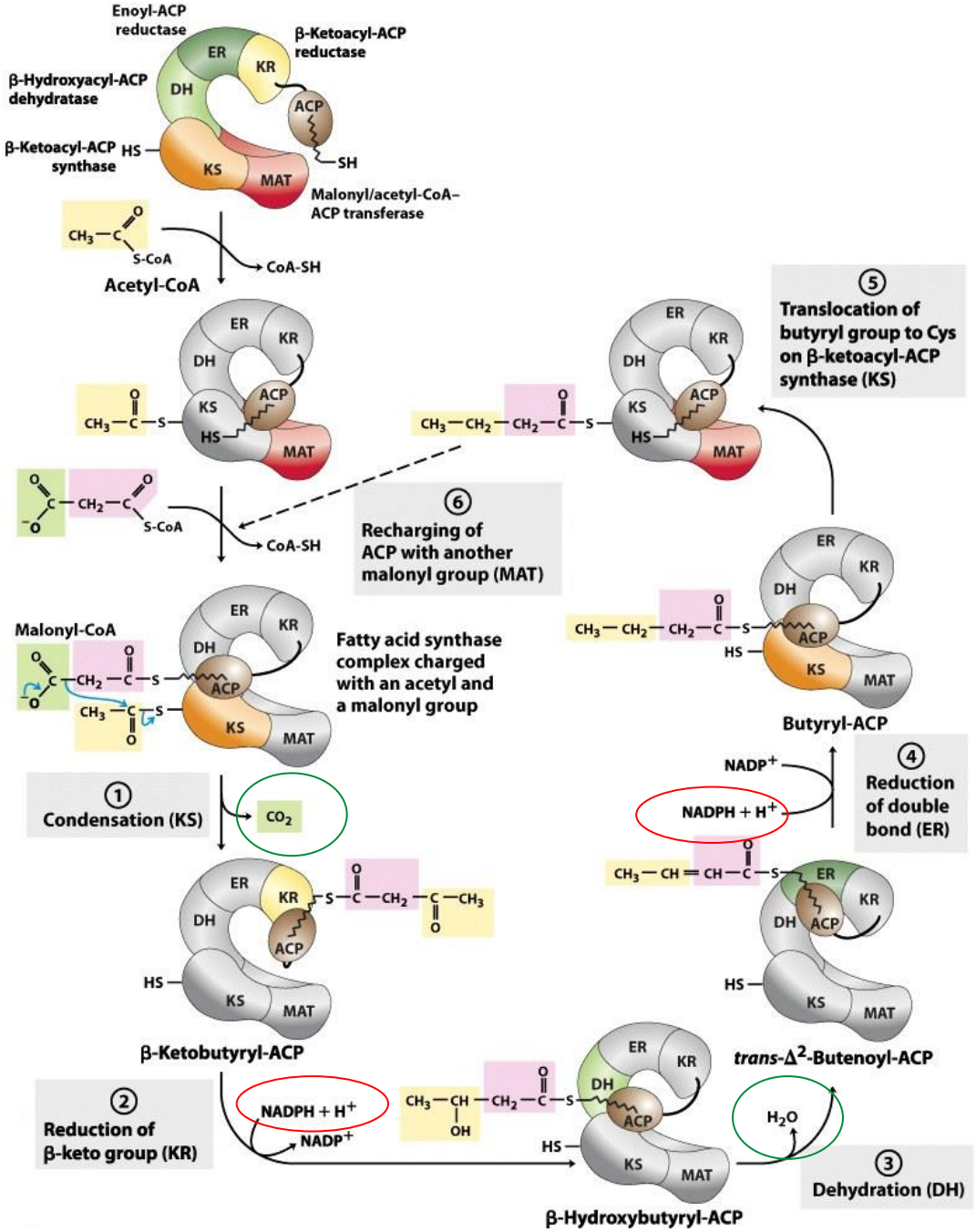
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Beginning of the second round of the fatty acid synthesis cycle.



Sequence of events during synthesis of a fatty acid

2 NADPH needed
1 CO_2 released
1 H_2O released



Overall reaction

$$7\ \text{acetyl-CoA} + 7CO_2 + 7\text{ATP} \rightarrow 7\ \text{malonyl-CoA} + 7\text{ADP} + 7\text{Pi} \quad (1)$$

$$\text{acetyl-CoA} + 7\ \text{malonyl-CoA} + 14\text{NADPH} + 14H^+ \rightarrow \text{palmitate} + 7CO_2 + 8\text{CoA} + 14\text{NADP}^+ + 6H_2O \quad (2)$$

(1+2)

$$8\ \text{acetyl-CoA} + 7\text{ATP} + 14\text{NADPH} + 14H^+ \rightarrow \text{palmitate} + 8\text{CoA} + 7\text{ADP} + 7\text{Pi} + 14\text{NADP}^+ + 6H_2O$$

1) β-oxidation of saturated FA's

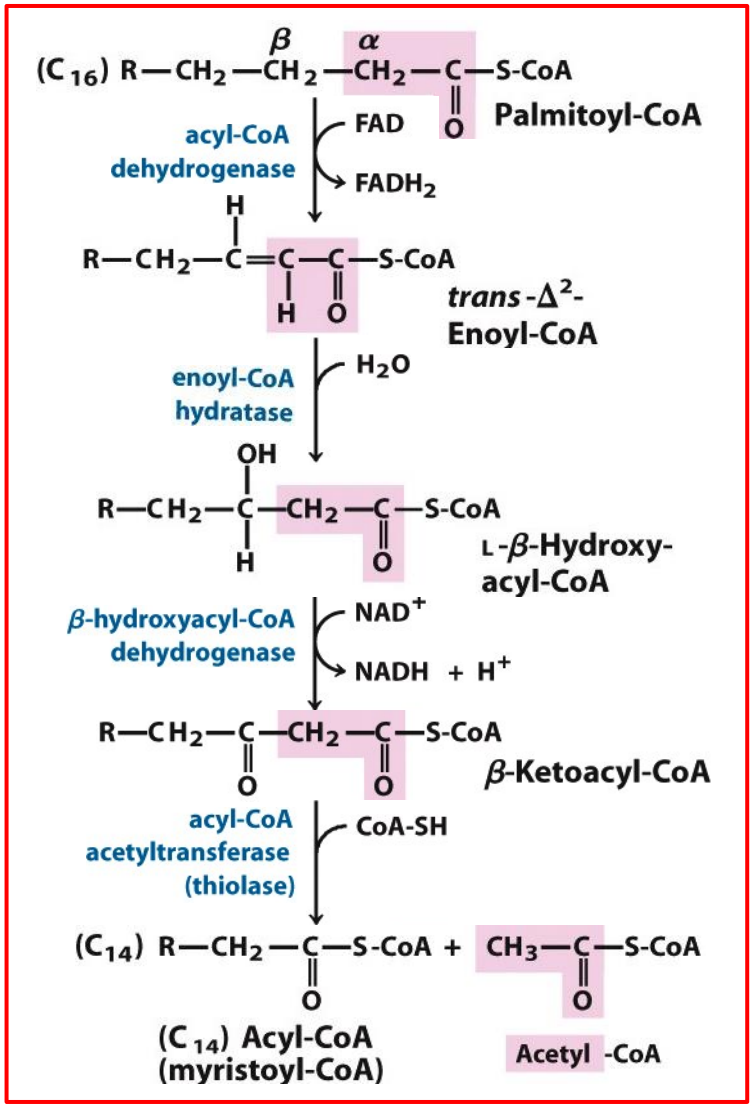
- 4 basic steps

① Dehydrogenation of C_n acyl-CoA

② Water addition

③ Dehydrogenation: **specific for L isomer**

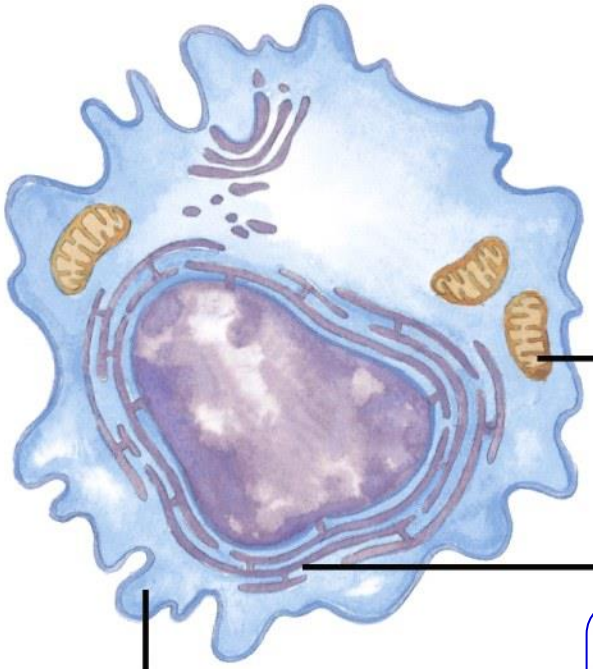
④ Thiolysis: **C_{n-2} acyl-CoA + Acetyl CoA.**



Palmitoyl-CoA + 7CoA + 7FAD + 7NAD^+ + 7H_2O →
8 acetyl-CoA + 7$FADH_2$ + 7NADH + 7H^+

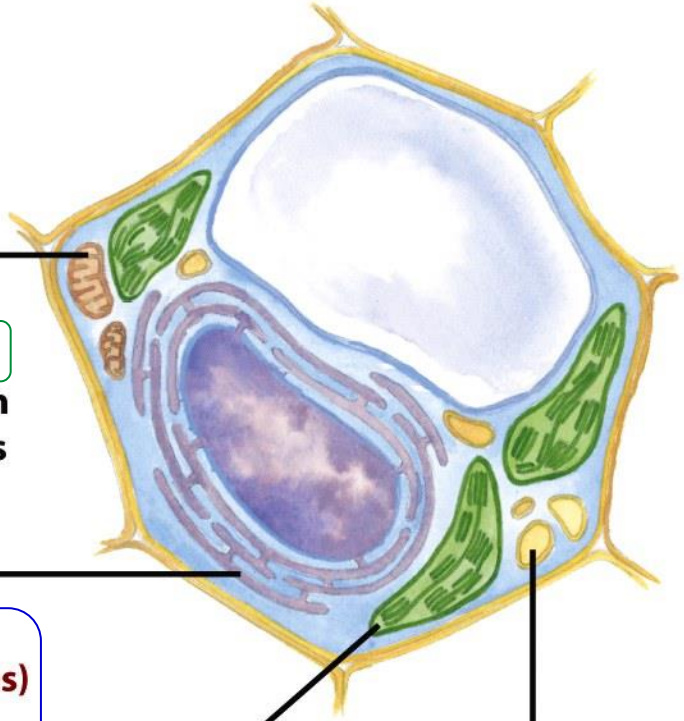
Where fatty acid synthesis take place ?

Subcellular localization of lipid metabolism



Animal cells, yeast cells

Plant cells



Mitochondria

- No fatty acid oxidation
- Fatty acid oxidation
- Acetyl-CoA production
- Ketone body synthesis
- Fatty acid elongation

Endoplasmic reticulum

- Phospholipid synthesis
- Sterol synthesis (late stages)
- Fatty acid elongation
- Fatty acid desaturation

Cytosol

- NADPH production (pentose phosphate pathway; malic enzyme)
- [NADPH]/[$NADP^+$] high
- Isoprenoid and sterol synthesis (early stages)
- Fatty acid synthesis

Chloroplasts

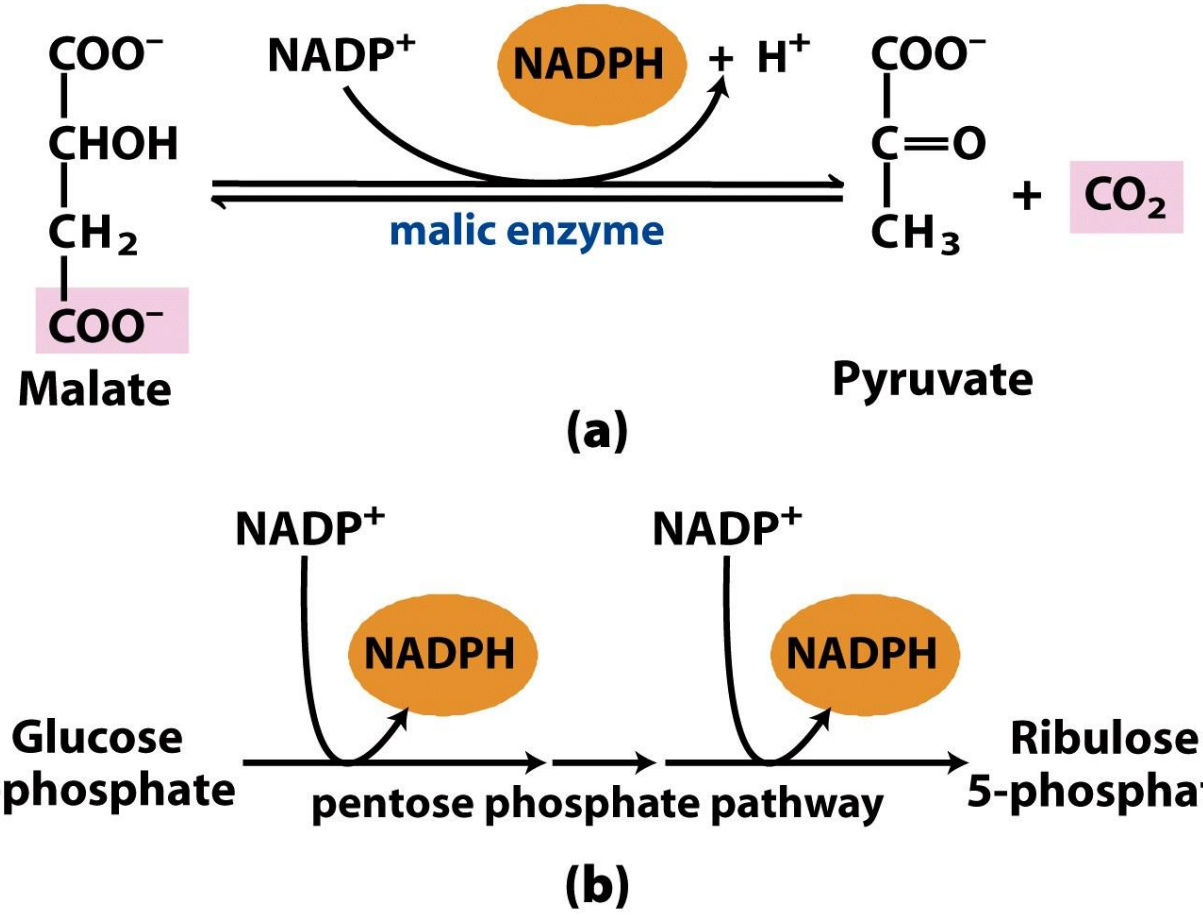
- NADPH, ATP production
- [NADPH]/[$NADP^+$] high
- Fatty acid synthesis

Peroxisomes

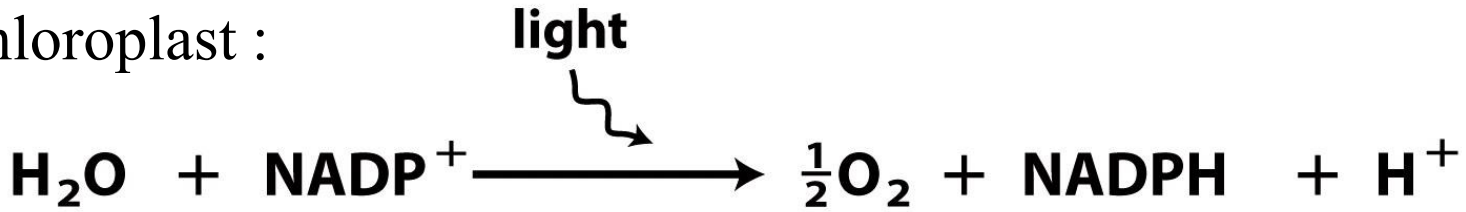
- Fatty acid oxidation ($\longrightarrow H_2O_2$)
- Catalase, peroxidase: $H_2O_2 \longrightarrow H_2O$

How NADPH supplied ?

In animal cytosol: [NADPH]/[NADP+] is high by …



In chloroplast :



How about acetate ?

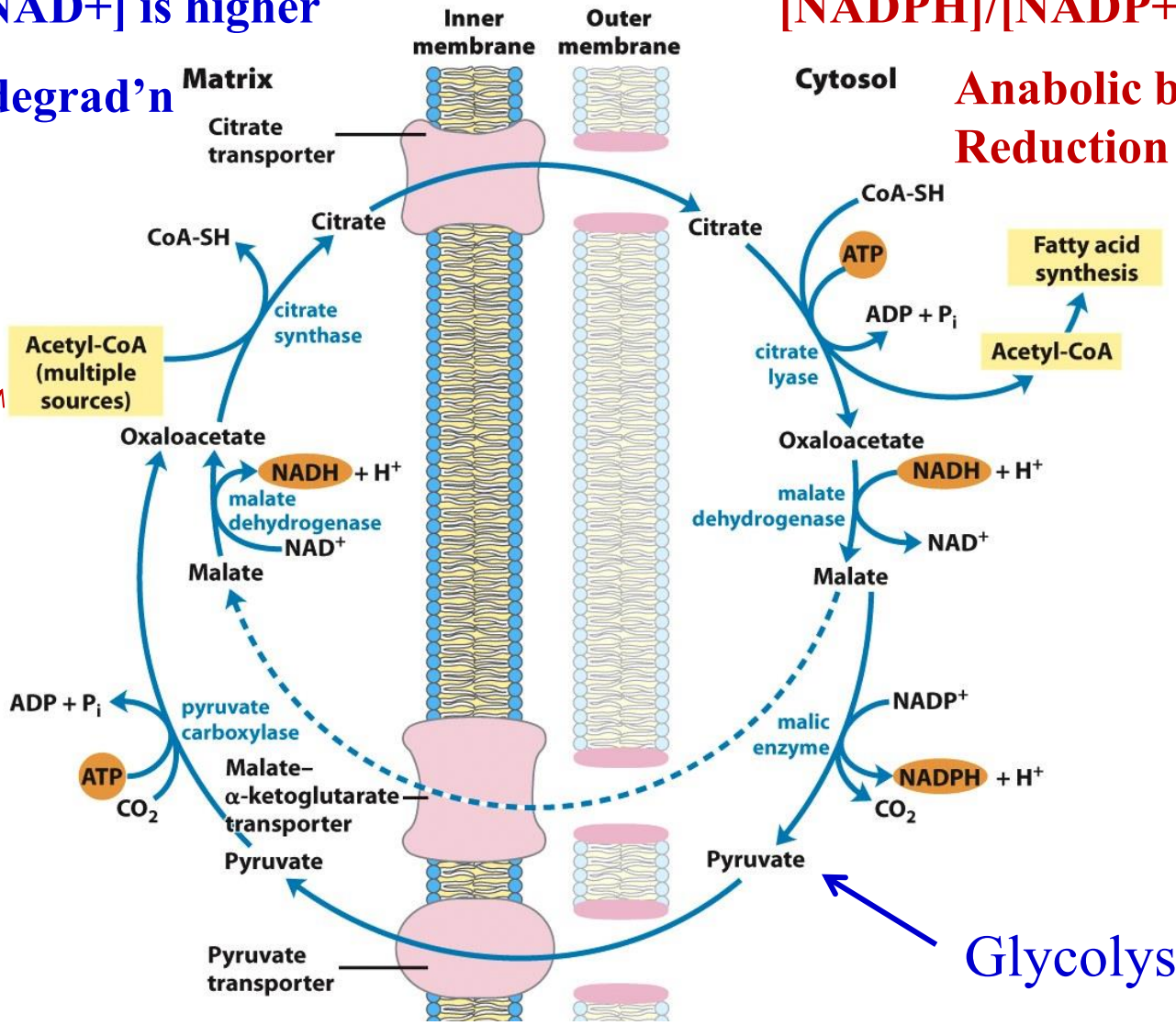
Shuttle for transfer of acetyl groups from mitochondria to the cytosol

[NADH]/[NAD+] is higher

Catabolic degrad'n
Oxidation

[NADPH]/[NADP+] is higher

Anabolic biosynthesis
Reduction



Mainly from pyruvate and amino acids

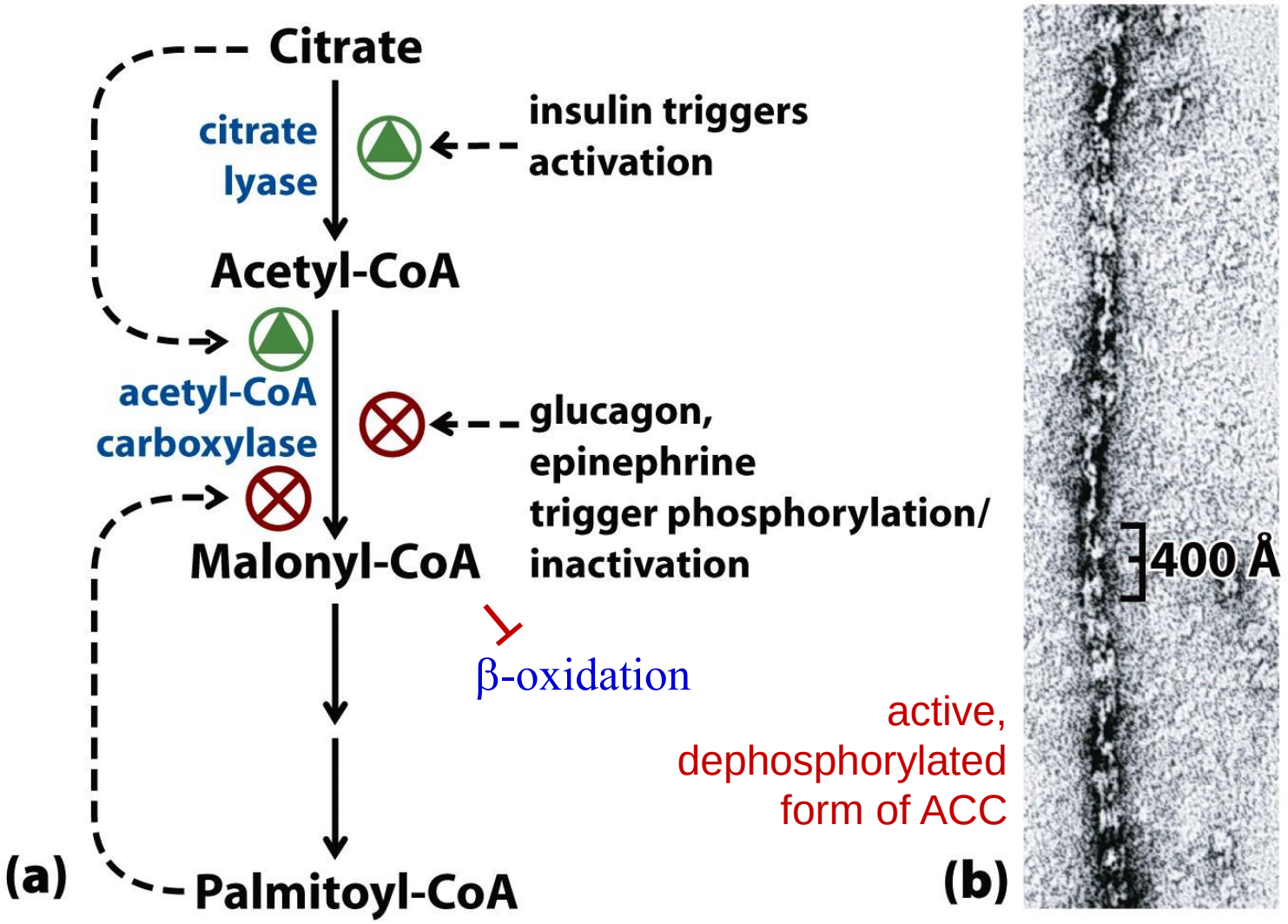
(Not from FA oxidation)

Glycolysis

FA synthesis and oxidation are reciprocally regulated

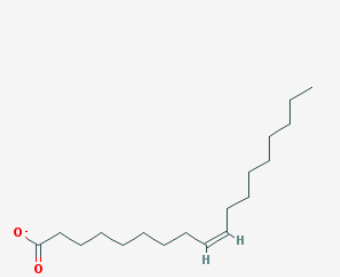
Regulation of FA synthesis

Regulation of fatty acid synthesis

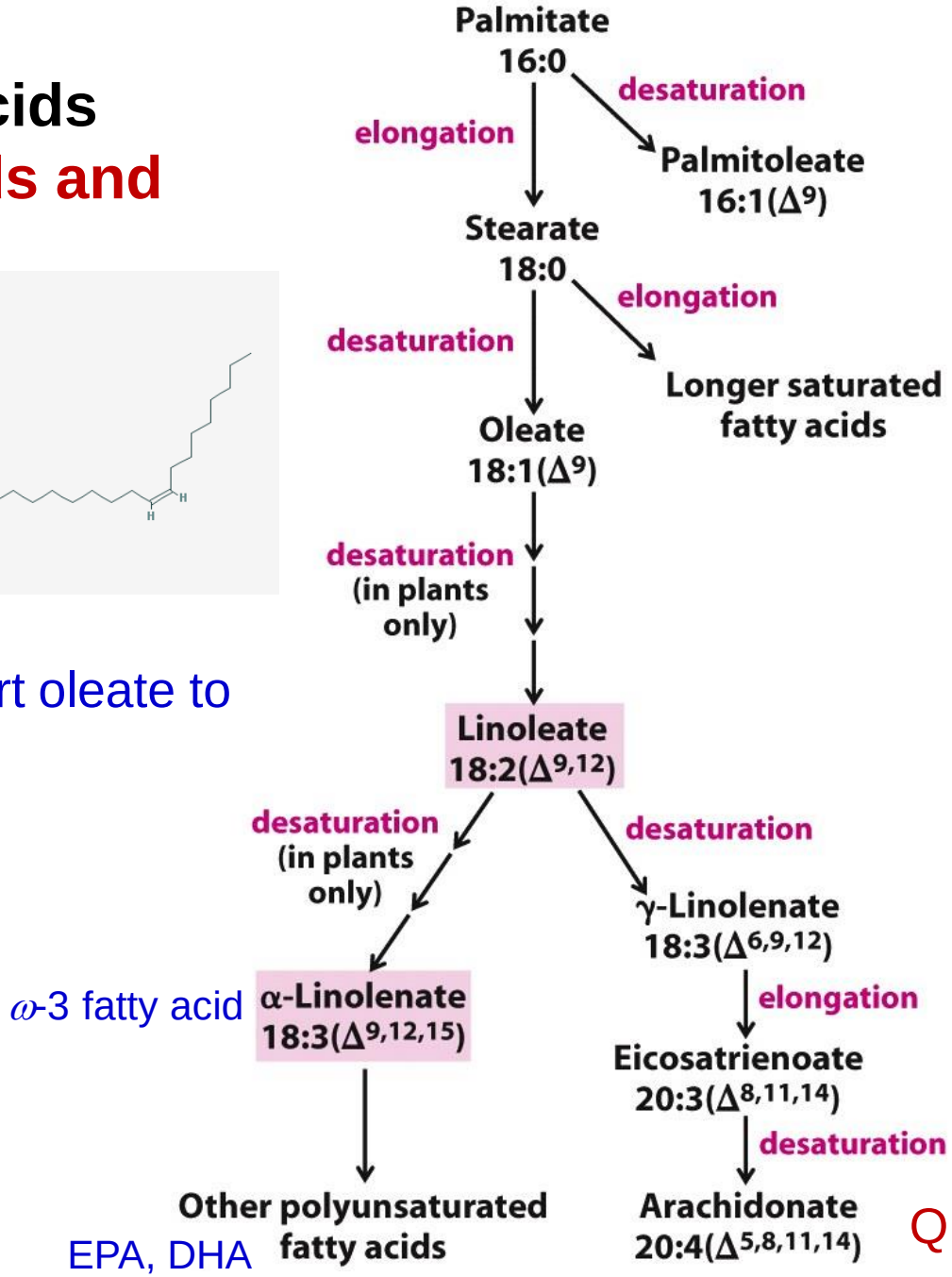


Elongation & Desaturation

Desaturation of fatty acids produces ω-3 fatty acids and arachidonate

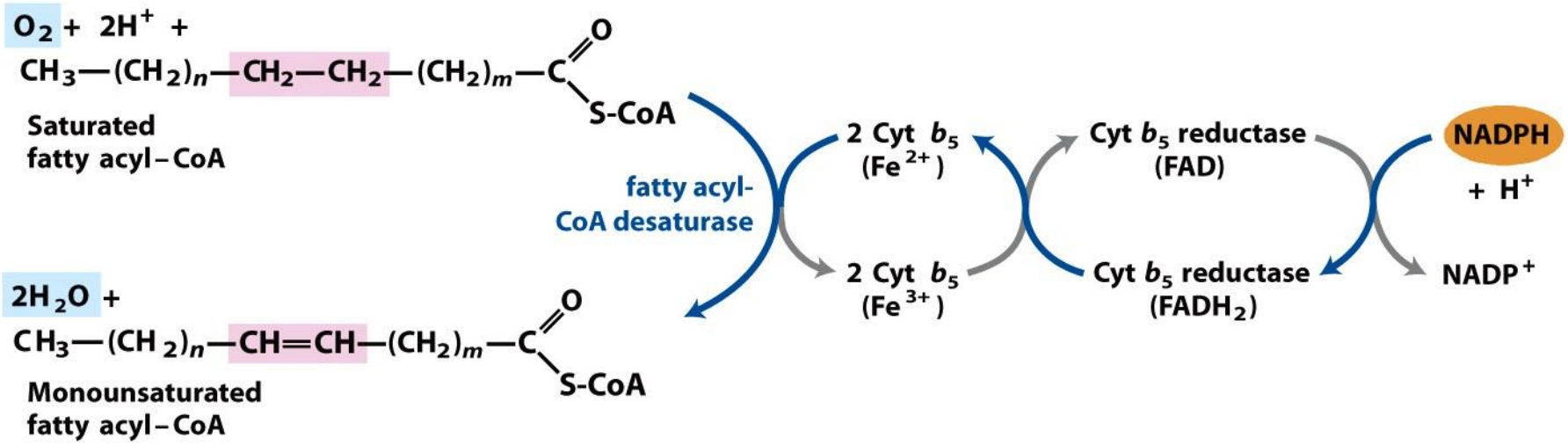


Mammals cannot convert oleate to linoleate or α-linolenate



Q

Electron transfer in the desaturation of fatty acids in vertebrates

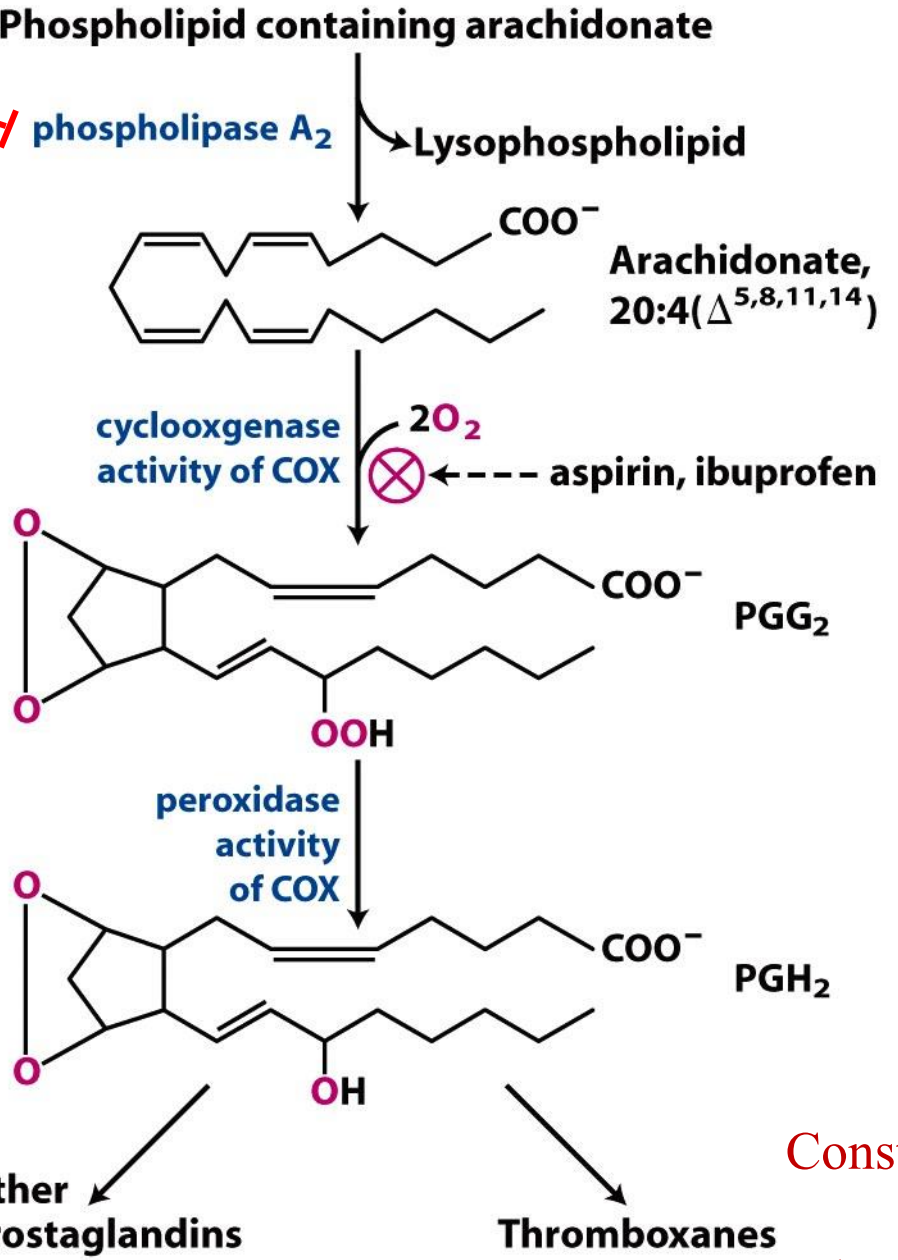


Desaturase simultaneously oxidize the fatty acid and NADPH (Mixed-function oxidase : they oxidize two different substrates simultaneously – fatty acyl-CoA & NADPH)

20 C eicosanoids synthesis (in ER)

Steroid H

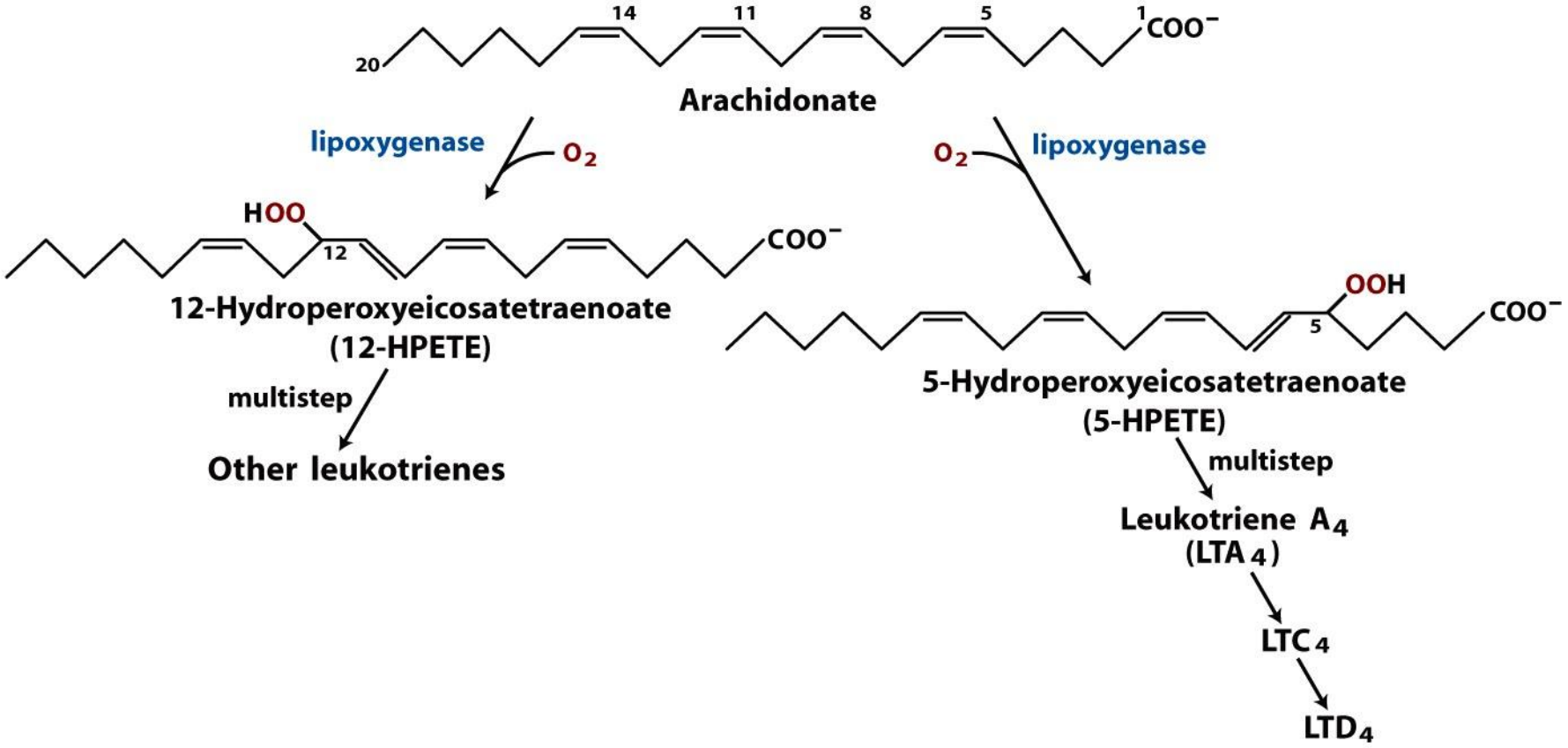
The "cyclic" pathway from arachidonate to prostaglandins and thromboxanes



Fever
Inflammation
pain

Constriction of blood vessels
Platelet aggregation

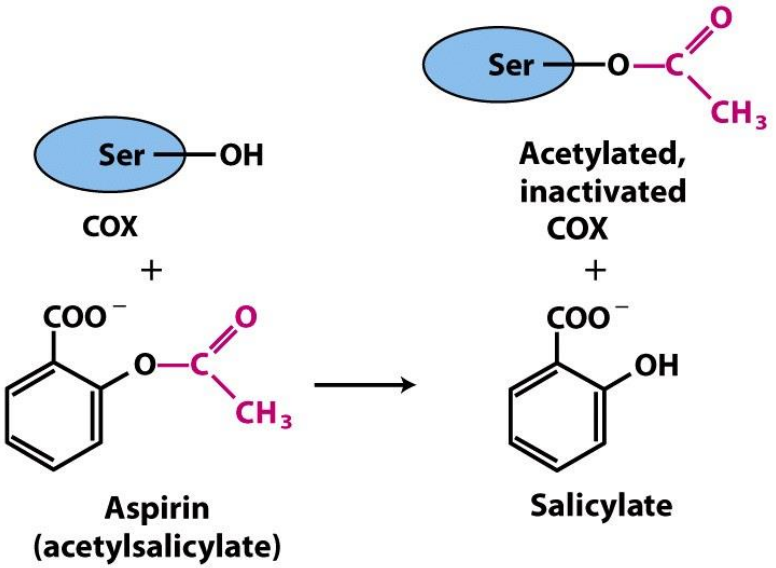
The "linear" pathway from arachidonate to leukotrienes



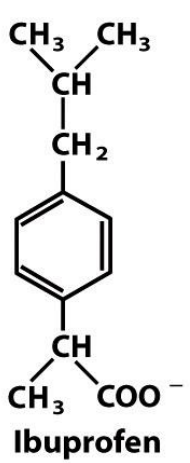
Leukotrienes are fatty signaling molecules

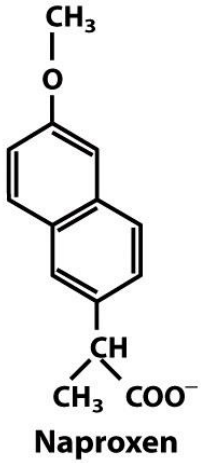
They were first found in leukocytes (hence their name). One of their roles (specifically, leukotriene D4) is to trigger contractions in the smooth muscles lining the trachea; their overproduction is a major cause of inflammation in asthma and allergic rhinitis

COX inhibitors (NSAIDS)

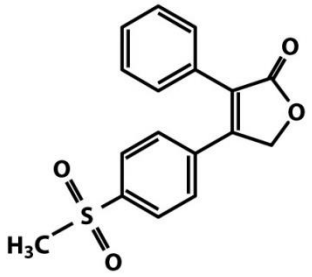


Side effects: stomach irritation

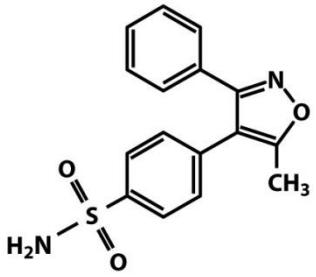




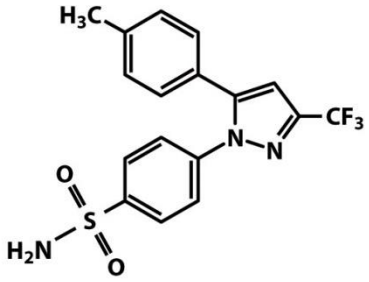
COX-2 inhibitors



(withdrawn)



(withdrawn)



(still on the market)

Side effects: heart attack, stroke

Summary

1. Malonyl-CoA is formed from acetyl-CoA and bicarbonate
2. Fatty acid synthesis proceeds in a repeating reaction sequence
3. Fatty acid synthesis occurs in the cytosol of many organisms but in the chloroplasts of plants
4. Acetate is shuttled out of mitochondria as citrate
5. Desaturation of fatty acids requires a mixed-function oxidase
6. Eicosanoids are formed from 20-carbon polyunsaturated fatty acids

Lipid biosynthesis

21.2 Biosynthesis of triacylglycerols

21.3 Biosynthesis of membrane phospholipids

21.4 Biosynthesis of cholesterol, steroid, and isoprenoids

- Lipid transport

How much fuel do you have ?

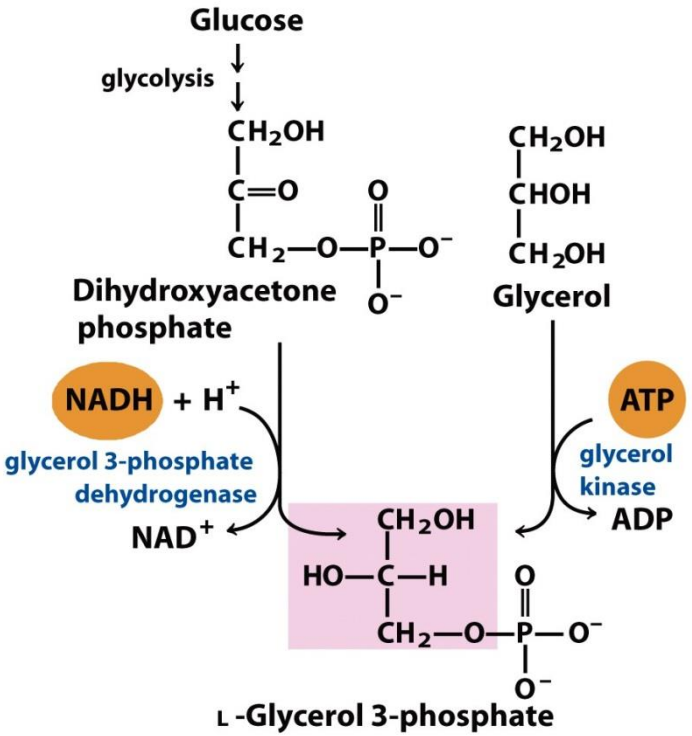
70 kg man

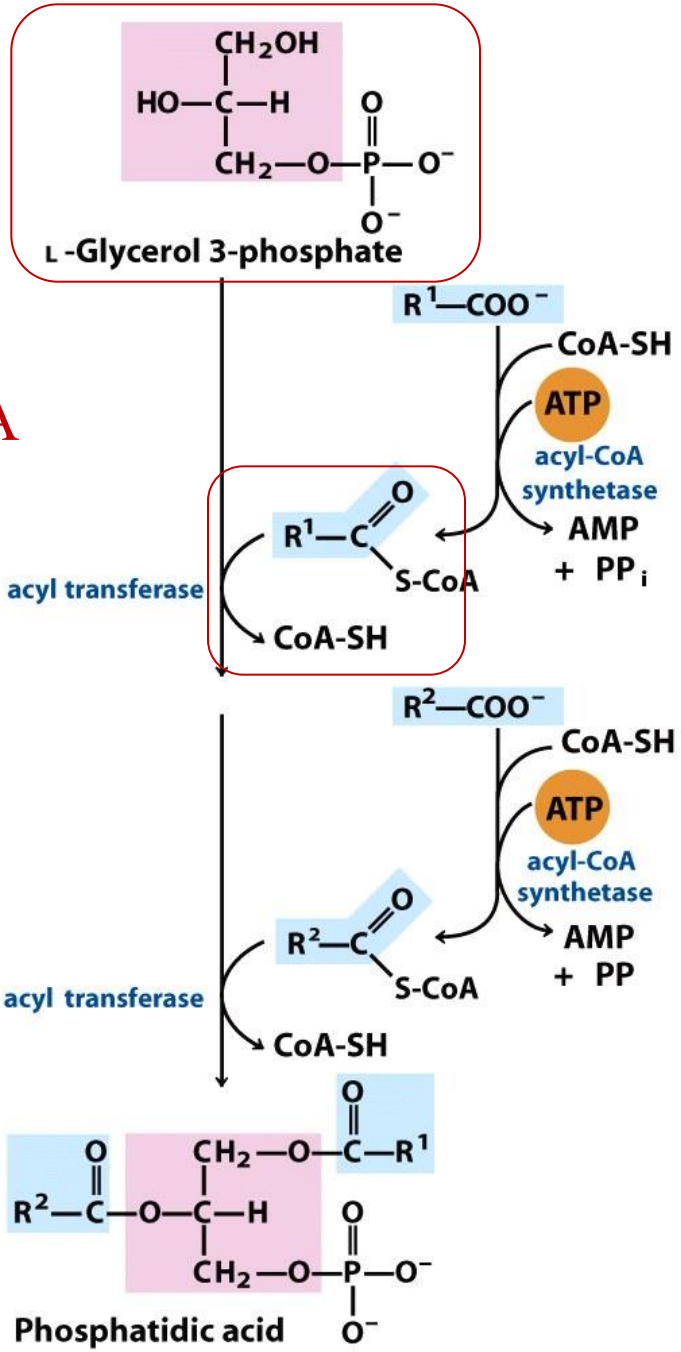
15 kg triacylglycerol

Energy for 12 weeks

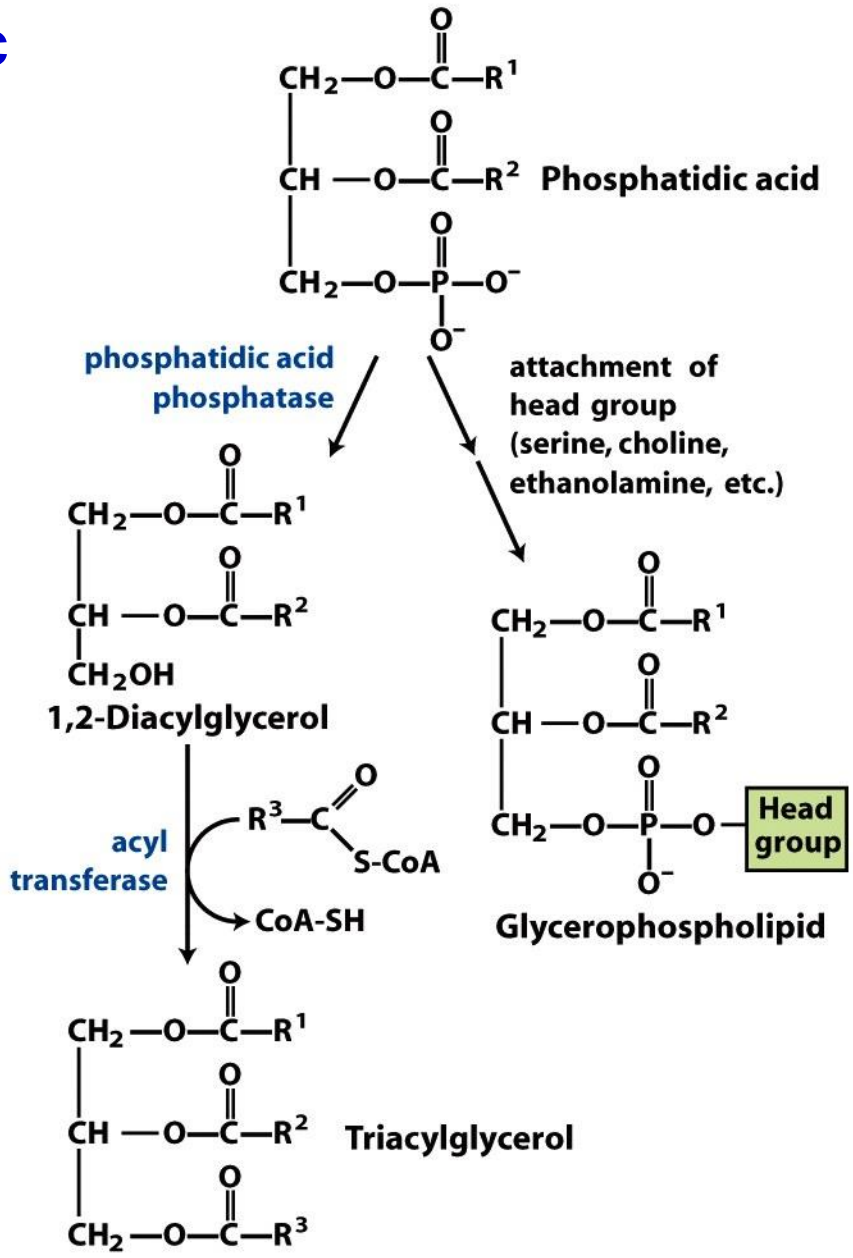
Biosynthesis of phosphatidic acid

Two precursors:
L-glycerol 3-phosphate & fatty acyl-CoA





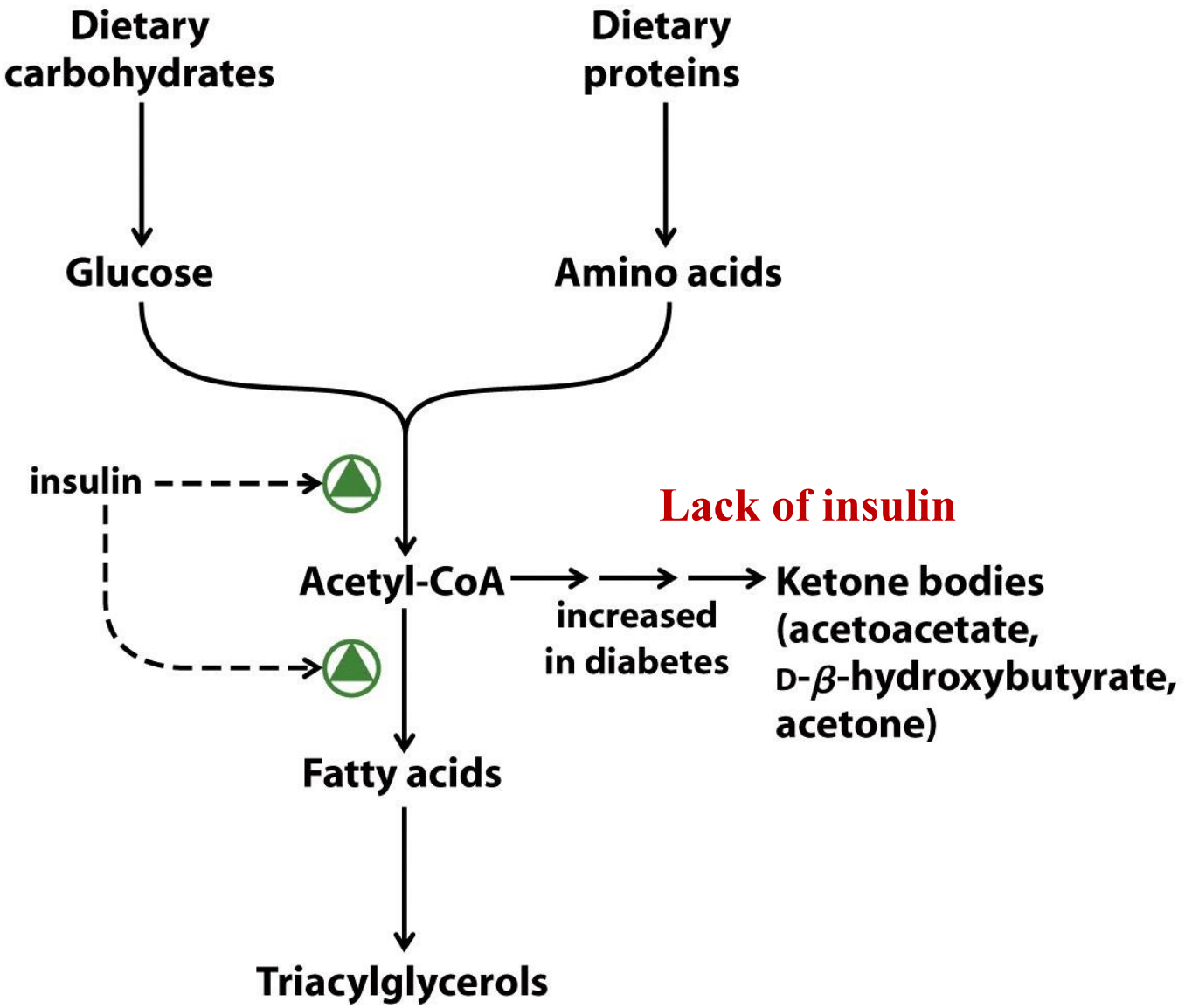
Phosphatidic acid in lipid biosynthesis



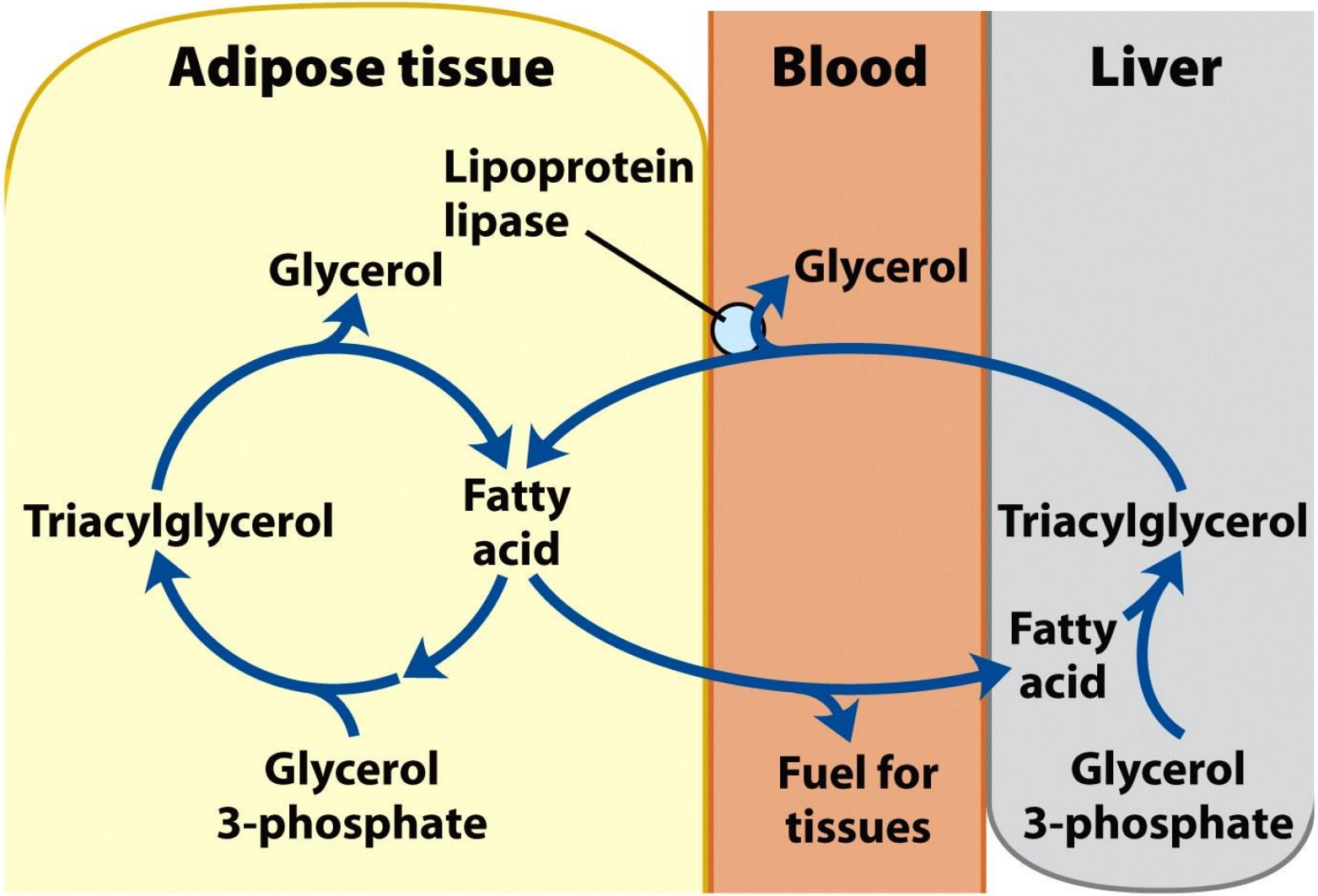
Membrane (growth)

Storage (fat)

Regulation of triacylglycerol synthesis by insulin



The triacylglycerol cycle (increased during starvation)



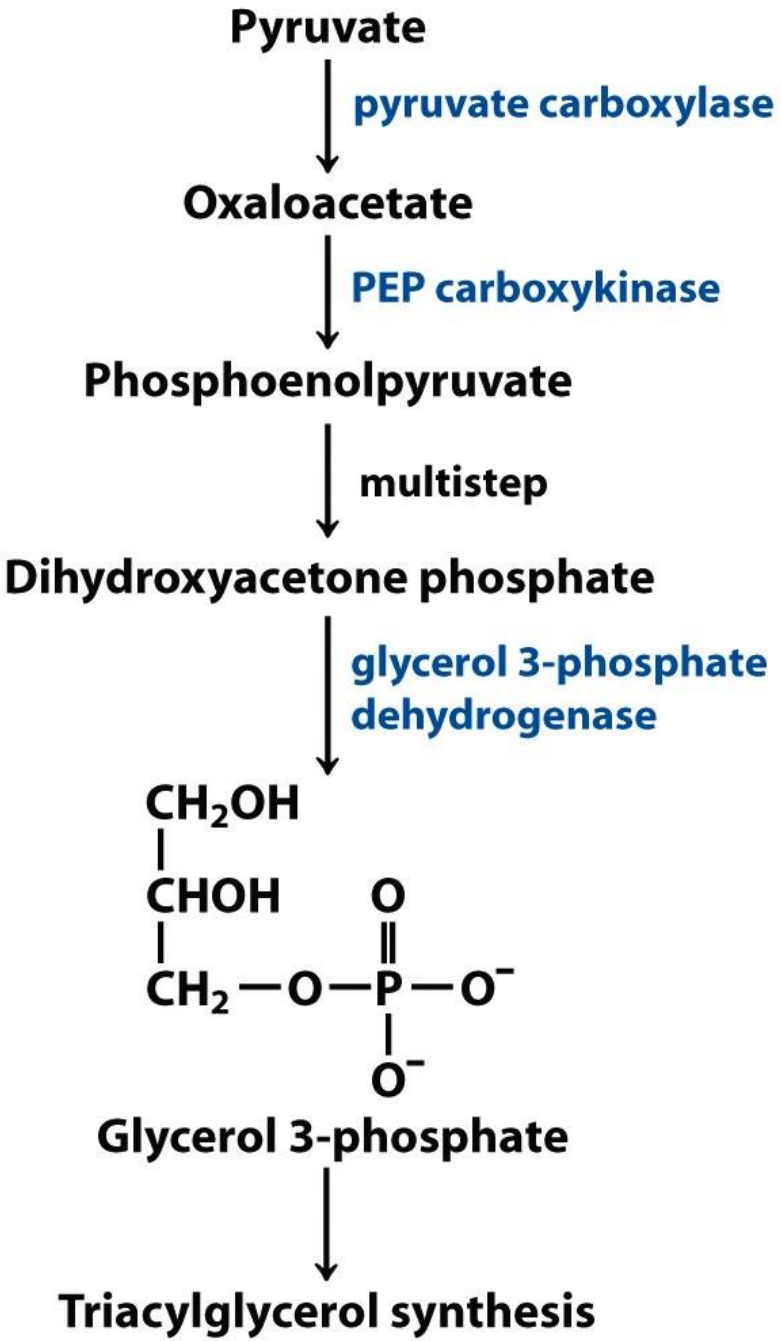
Reesterification: ~75%

Generation of glycerol-3-phosphate in adipocyte

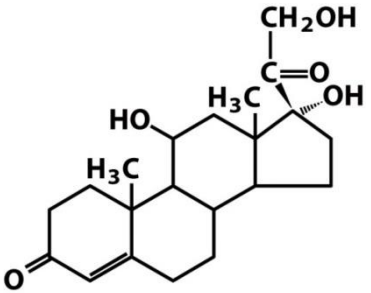
Glyceroneogenesis in adipose tissue (source of glycerol 3-phosphate)

→ Controls the rate of fatty acid release to the blood

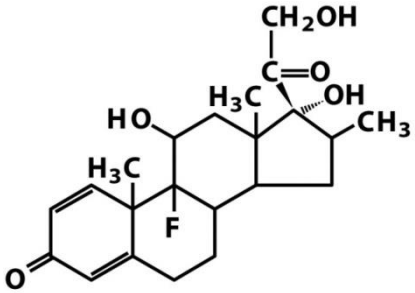
→ Shortened version of gluconeo-genesis



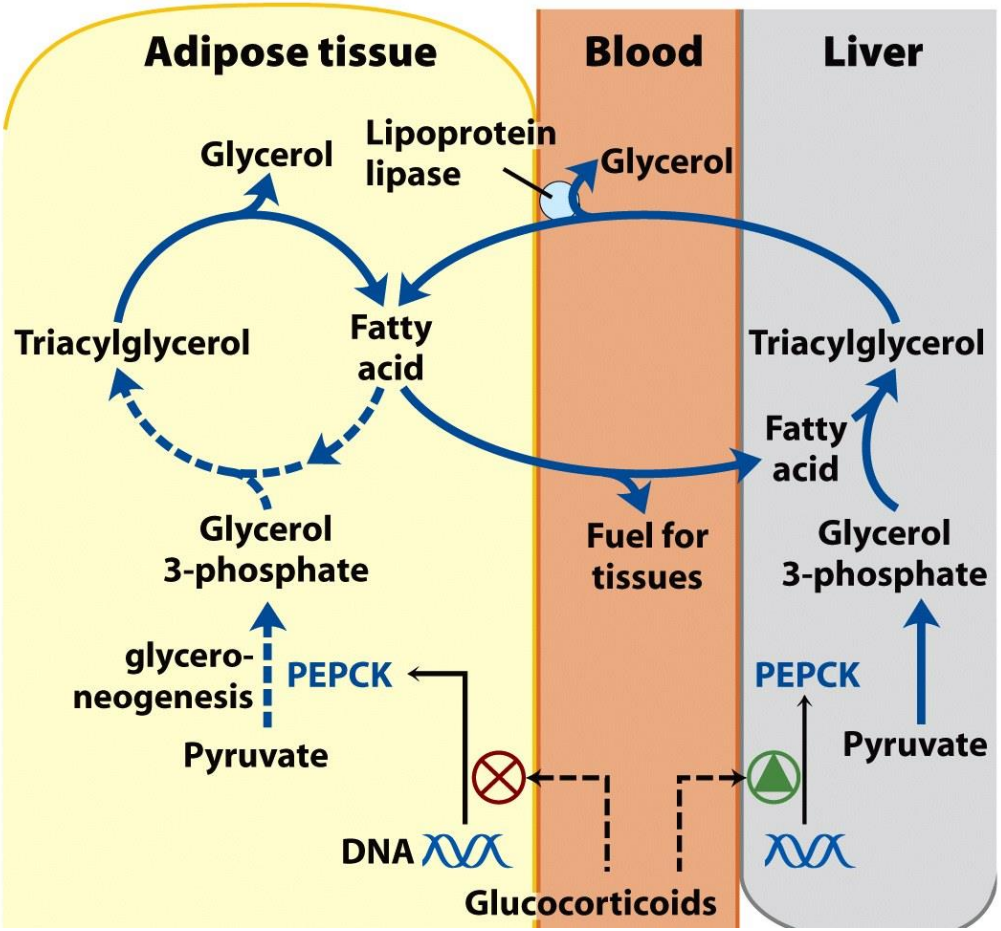
Glucocorticoid hormones stimulate glyceroneogenesis in the liver, but suppress it in adipose tissue



Cortisol



Dexamethasone



→ Decreased glyceroneogenesis in adipocyte

→ Recycling of FFA declines → **more FFA released in blood**

→ interfere glucose utilization, promote insulin resistance

Glucose/glycogen turnover
Gluconeogenesis increased

Cortisol– FFA– insulin resistance

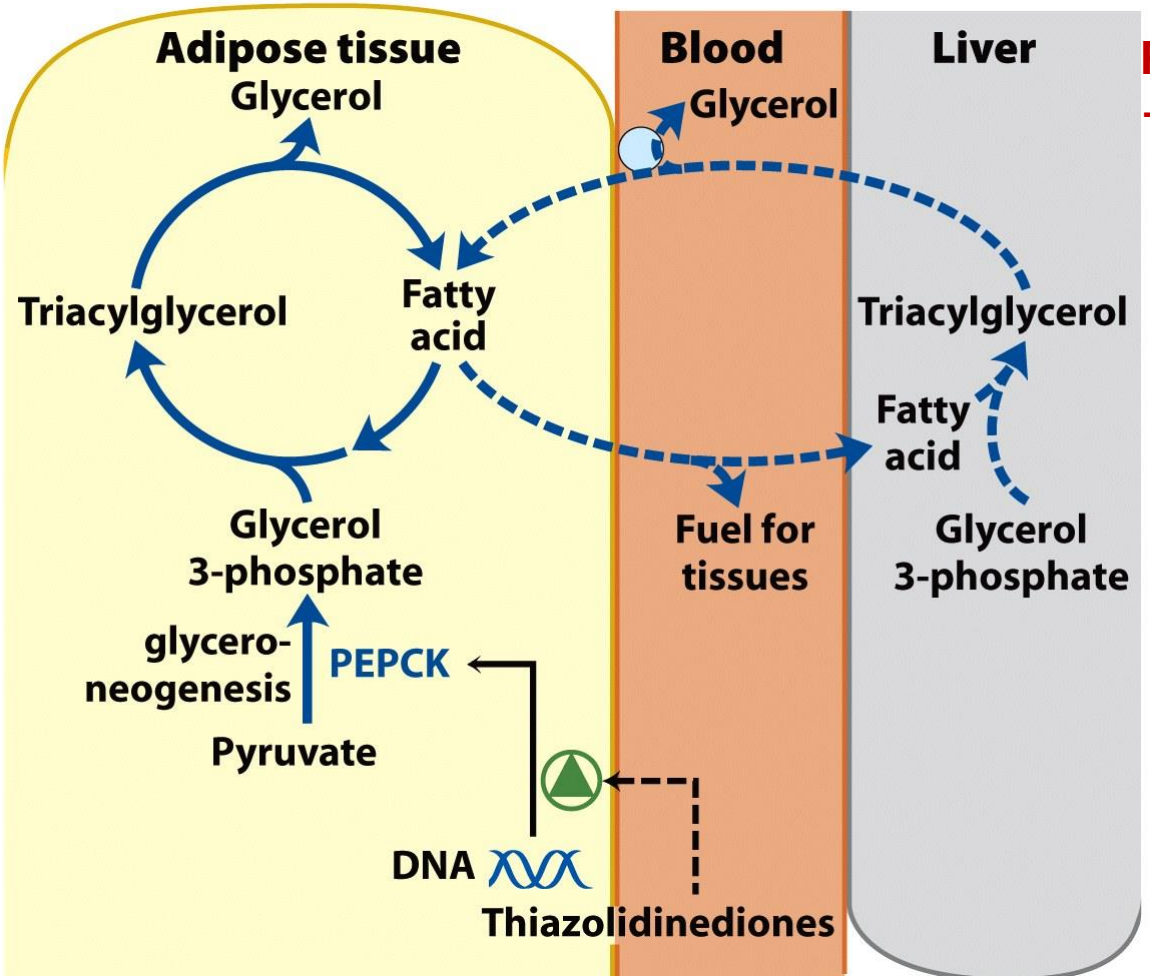


Depression

Decrease of immune activity

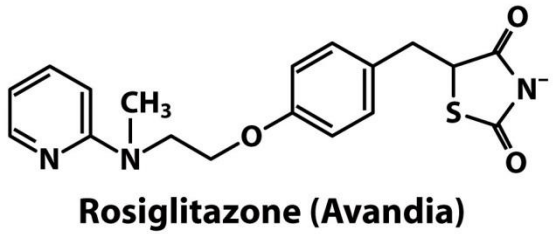
Chronic fatigue syndrom

Mechanism of thiazolidinediones (PPAR γ agonist) in type 2 diabetes

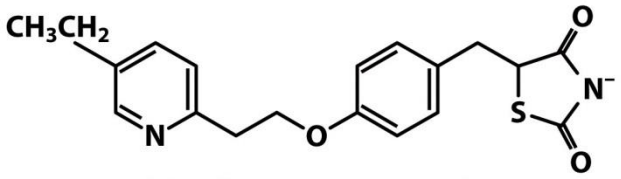


Reduced FA release

→ Increase sensitivity to insulin



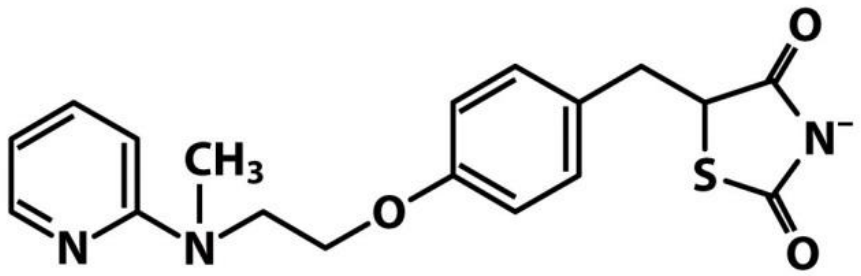
Rosiglitazone (Avandia)



Pioglitazone (Actos)

Thiazolidinediones

Unnumbered 21 p824
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Rosiglitazone (Avandia)

Annual sales peaked at approximately $2.5bn in 2006, but declined after reports of adverse effects.

The drug is currently the subject of over 13,000 lawsuits against GSK

A study in 2007 has been widely cited as finding that Avandia may increase the risk of heart attack by as much as 43%

In February 2010, David Graham, the FDA's associate director of drug safety, recommended rosiglitazone be taken off the market.

21.3 Biosynthesis of membrane phospholipids

Fluid mosaic model for membrane structure

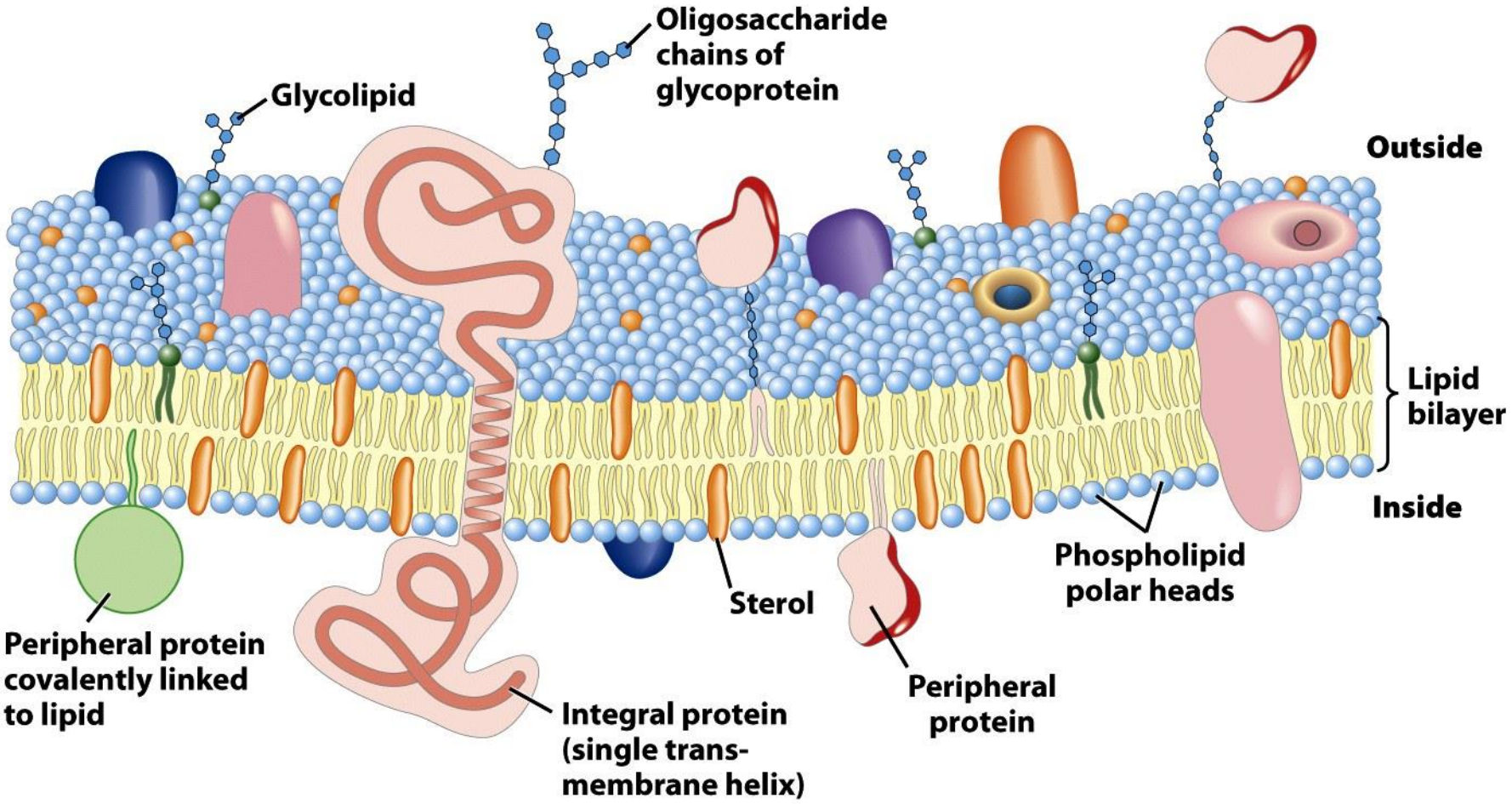
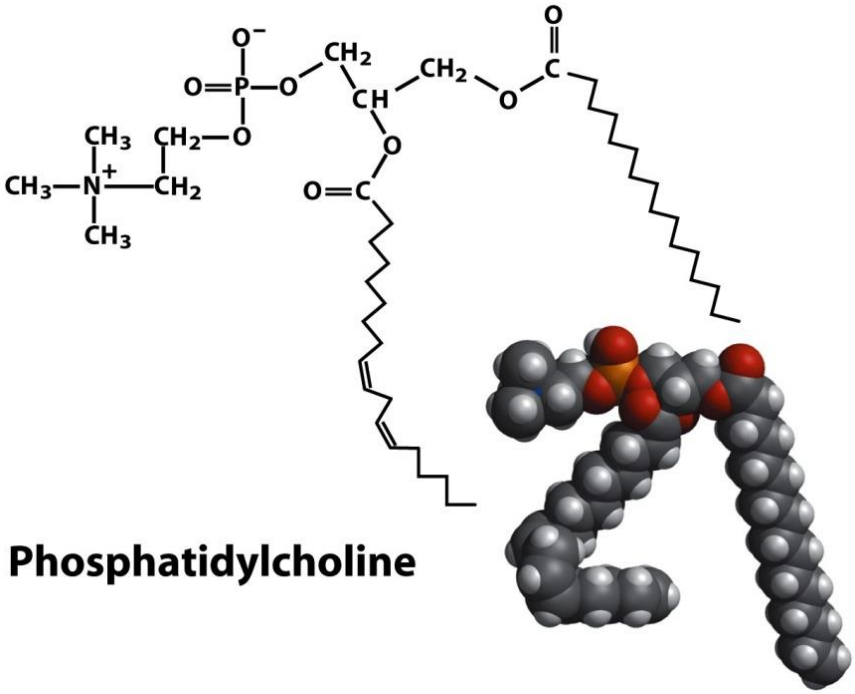


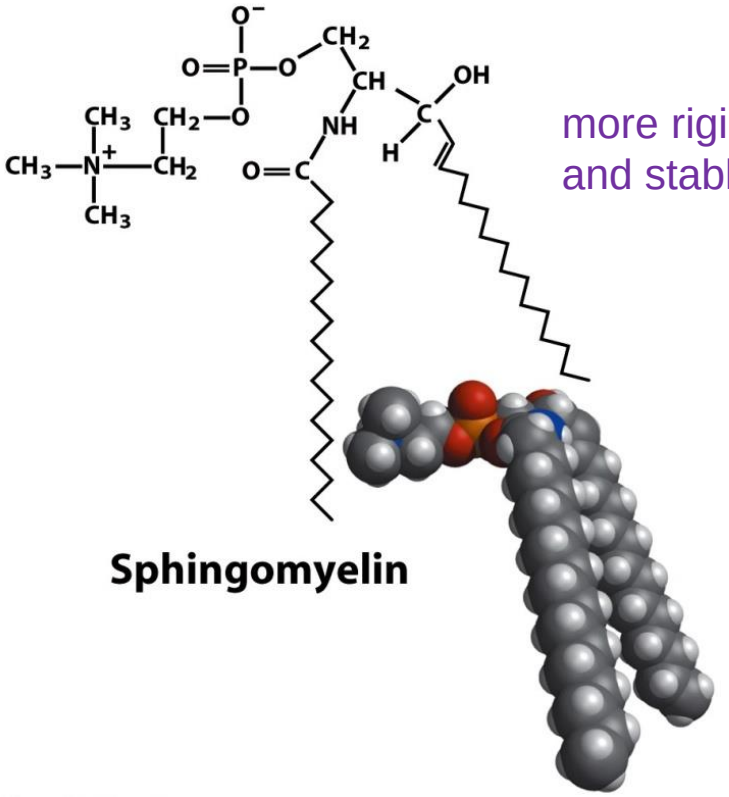
Figure 11-3
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Both proteins and lipids are free to move laterally in the plane of the bilayer, but movement of either from one leaflet of the bilayer to the other is restricted.

Glycerophospholipids



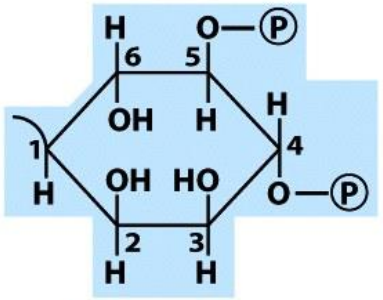
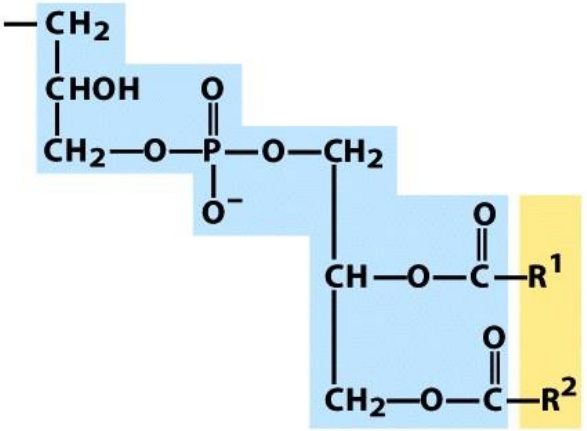
Sphingolipids



more rigid
and stable

protect the cell surface against
harmful environmental factors

Glycerophospholipids

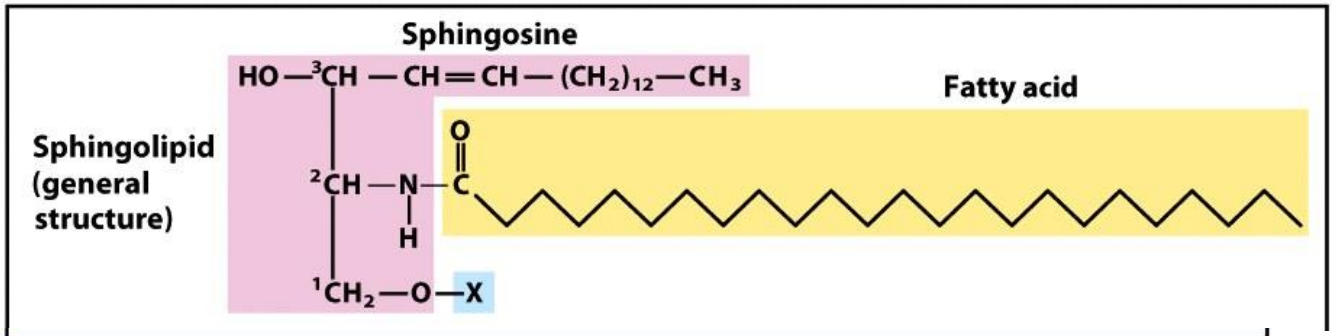
Name of glycerophospholipid	Name of X	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—	—H	−1
Phosphatidylethanolamine	Ethanolamine	$-CH_2-CH_2-\overset{+}{N}H_3$	0
Phosphatidylcholine	Choline	$-CH_2-CH_2-\overset{+}{N}(CH_3)_3$	0
Phosphatidylserine	Serine	$-CH_2-\underset{\displaystyle COO^-}{\underset{\displaystyle |}{CH}}-\overset{+}{N}H_3$	−1
Phosphatidylglycerol	Glycerol	$-CH_2-\underset{\displaystyle OH}{\underset{\displaystyle |}{CH}}-CH_2-OH$	−1
Phosphatidylinositol 4,5-bisphosphate	*myo*-Inositol 4,5-bisphosphate		−4
Cardiolipin	Phosphatidyl-glycerol		−2

Negatively charged !

Sphingolipids



"Sphingo"
Compounds isolated from the brain and nervous tissue

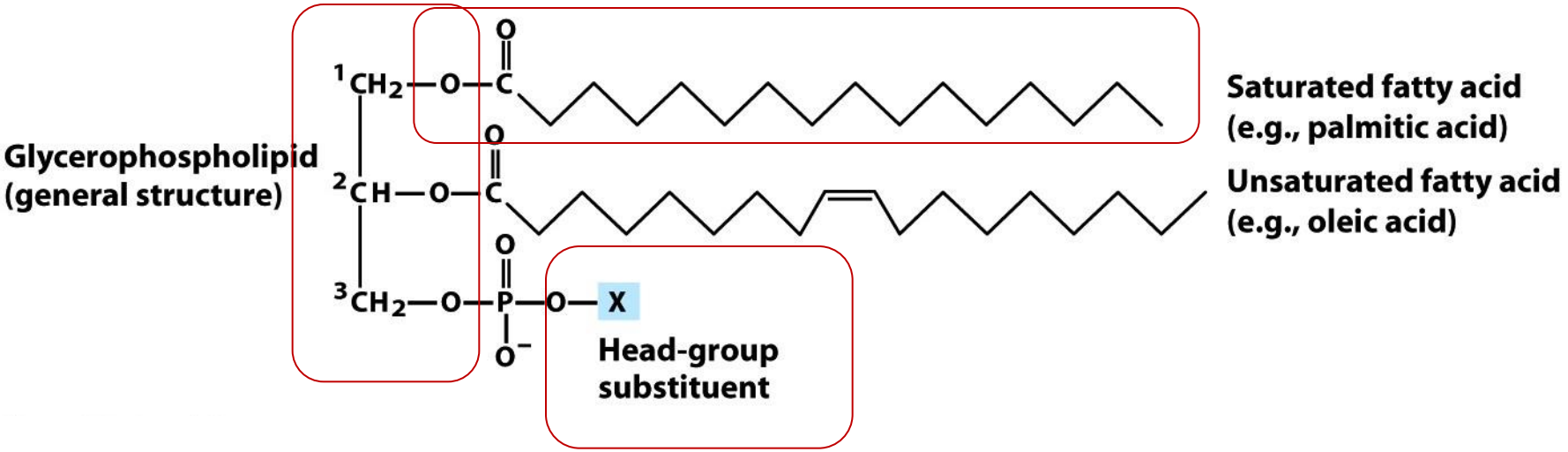


Name of sphingolipid	Name of X—O	Formula of X
Ceramide	—	—H
Sphingomyelin	Phosphocholine	
Neutral glycolipids Glucosylcerebroside	Glucose	
Lactosylceramide (a globoside)	Di-, tri-, or tetrasaccharide	
Ganglioside GM2	Complex oligosaccharide	

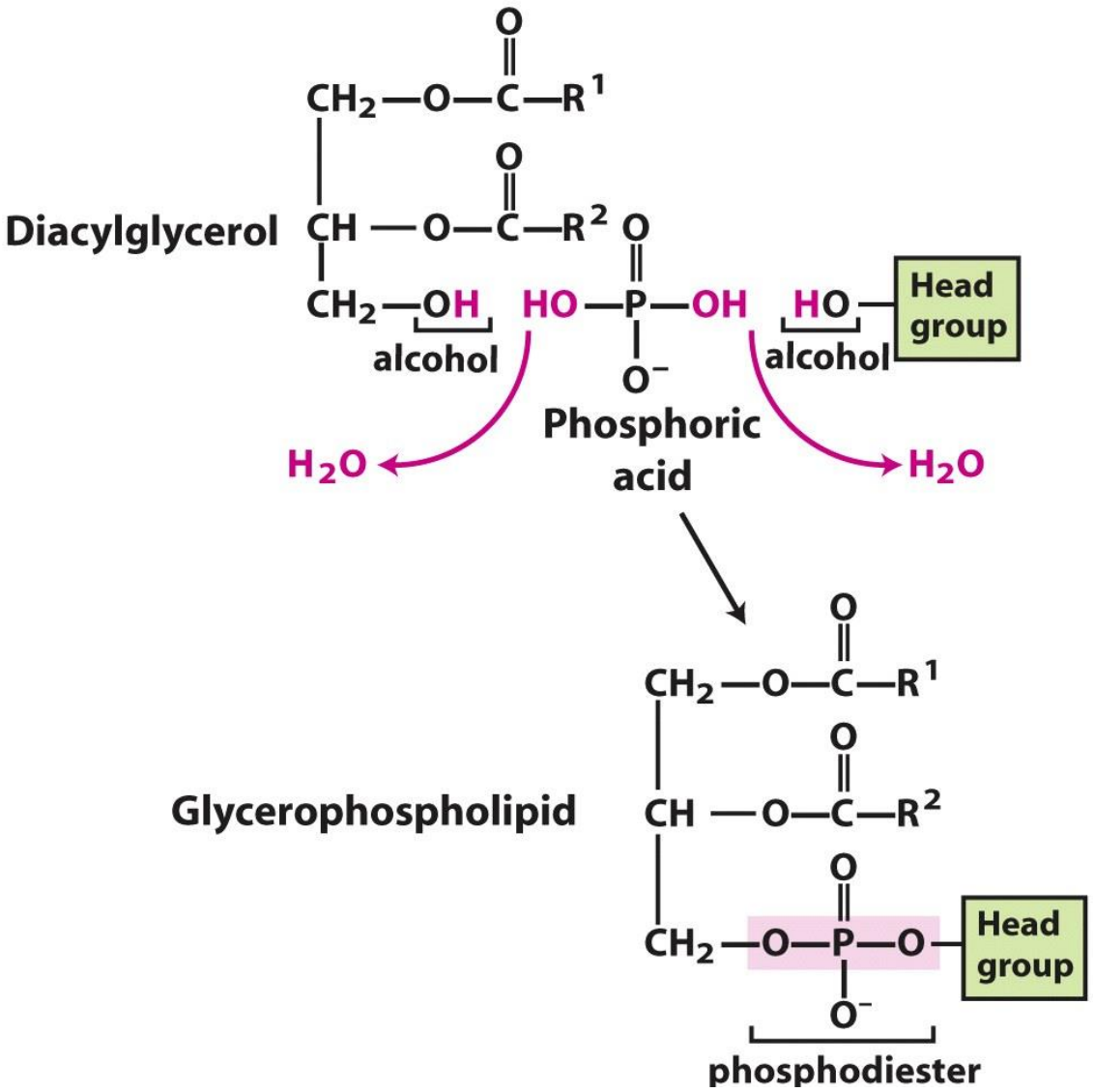
ganglioside :
glycosphingolipid with sialic acids

Biosynthetic pathways of membrane phospholipids

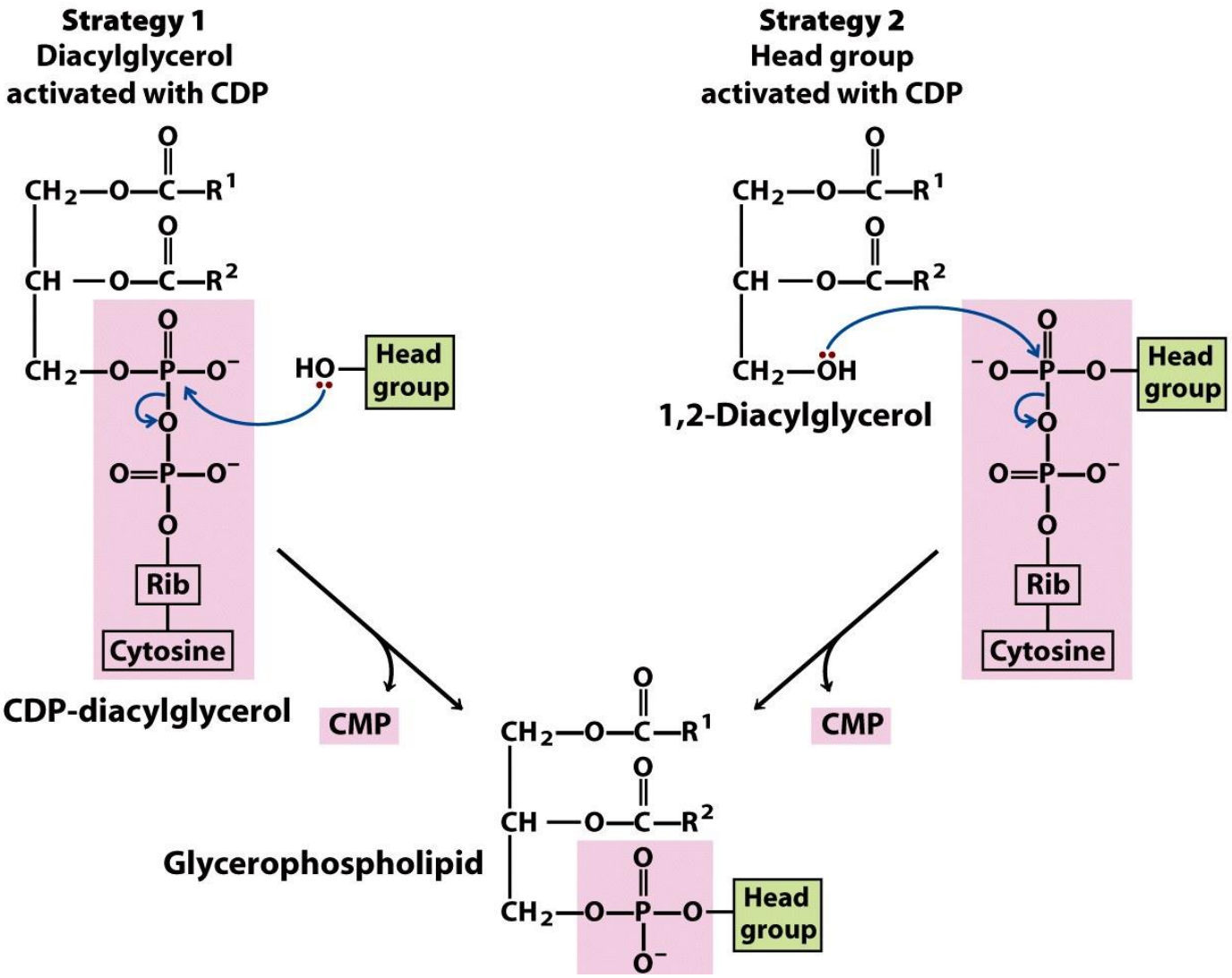
(1) Synthesis of the backbone molecule (glycerol or sphingosine)
(2) Attachment of fatty acids
(3) Addition of a hydrophilic head group
(4) Alteration or exchange of the head group



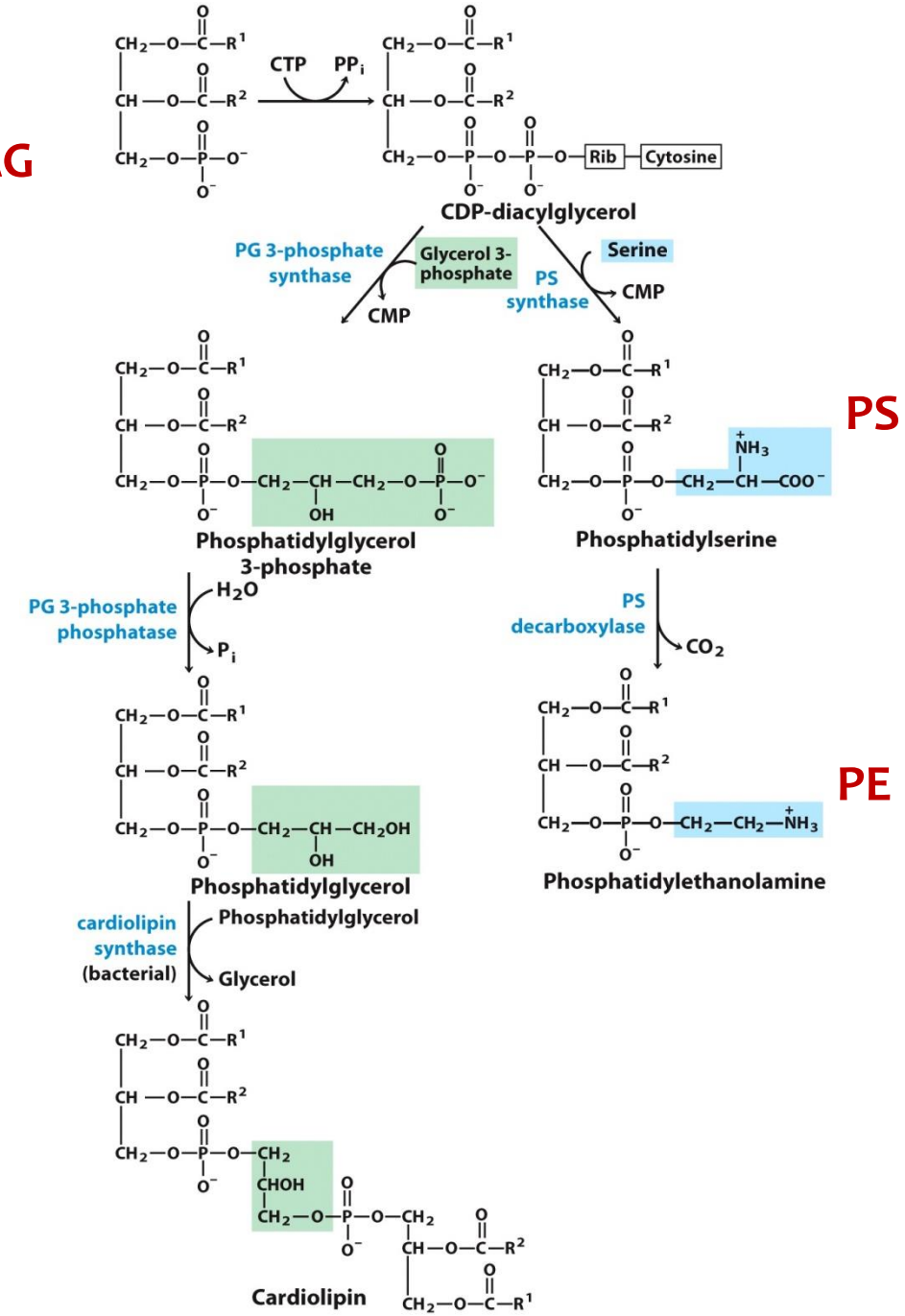
Head-group attachment



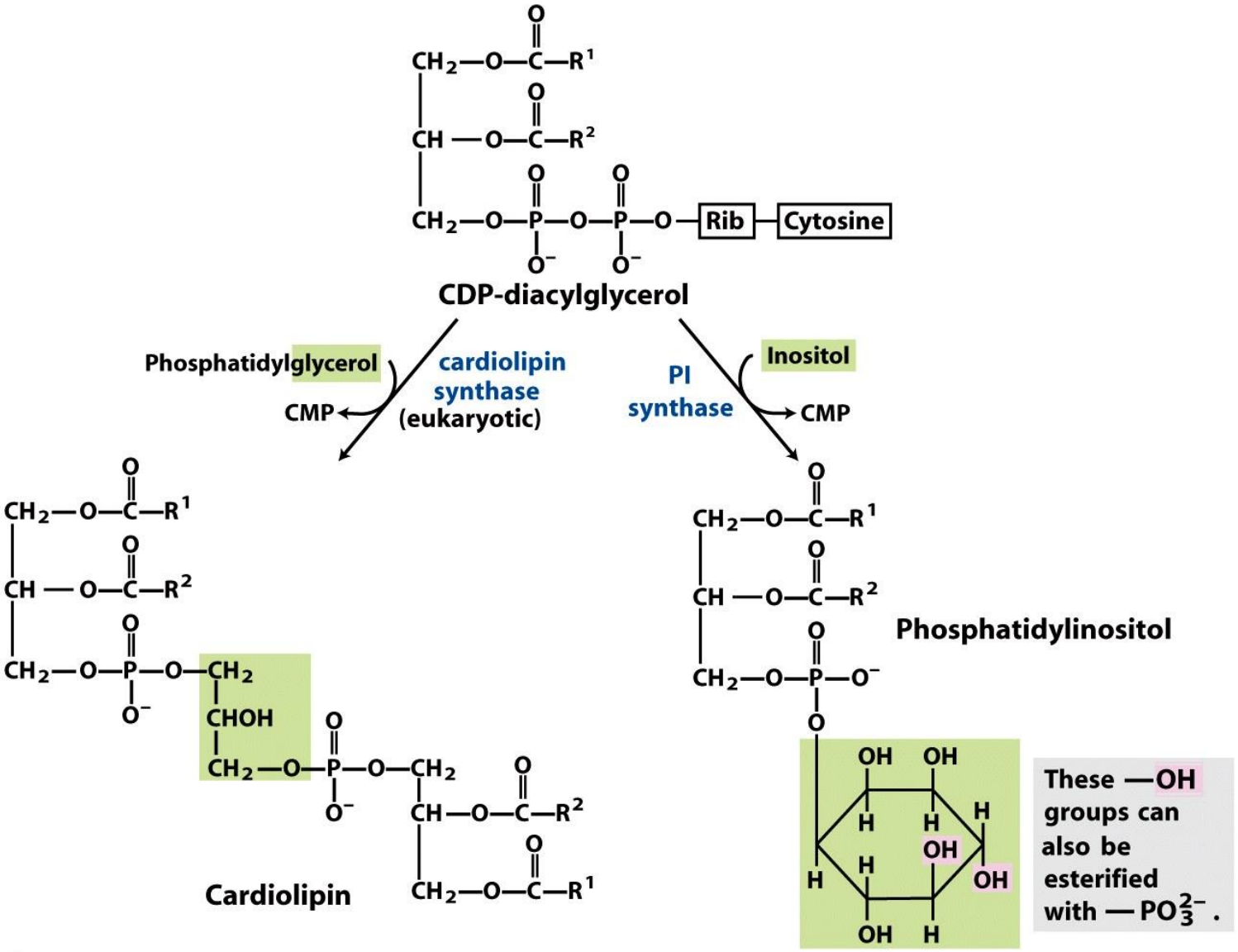
Two general strategies for forming the phosphodiester bond of phospholipids



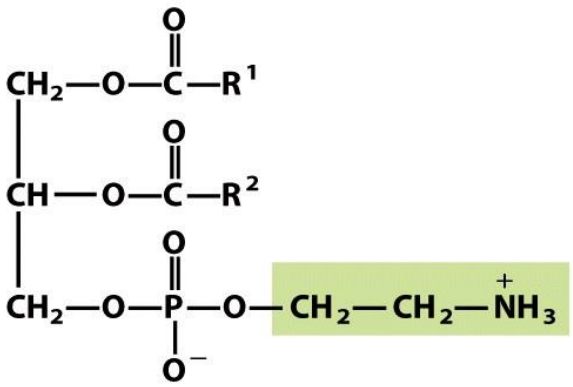
Origin of the polar head groups of phospholipids in *E. coli*.



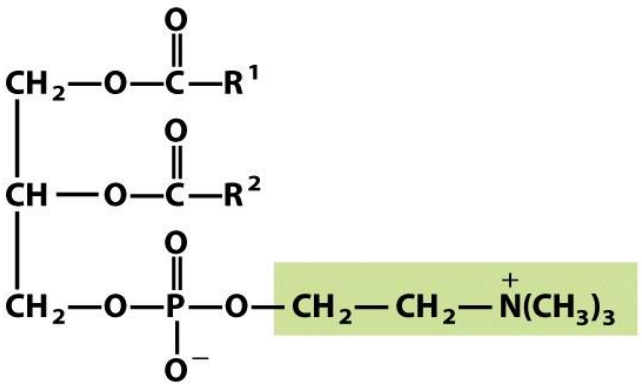
Synthesis of cardiolipin and phosphatidylinositol in eukaryotes



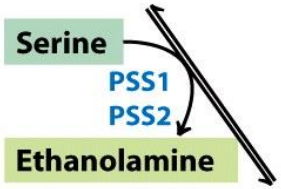
Pathways for phosphatidylserine and phosphatidylcholine synthesis in mammals

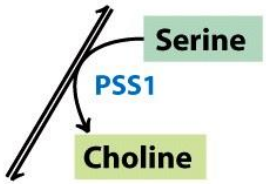


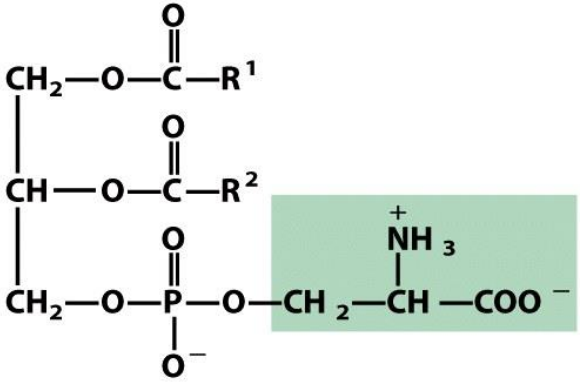
Phosphatidylethanolamine



Phosphatidylcholine

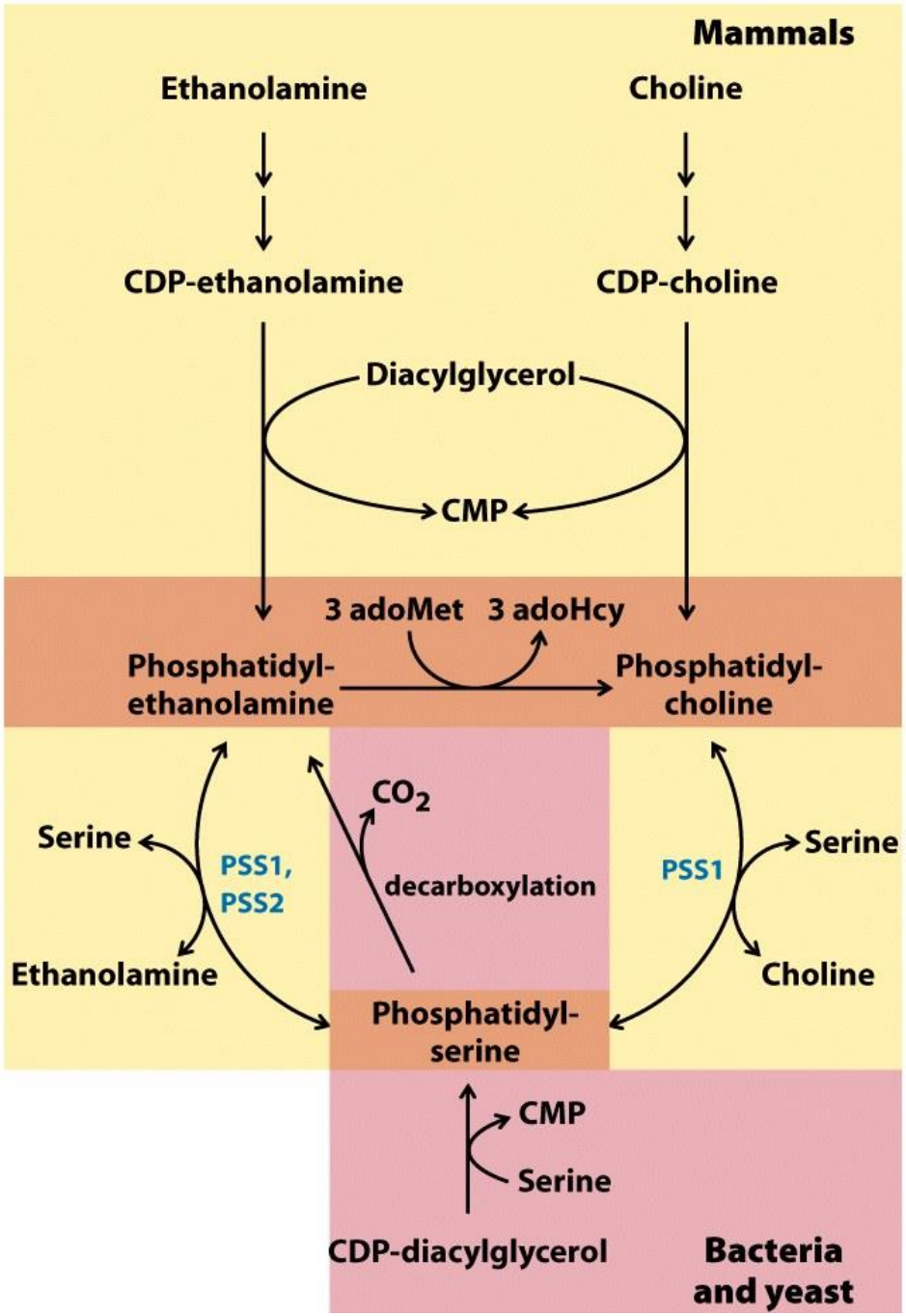




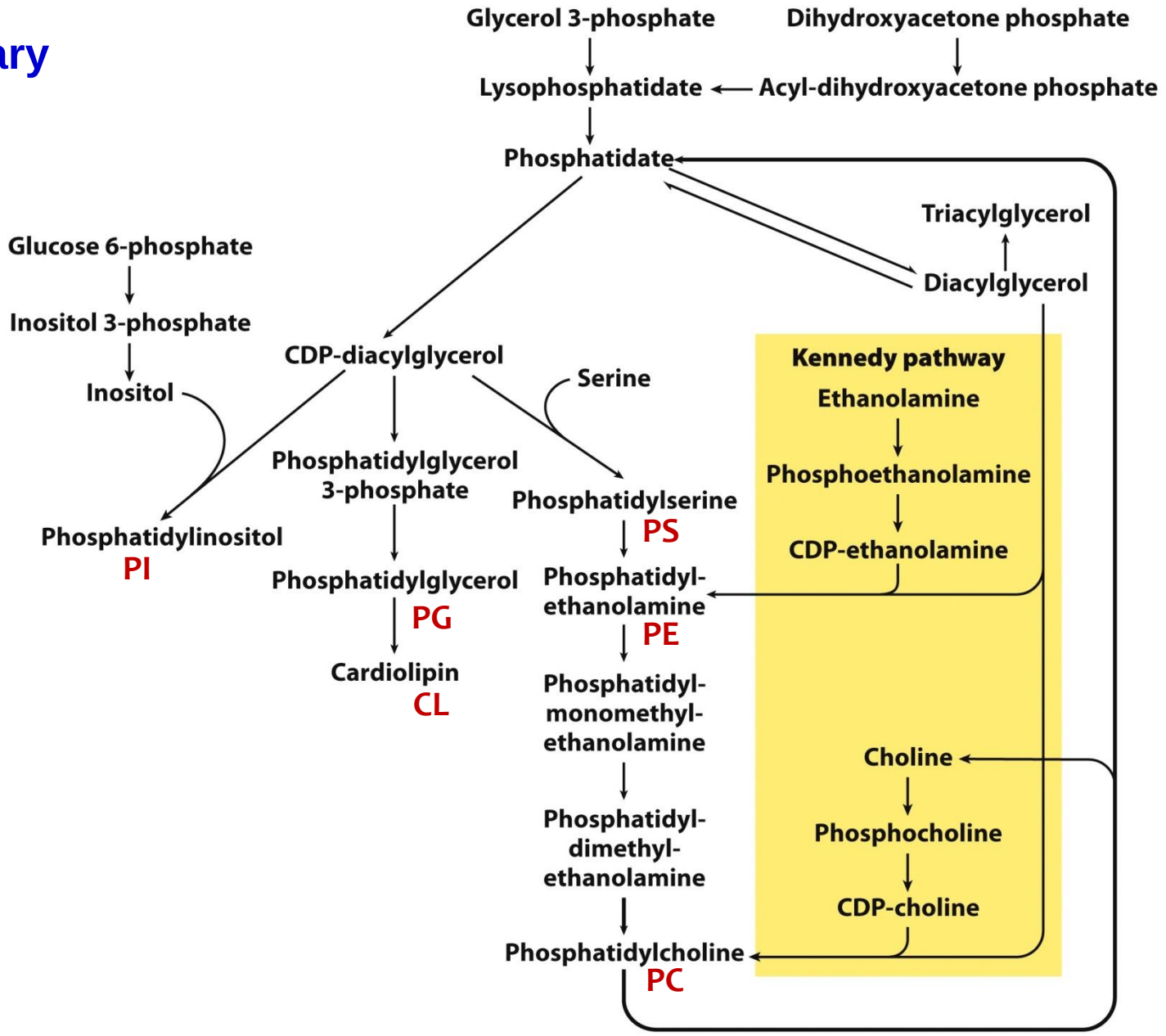


Phosphatidylserine

Summary of the pathways for synthesis of major phospholipids

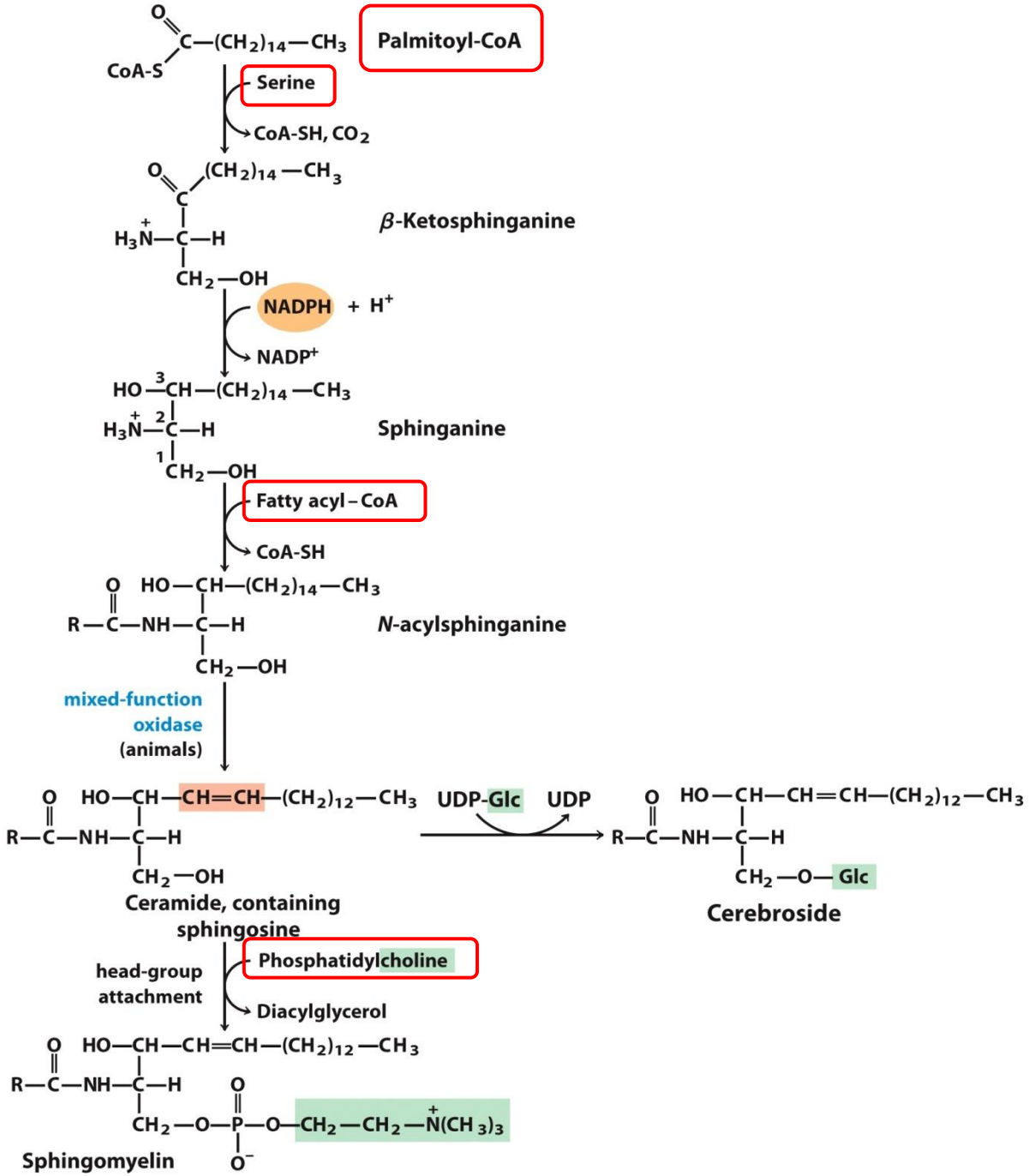


Summary

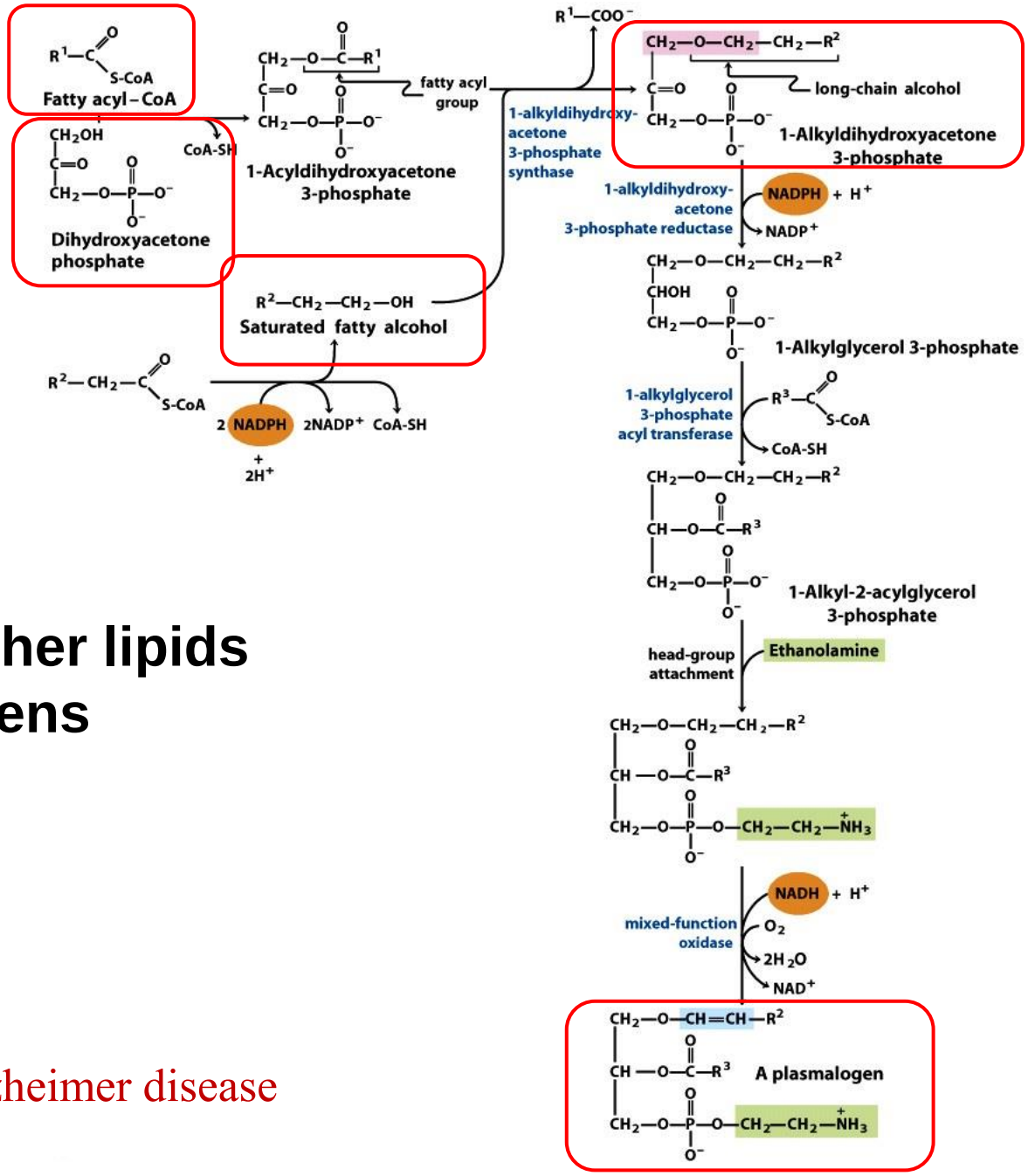


Kennedy pathway : de novo pathway for PE and PC synthesis

Biosynthesis of sphingolipids

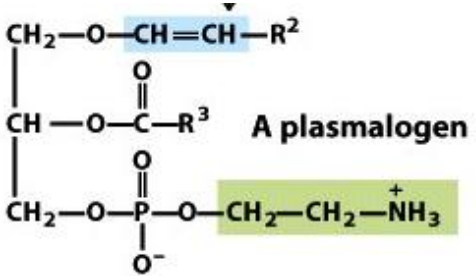


Synthesis of ether lipids and plasmalogens



Nervous system & Alzheimer disease

What is plasmalogen ?



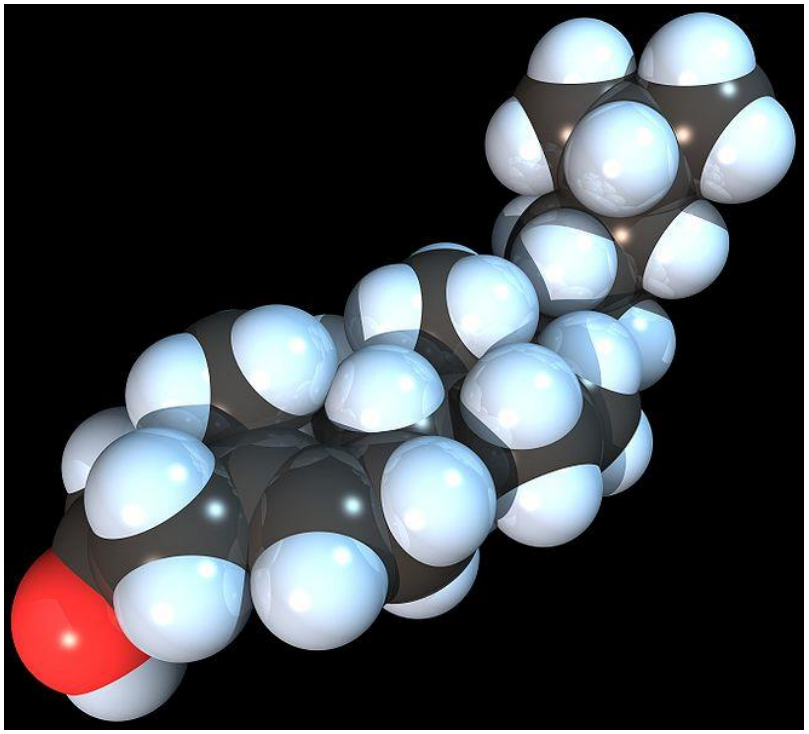
Plasmalogens are a type of ether phospholipid with a vinyl ether linkage and an ester linkage.

In human heart tissue, nearly 30–40% of choline glycerophospholipids are plasmalogens. Even more striking is the fact that almost 30% of the glycerophospholipids in the adult human brain and up to 70% of myelin sheath ethanolamine glycerophospholipids are plasmalogens

It has been demonstrated that they can protect mammalian cells against the damaging effects of reactive oxygen species

21.4 Biosynthesis of cholesterol, steroid, and isoprenoids

cholesterol



As a precursor for …

1. Steroid hormones
2. Vitamin D
3. Bile acids

Also used in …

1. Cellular membrane
2. Plasma lipoproteins

Origin of the carbon atoms of cholesterol

… from simple acetate

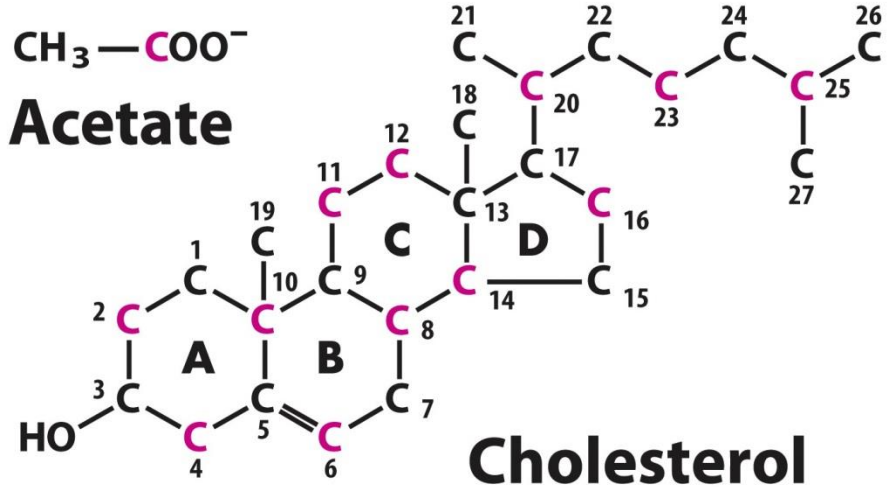
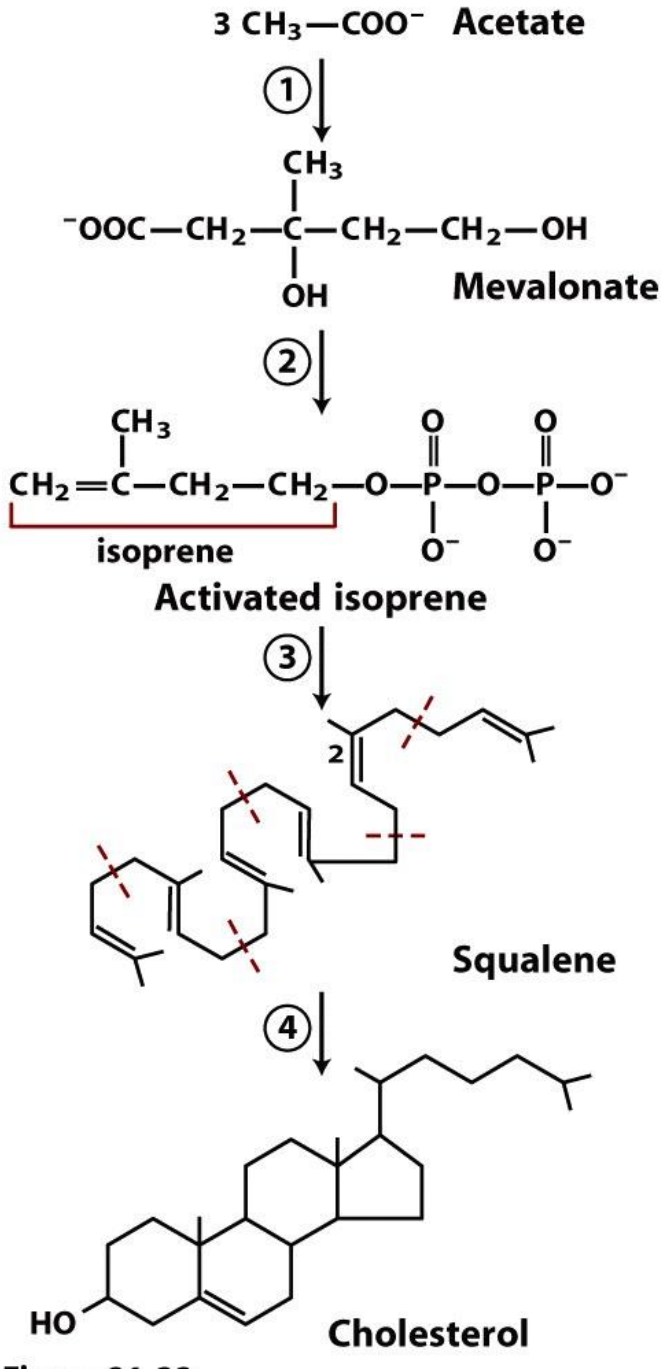
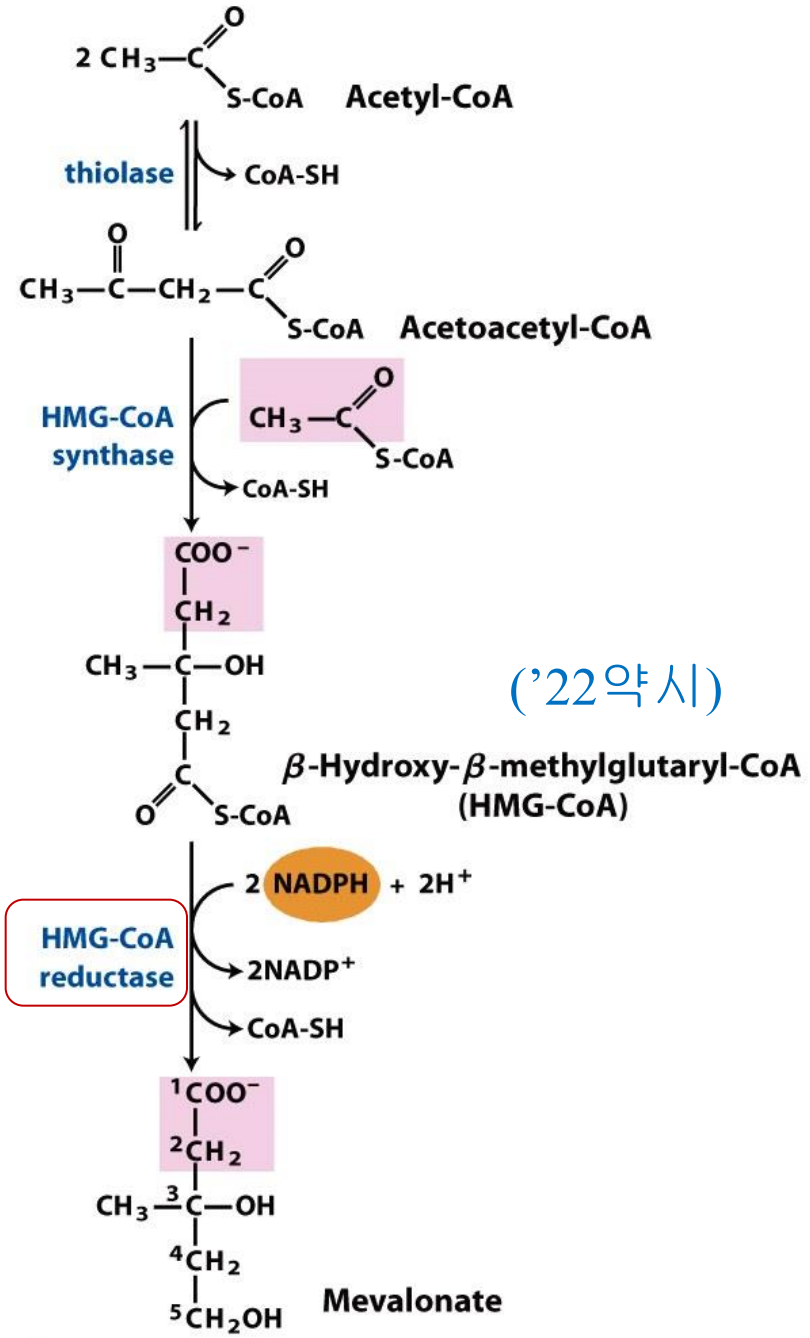


Figure 21-32
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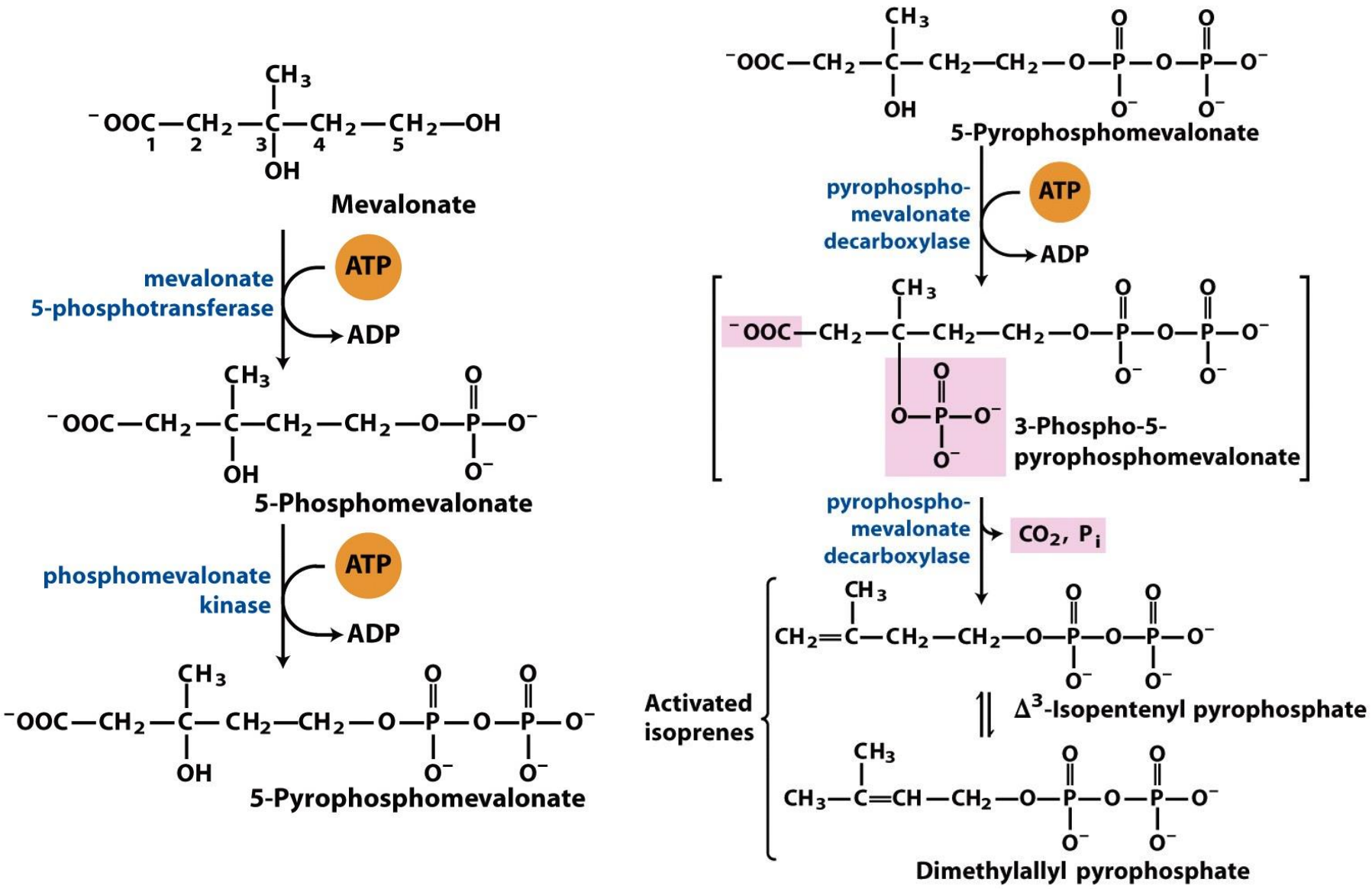


(1) Formation of mevalonate from acetyl-CoA

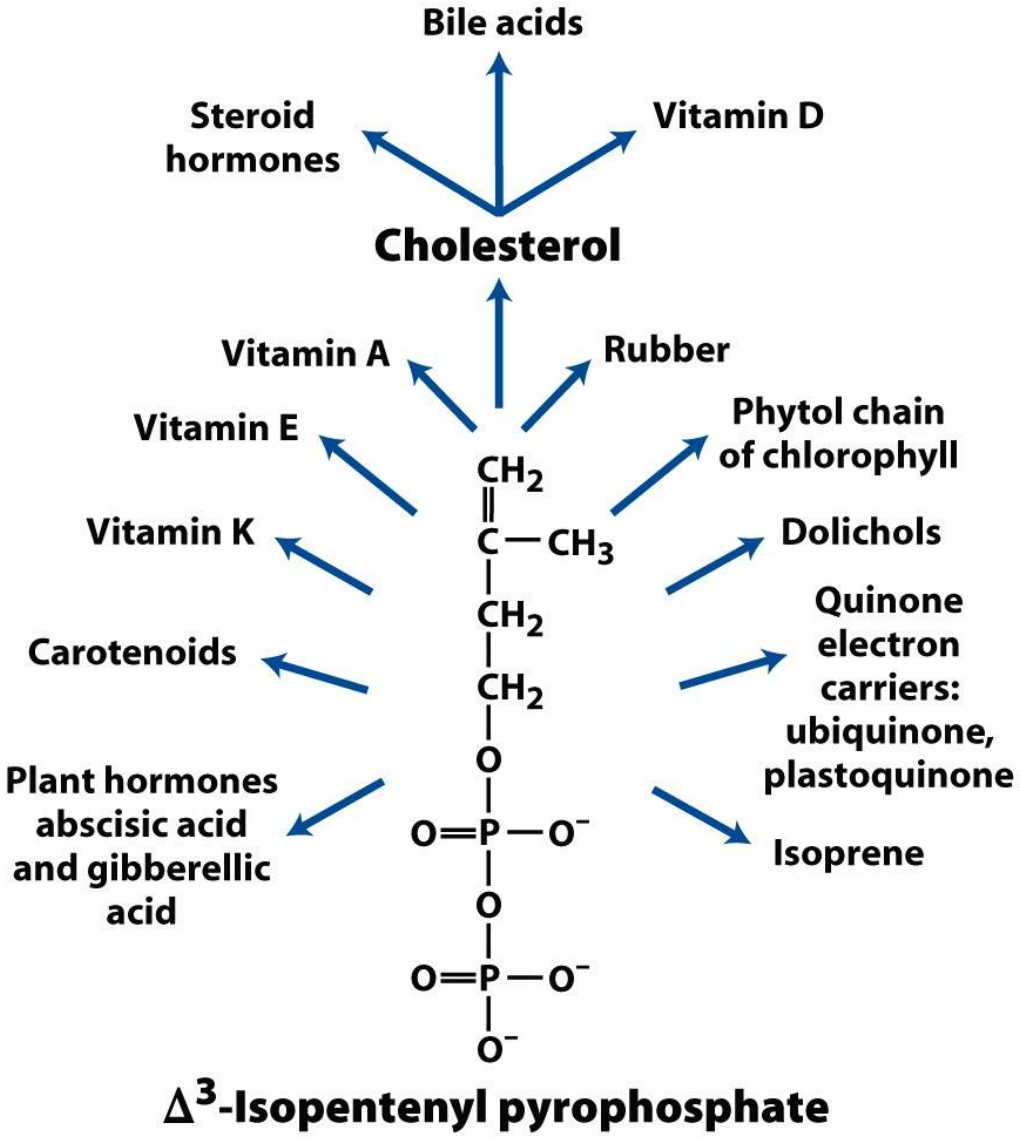


('22약시)

(2) Conversion of mevalonate to activated isoprene units

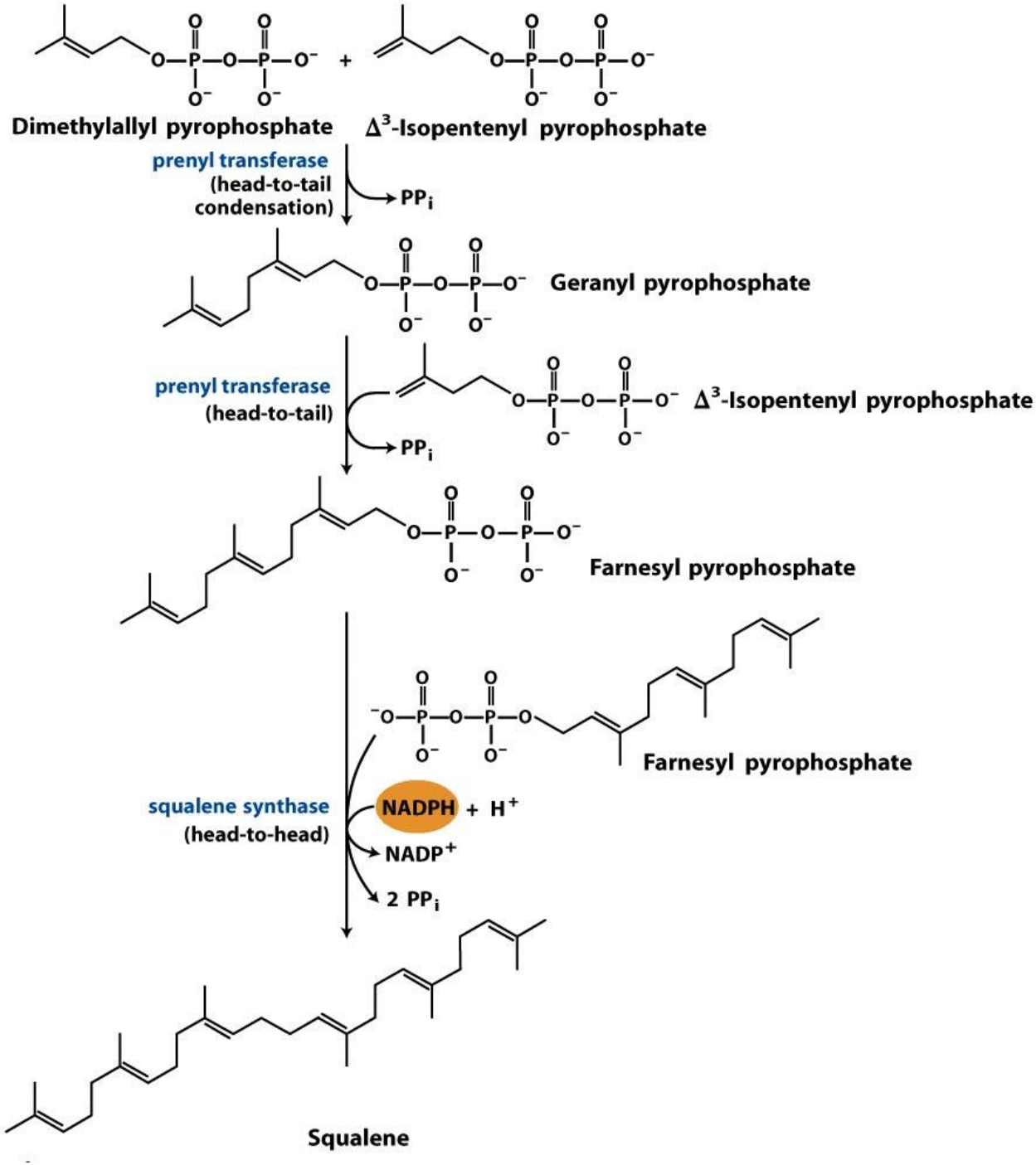


Overview of isoprenoid biosynthesis



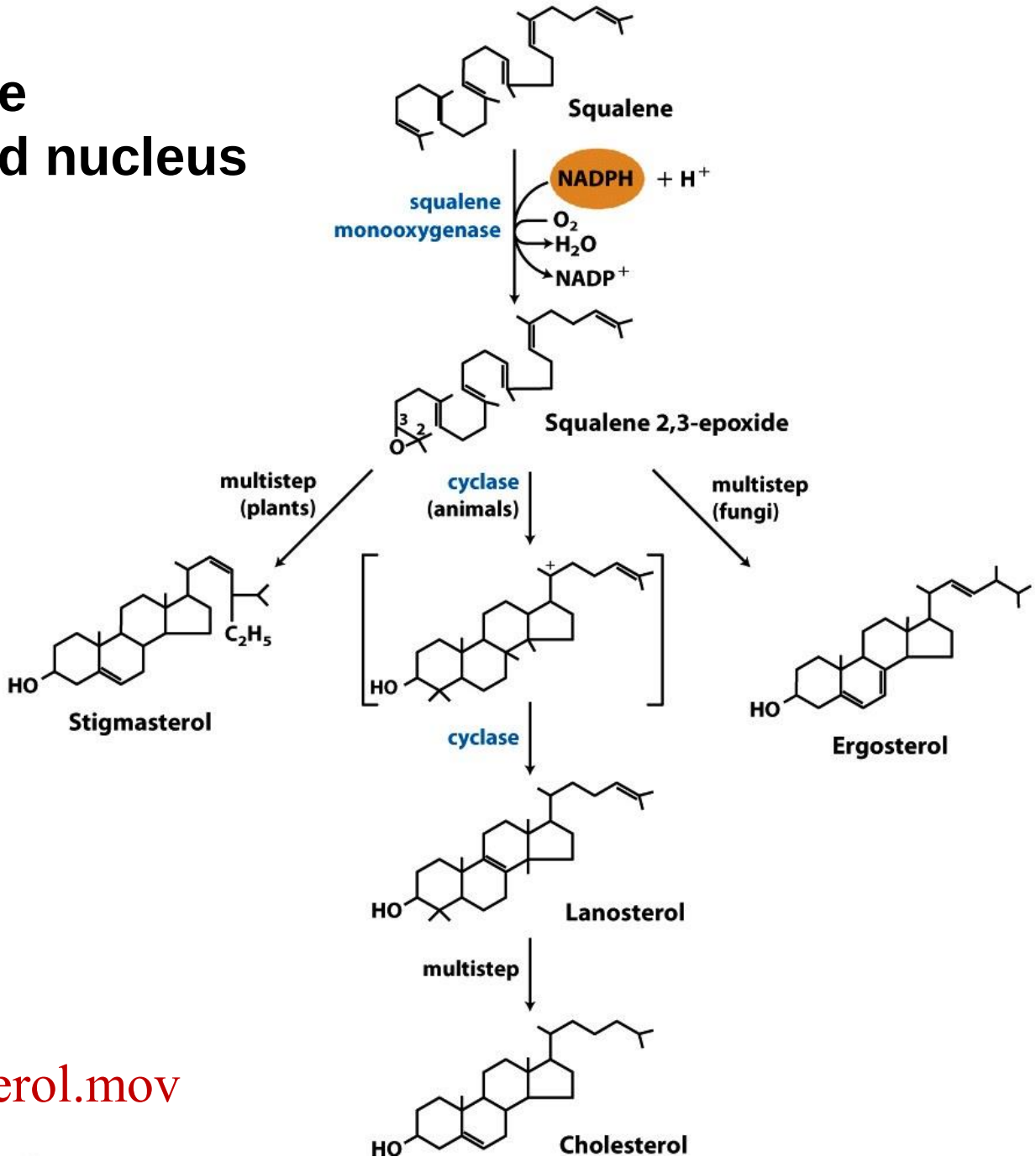
('22약시)

(3) Formation of squalene



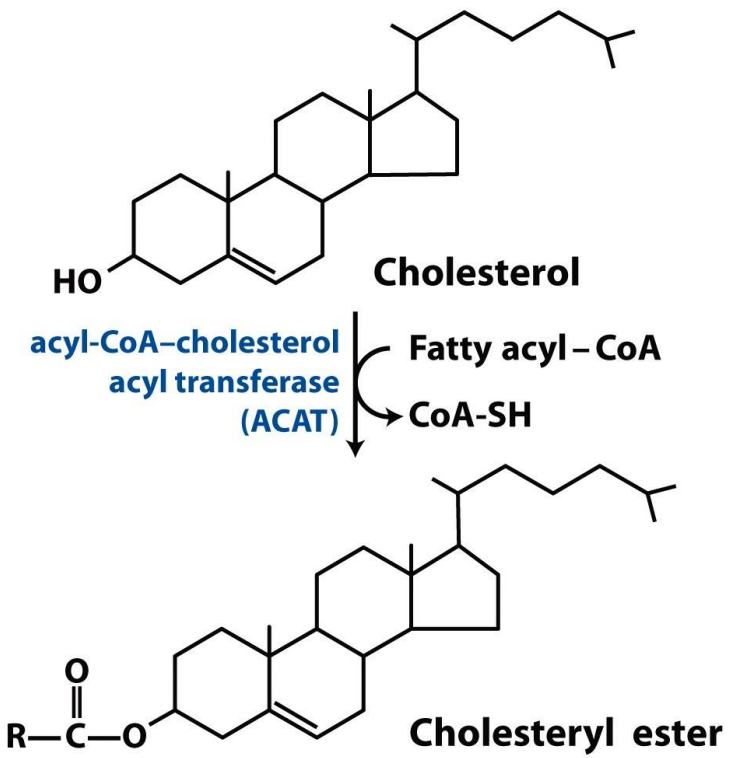
biosynthesis_squalene.mov

(4) squalene to the condensed steroid nucleus



squalene_to_cholesterol.mov

Synthesis of cholesteryl esters
(more hydrophobic form for storage and transport)

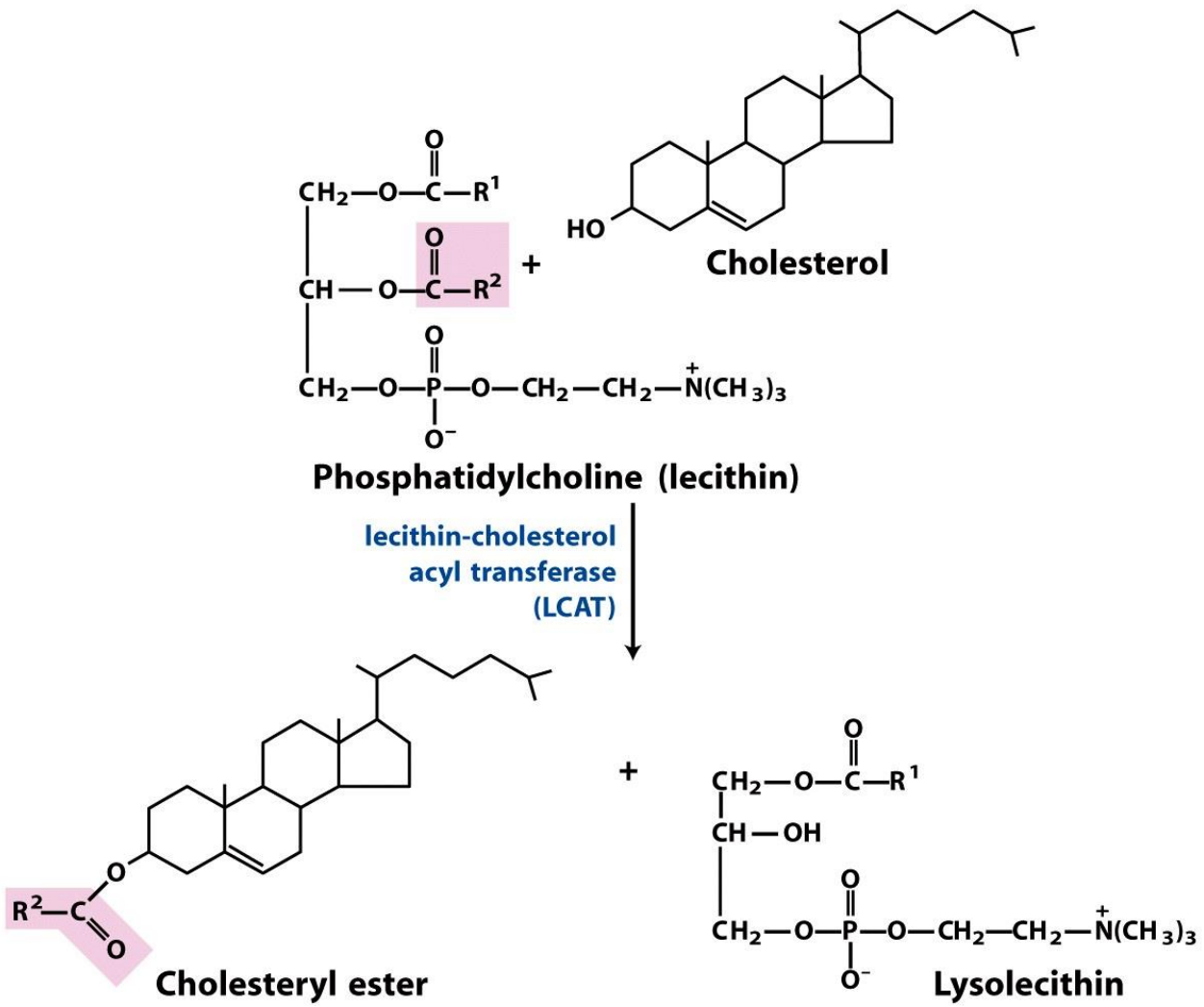


Cellular membrane

Bile cholesterol

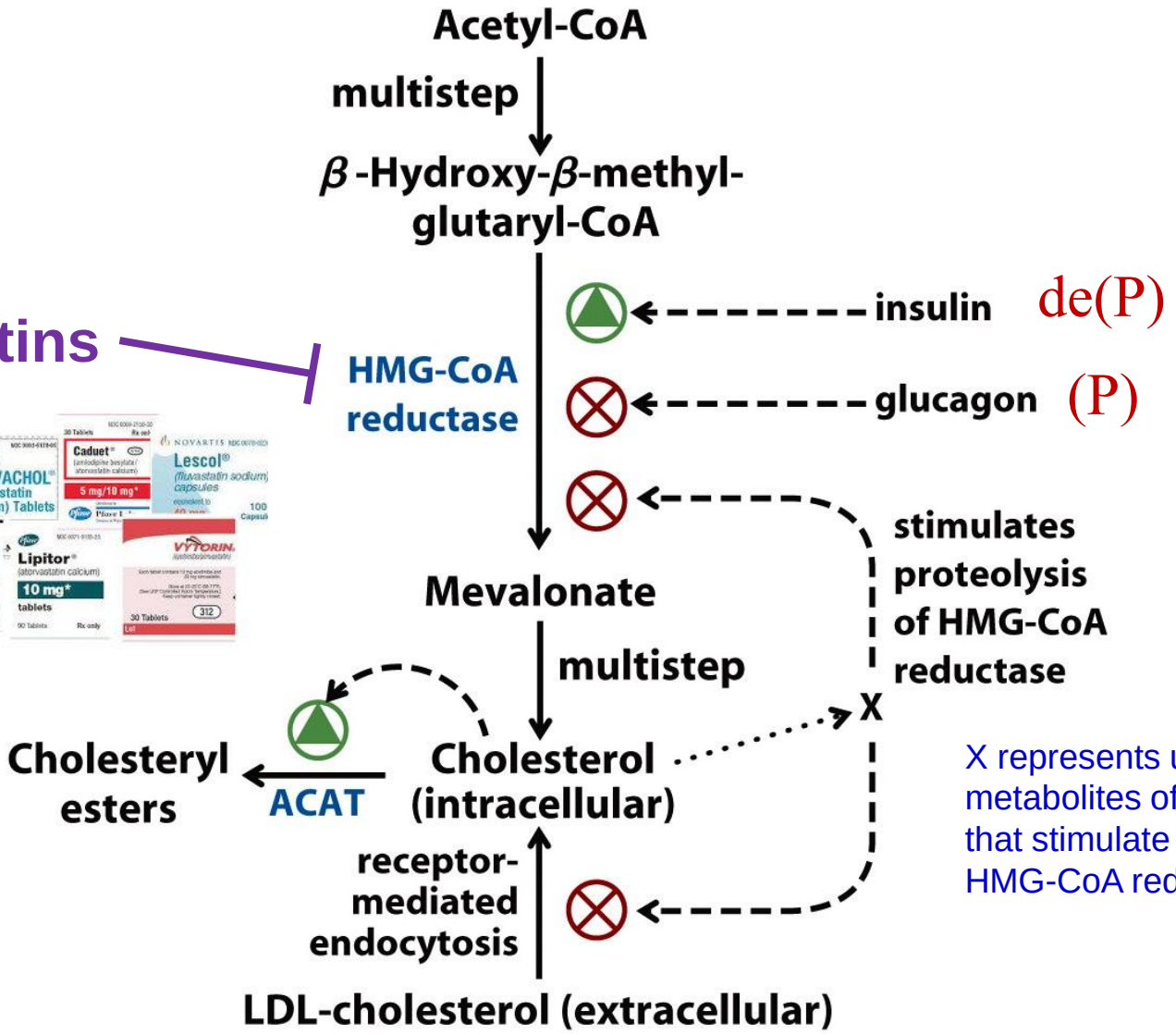
Bile acids

Reaction catalyzed by lecithin-cholesterol acyl transferase (LCAT) in HDL



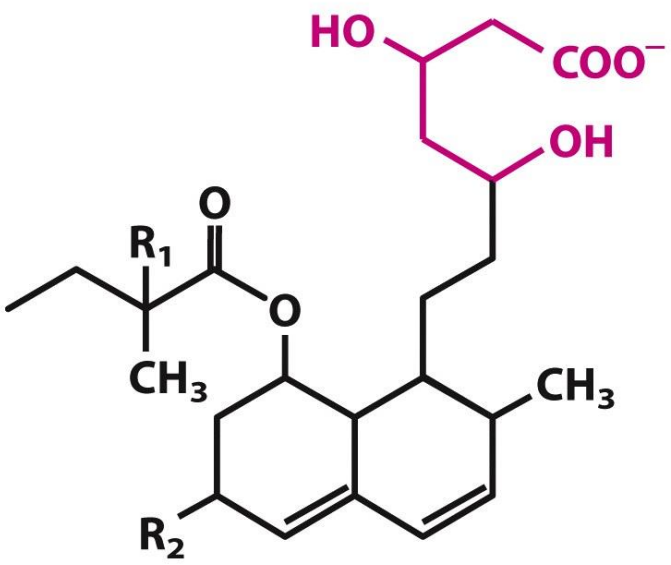
Regulation of cholesterol formation balances synthesis with dietary uptake

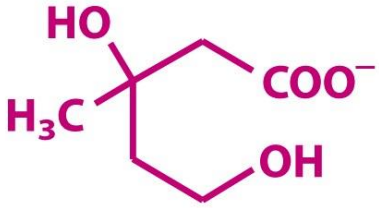




X represents unidentified metabolites of cholesterol that stimulate proteolysis of HMG-CoA reductase.

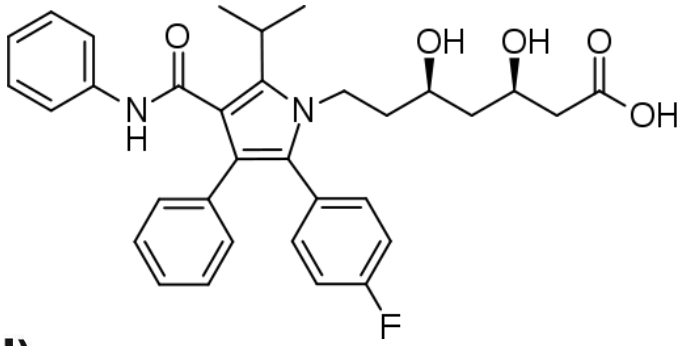
Statins as inhibitors of HMG-CoA reductase





Mevalonate

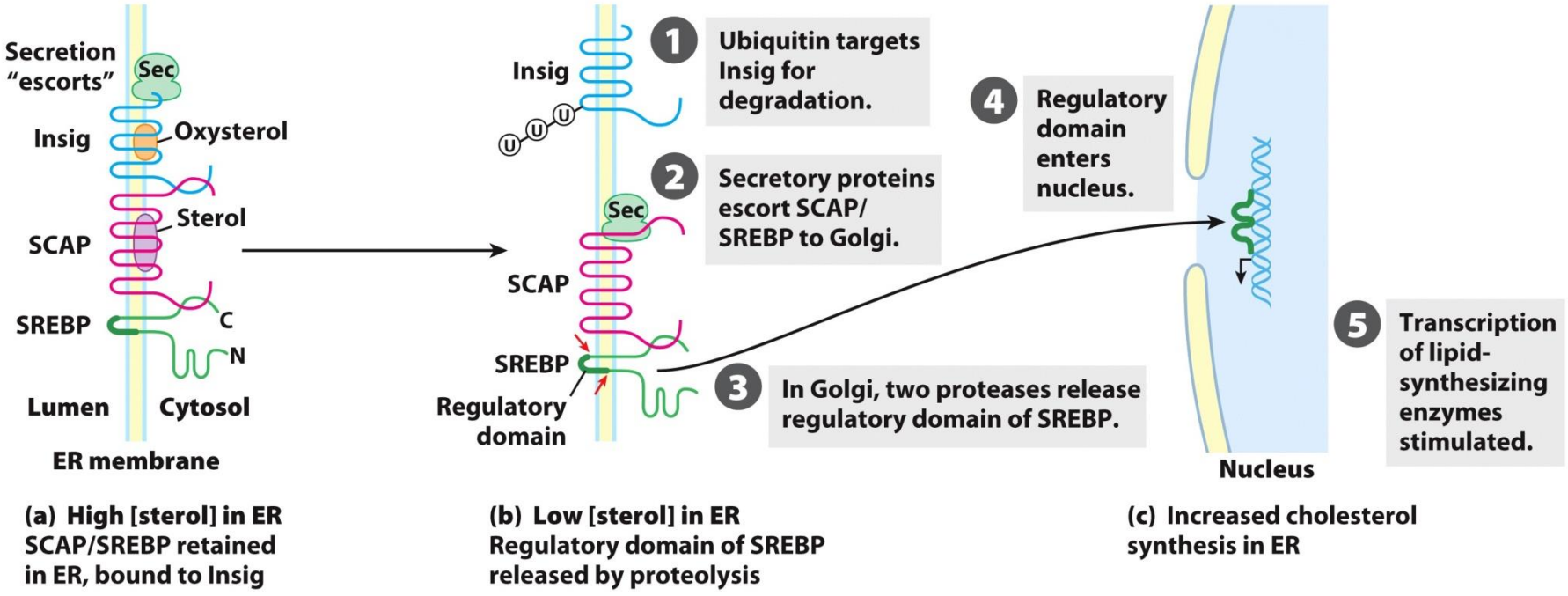
$R_1 = H$	$R_2 = H$	**Compactin**
$R_1 = CH_3$	$R_2 = CH_3$	**Simvastatin (Zocor)**
$R_1 = H$	$R_2 = OH$	**Pravastatin (Pravachol)**
$R_1 = H$	$R_2 = CH_3$	**Lovastatin (Mevacor)**



Atrovastatin: Lipitor
$ 12B/yr in sales

Box 21-3 figure 1
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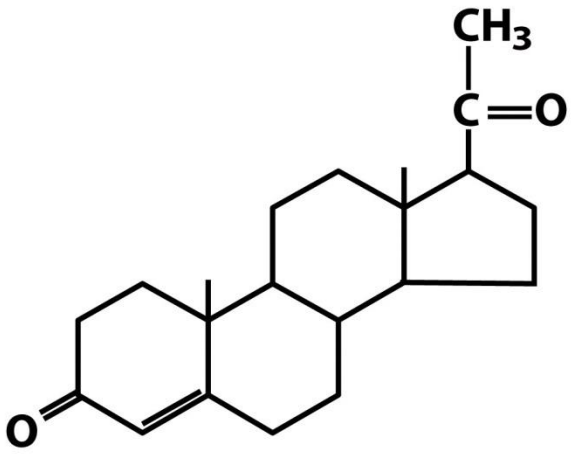
When cholesterol is scarce… SREBP activation



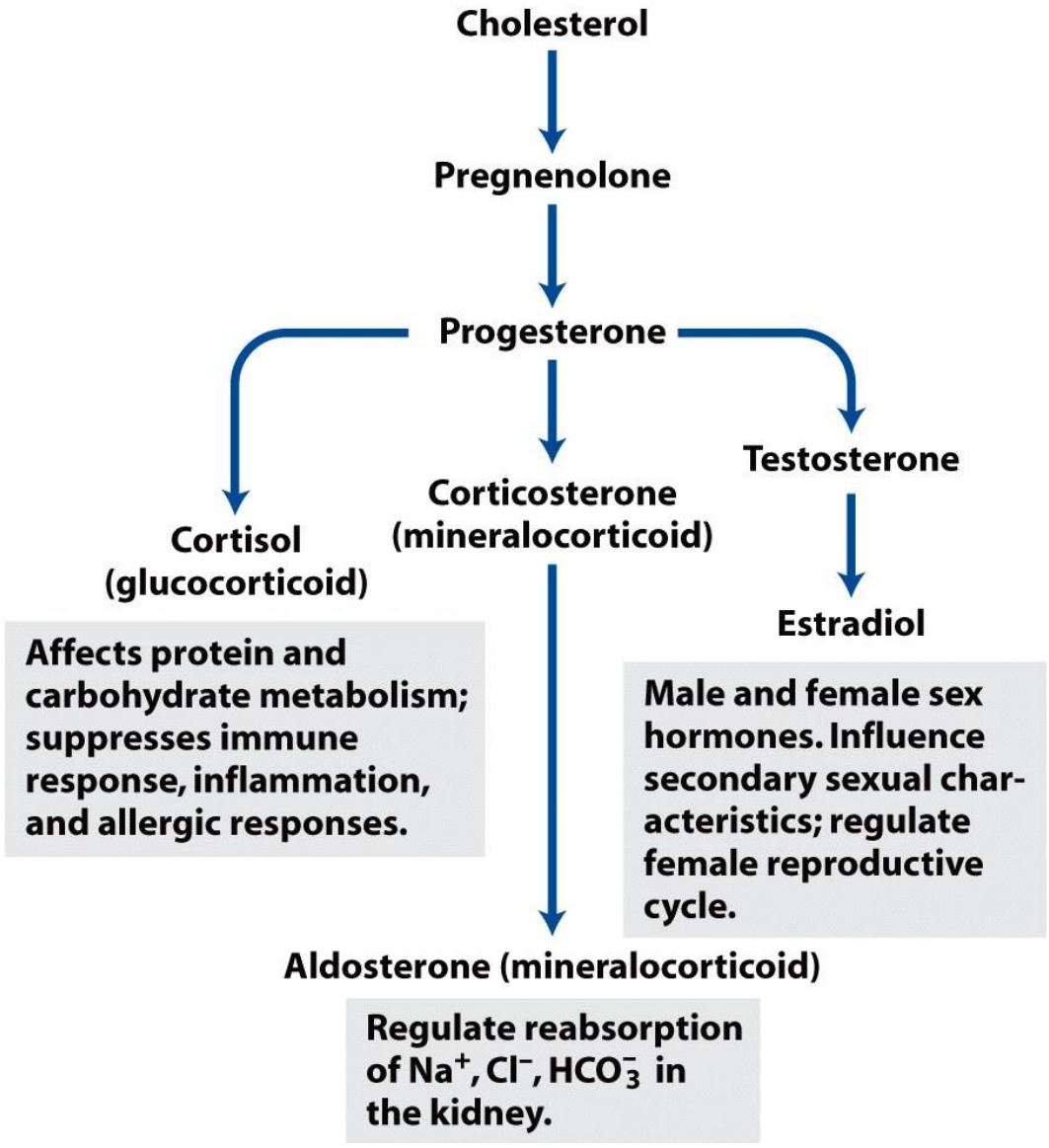
HMG-CoA reductase ↑

steroid hormones

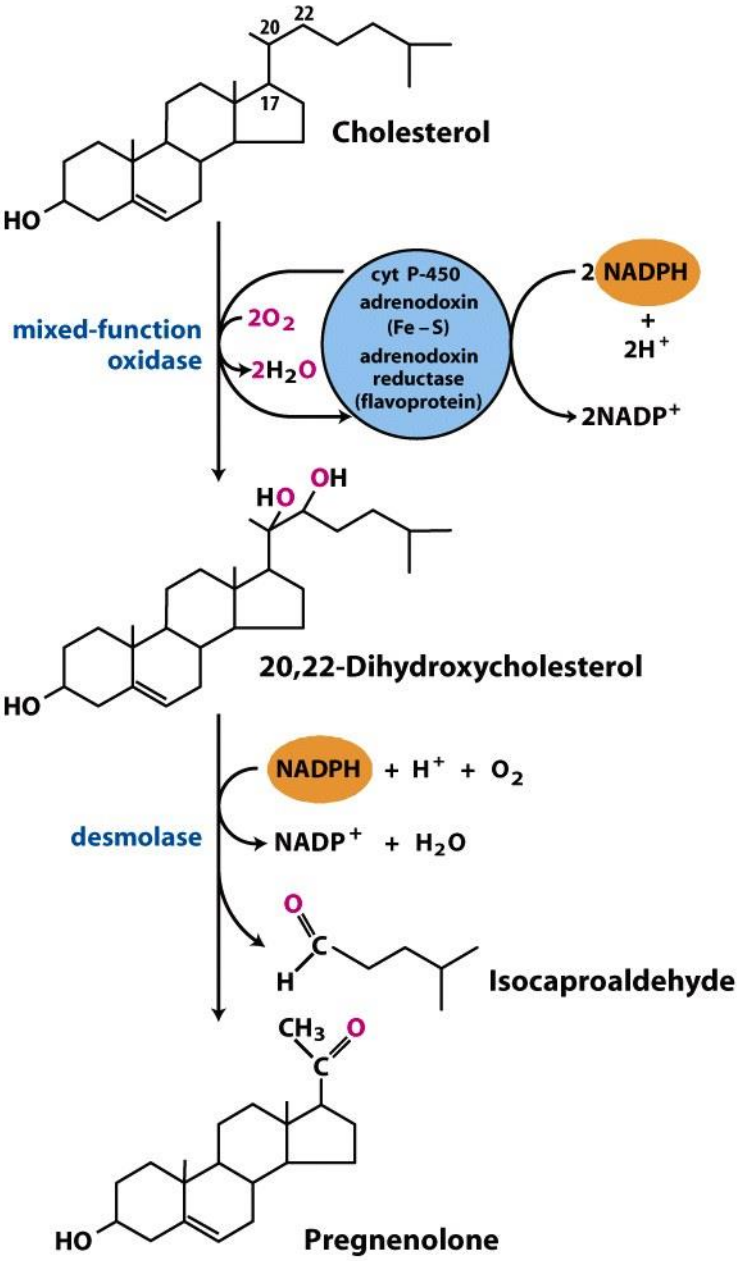
Some steroid hormones derived from cholesterol



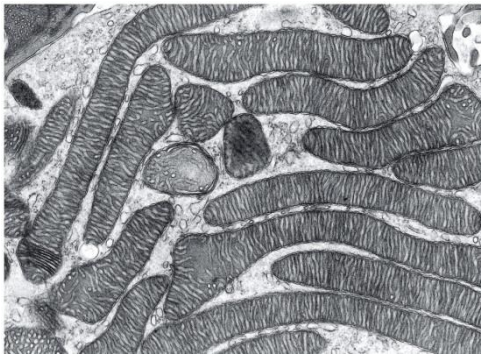
Progesterone



Side-chain cleavage in the synthesis of steroid hormones



Mitochondrial P-450 oxygenases catalyze steroid hydroxylations



Mitochondria of adrenal gland, specialized for steroid synthesis

Cholesterol

Cholesterol side-chain cleavage enzyme

Progestagens (21 carbons)

Pregnenolone

Progesterone

Deoxy-corticosterone

Corticosterone

17α-hydroxylase

3-beta-hydroxysteroid dehydrogenase (3β-HSD)

21α-hydroxylase

11β-hydroxylase

Aldosterone synthase

Mineralocorticoids (21 carbons)

Aldosterone

17α-hydroxy pregnenolone

17α-hydroxy progesterone

11-deoxycortisol

Cortisol

Glucocorticoids (21 carbons)

17,20 lyase

Androgens (19 carbons)

Dehydroepi-androsterone

Androste-nedione

17β-HSD

Androstenediol

Testosterone

5α-reductase

Dihydrotestosterone

Aromatase

Estrone

Estradiol

Estrogens (18 carbons)

(liver and placenta)

Estriol

Cellular location of enzymes

Mitochondria

Smooth endoplasmic reticulum

Lipid transport

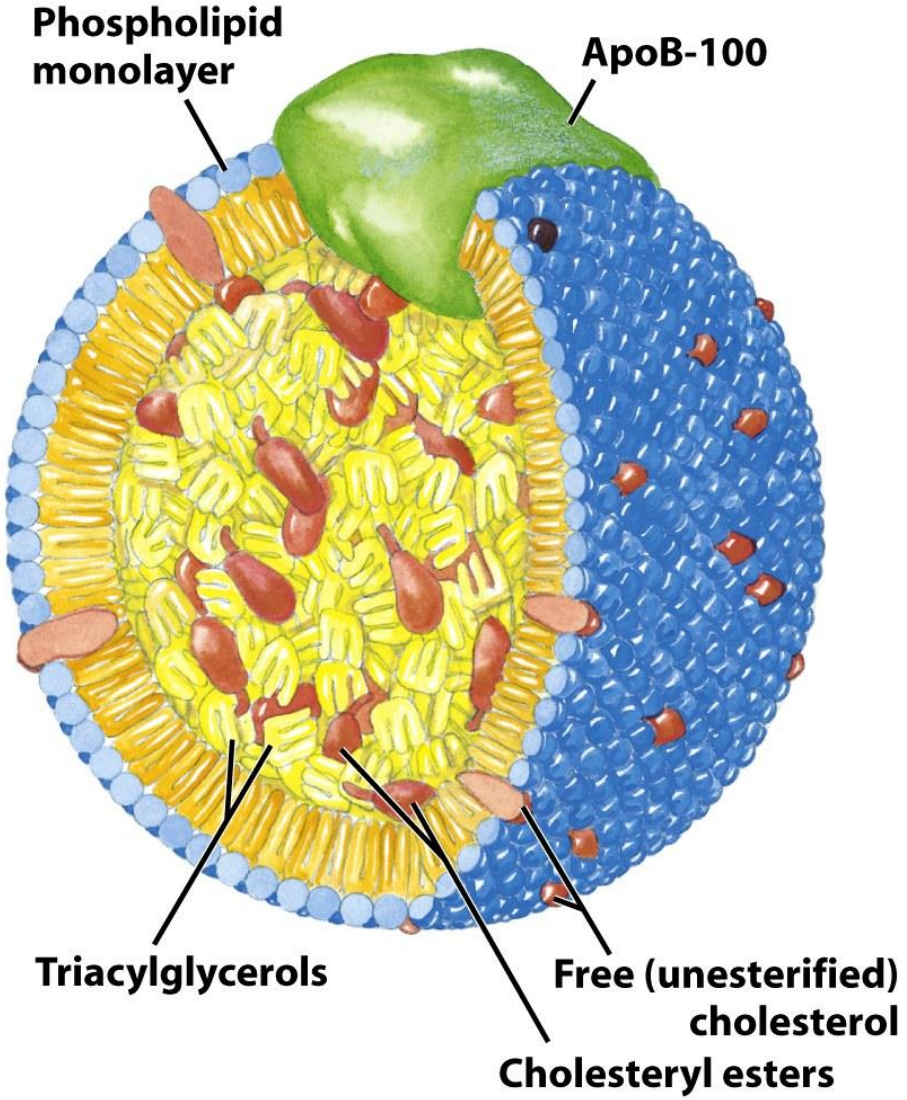


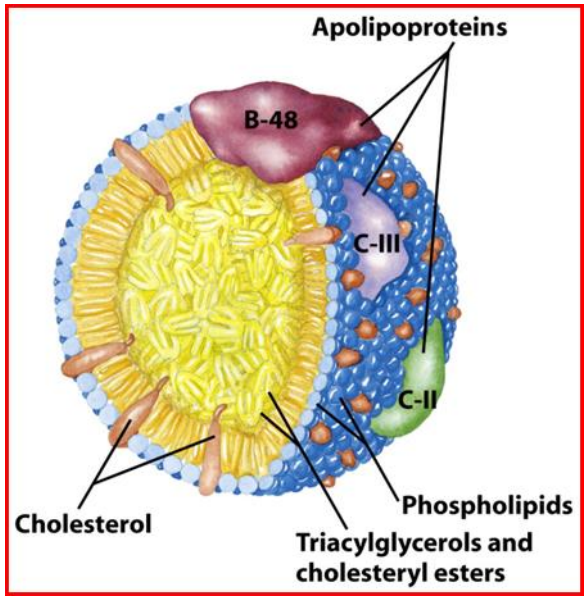
Blood plasma samples collected after a fast (left) and after a high-fat meal (right). Chylomicrons produced after a fatty meal give the plasma a milky appearance.



Cholesterol and other lipids are carried on plasma lipoproteins

Structure of a low-density lipoprotein (LDL)



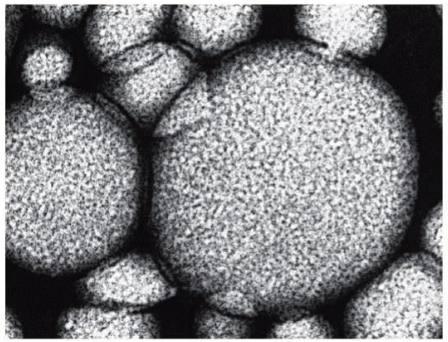


Chylomicron

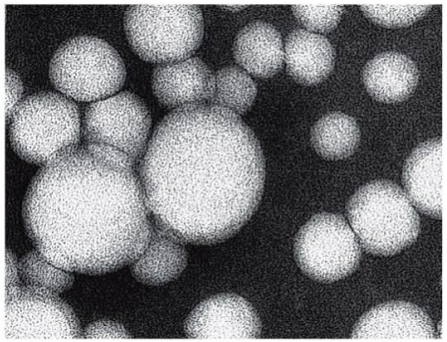
TABLE 21–1 Major Classes of Human Plasma Lipoproteins: Some Properties

Lipoprotein	Density (g/mL)	Composition (wt %)				
		Protein	Phospholipids	Free cholesterol	Cholesteryl esters	Triacylglycerols
Chylomicrons	<1.006	2	9	1	3	85
VLDL	0.95–1.006	10	18	7	12	50
LDL	1.006–1.063	23	20	8	37	10
HDL	1.063–1.210	55	24	2	15	4

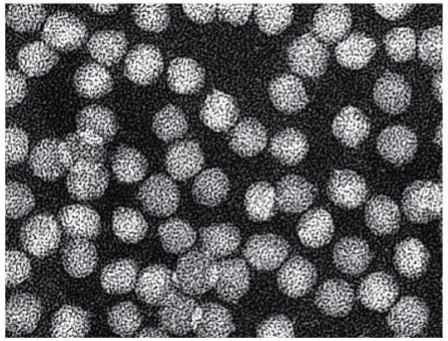
Source: Modified from Kritchevsky, D. (1986) Atherosclerosis and nutrition. *Nutr. Int.* 2, 290–297.



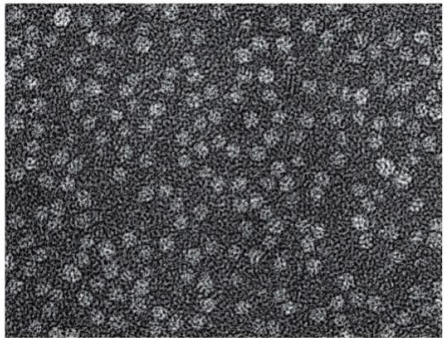
Chylomicrons (×60,000)



VLDL (×180,000)



LDL (×180,000)



HDL (×180,000)

바뀐 보험급여 기준 알아두자 (2014.05.13)

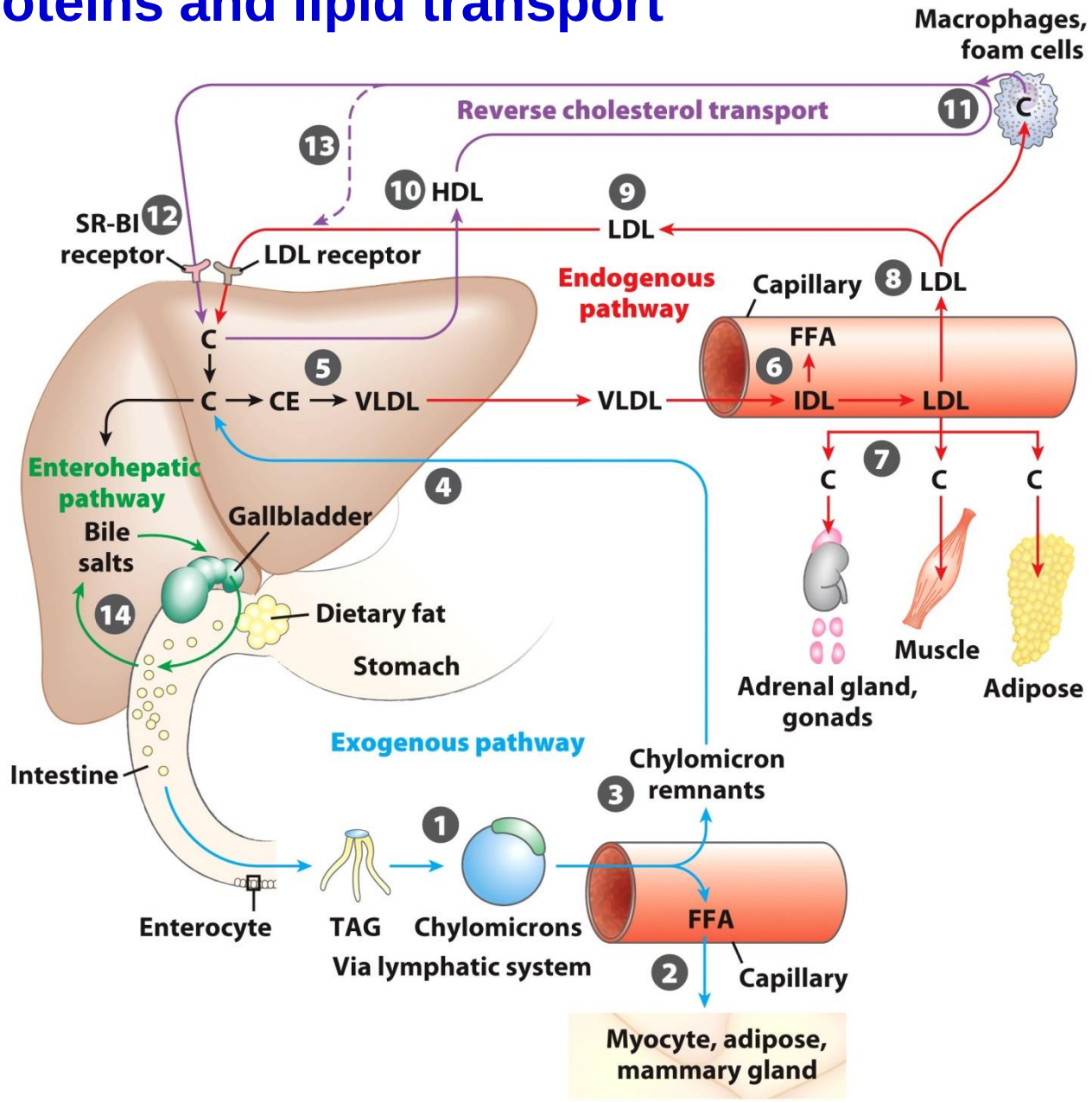
change 1 '총콜레스테롤 . LDL' 기준 변경
이상지질혈증 약을 처방할 때 기존에는 총콜레스테롤 기준(위험요인 없으면 250mg/dL 이상, 위험요인 있으면 220mg/dL 이상)으로 보험급여를 인정했다. 하지만 바뀐 기준에서는 LDL을 기준으로 보험급여를 인정하도록 투여 대상을 정했다. 그 기준은 ▷LDL 160mg/dL 이상(위험요인 1개 이하) ▷LDL 130mg/dL 이상(위험요인 2개 이상) ▷LDL 100mg/dL 이상(관상동맥질환 위험자) ▷LDL 70mg/dL 이상(급성관동맥증후군) 등이다. (건강검진시에는 70~130가 기준치, < 100 가 정상치)

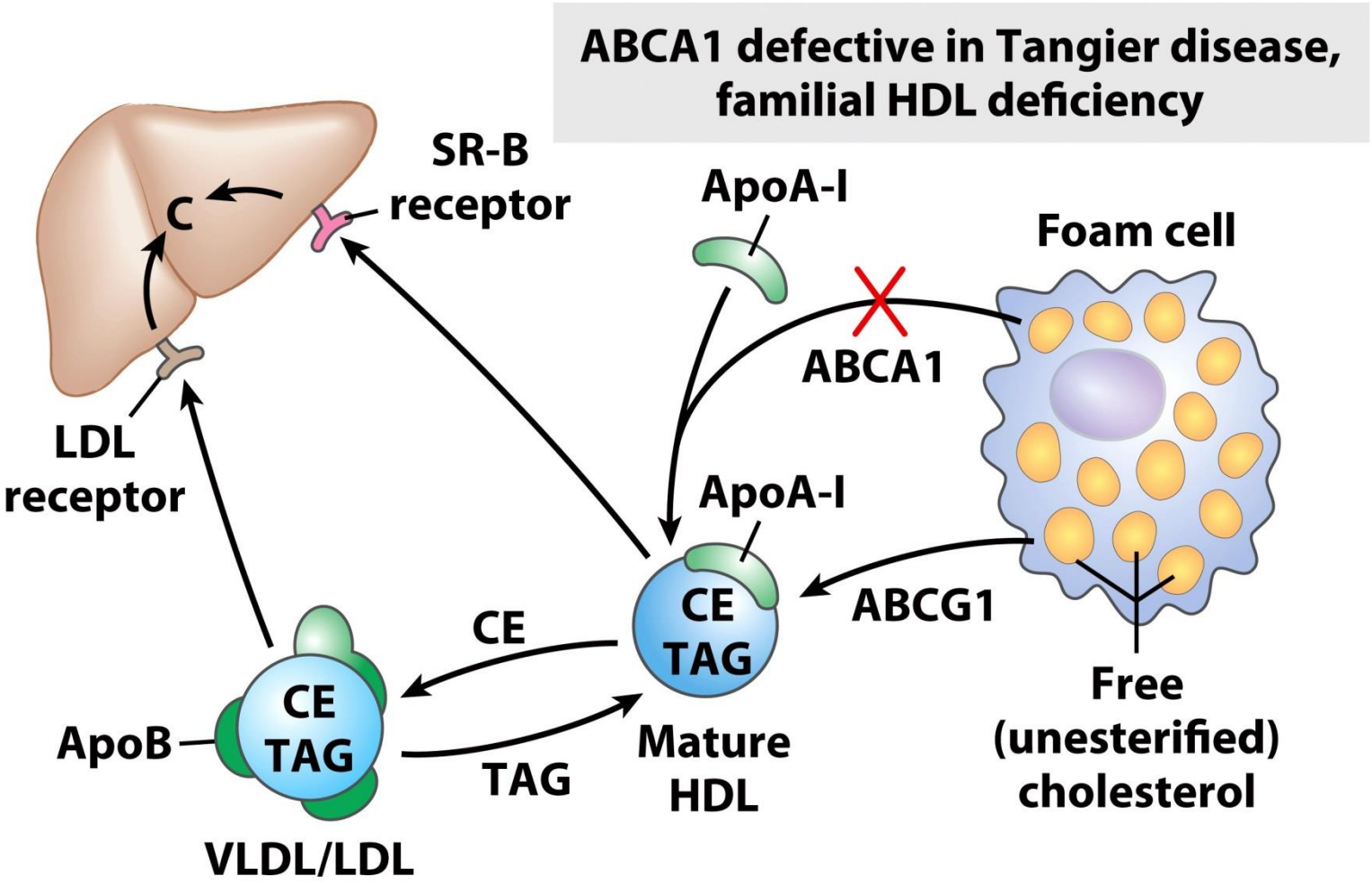
change 2 이상지질혈증 위험요인 규정
이번 기준 변경에서는 '국제 콜레스테롤 교육 프로그램(NCEP) ATP III' 기준에 따라 이상지질혈증의 위험요인들을 명시했다. 기존에는 심장질환자나 당뇨병 환자 등으로 명시했을 뿐, 이에 대한 구체적인 기준이 없었다. 위험요인은 ▷흡연 ▷고혈압(고혈압 약을 먹고 있는 경우이거나 140/90mmHg 이상) ▷낮은 고밀도지단백콜레스테롤(HDL 40mg/dL 미만) ▷관상동맥질환 조기 발병 가족력(부모, 형제, 자매 중 남자 55세 미만, 여자 65세 미만에서 관상동맥질환이 발병한 경우) ▷나이(남자 45세 이상, 여자 55세 이상) 등이다.

change3 고(高)중성지방 2회 검사 기준 제외
혈액 속 중성지방 수치가 높으면 약으로 낮춰야 하는데, 이 때의 급여 기준도 바뀌었다. 과거에는 두 번 연속 400mg/dL 이상(위험요인이 없을 때)인 경우나, 두 번 연속 200mg/dL 이상인 경우(위험요인 있을 때)에 보험급여를 인정했다. 하지만, 바뀐 기준에서는 '두 번 연속'의 조항이 빠지고 500mg/dL 이상이면 약을 보험급여로 처방하고, 위험요인이 있으면 200mg/dL 이상에서 처방하도록 했다.

change4 생활습관 변화 중요성 강조
개정된 기준에서는 생활습관 변화의 중요성을 직접 명시했다. "치료제를 처방할 때는 생활습관 변화를 병행해 치료 효과를 극대화시켜야 한다"고 되어 있다.

Lipoproteins and lipid transport

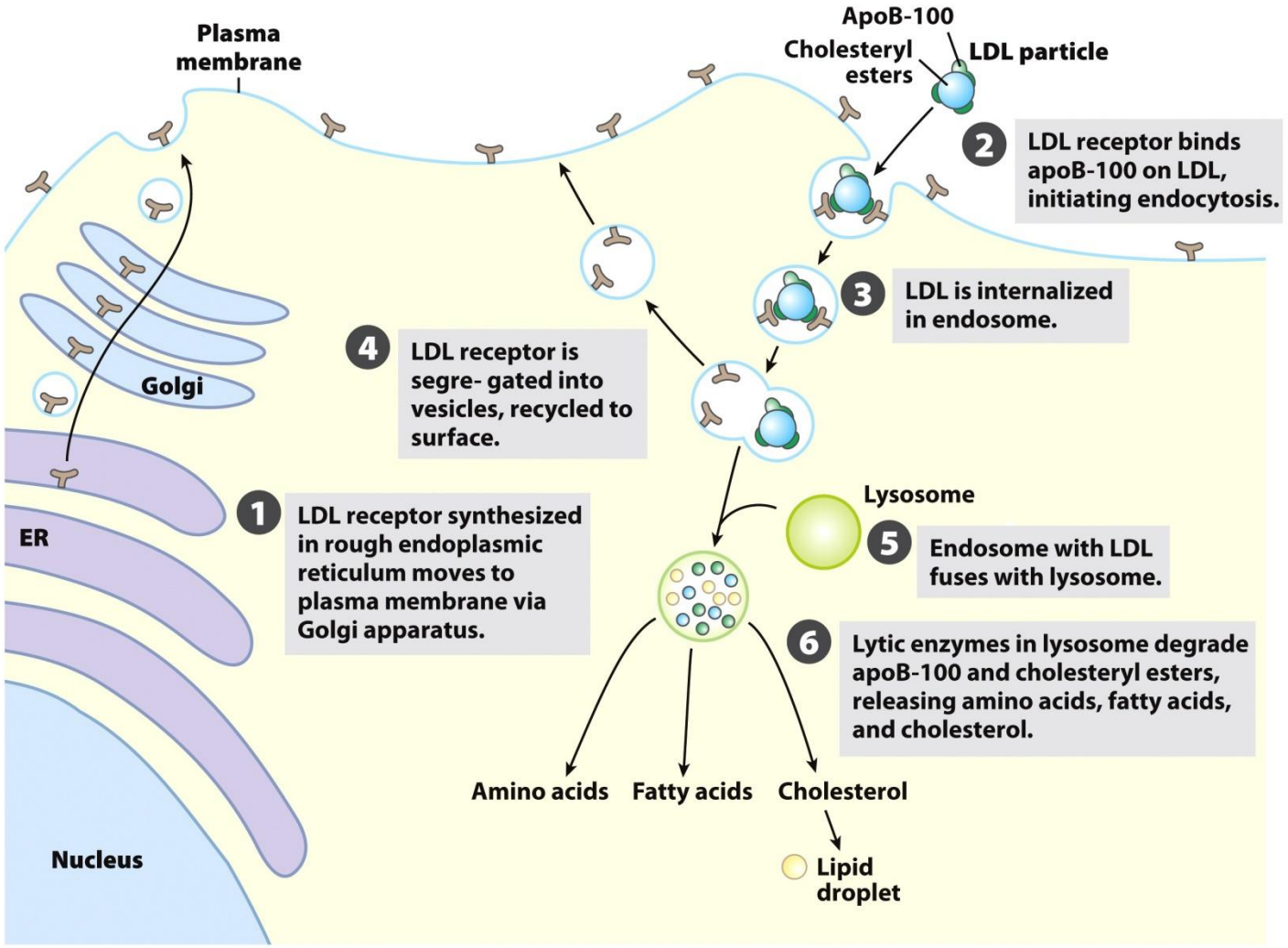




ABCA1 (ATP-binding cassette sub-family A member 1): also known as the cholesterol efflux regulatory protein (CERP)

ABCG1 (ATP-binding cassette sub-family G member 1) : cholesterol and phospholipids transport

Uptake of cholesterol by receptor-mediated endocytosis



High levels of LDL-bound cholesterol causes atherosclerosis



HDL clears cholesterol by reverse cholesterol transport

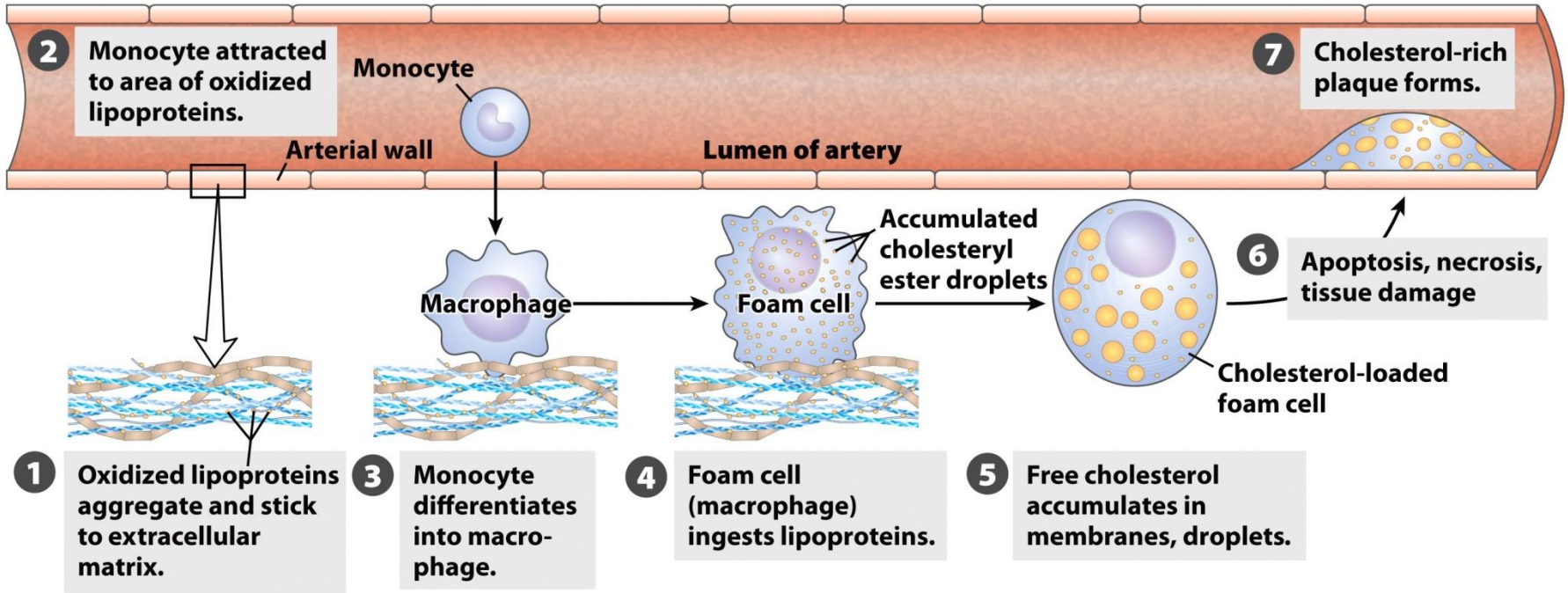
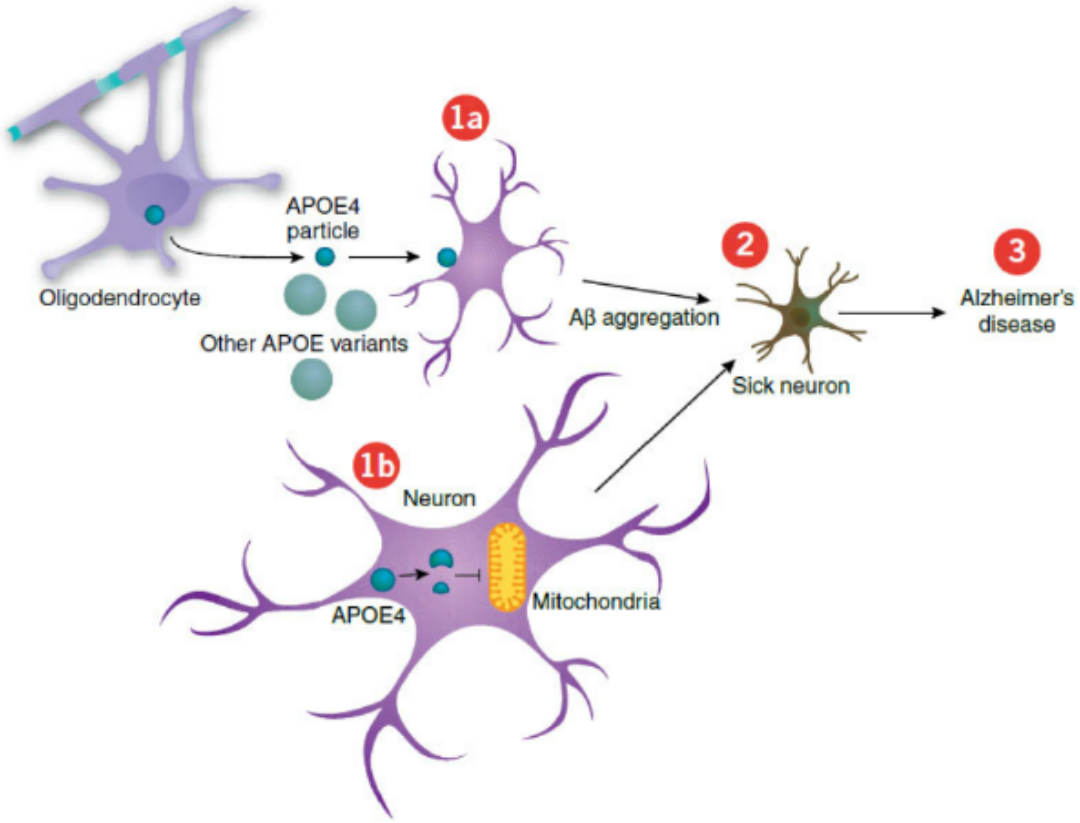


Figure 21-46
Lehninger Principles of Biochemistry, Sixth Edition
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apolipoprotein E4

the smaller E4 compete with full-sized APOE variants, leading to insufficient lipid delivery and higher levels of -amyloid Ab aggregation.



E4 originating from within neurons is proteolytically cleaved into fragments that interfere with mitochondrial activity , leading to neurodegeneration and AD

Summary

1. Cholesterol is made from acetyl-CoA in 4 stages
2. Cholesterol and other lipids are carried on plasma lipoproteins
3. Cholesteryl esters enter cell by receptor-mediated endocytosis
4. Steroid hormones are formed by side-chain cleavage and oxidation of cholesterol